Contact Angle Analysis

Contents

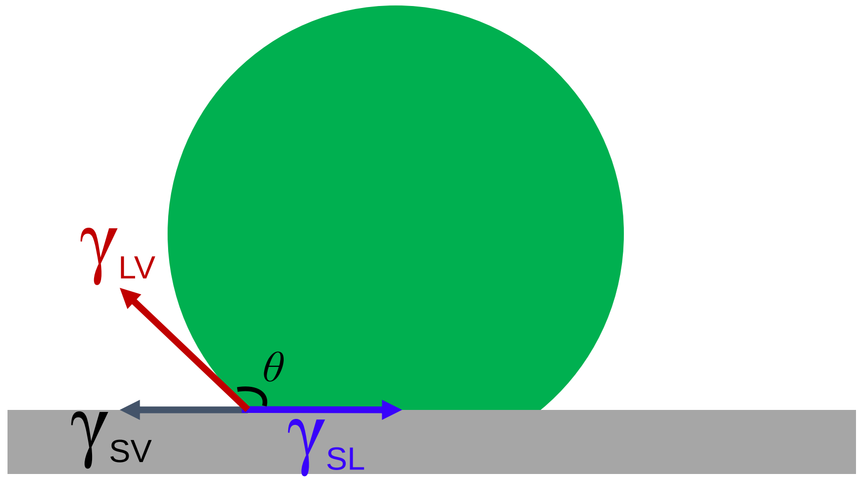
- Hydrophilic (亲水性) and hydrophobic (疏水性) in nature
- **Contact Angle and Wettability**
- **Young’s equation**
- **Sessile drop method (固着液滴法)**

Contact Angle and Wettability

Contact Angle is an angle formed by a liquid at the three-phase boundary where a liquid, gas and solid interest.

Case 1:

Case 2:



γ_{SV}: is the solid/vapor surface tension

γ_{SL}:is the solid/liquid surface tension

is the liquid/vapor surface tension

Θ is the contact angle

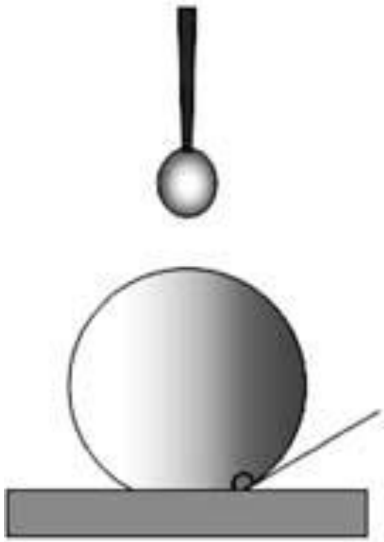
$$\gamma_{SV} - \gamma_{SL} - \gamma_{LV}\cos\theta = 0$$

(Young equation)

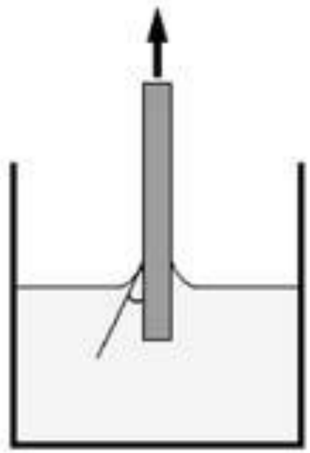
Only two measurable quantities:
1.the contact angle θ;
2.the liquid-vapor surface tension, γ_{LV}.

- θ, can be determined during the test
- γ_{LV} can be determined and calculated using the pendant drop method （吊坠法） by fitting the shape of the suspended droplet <u>(in a captured video image)</u> to the Young–Laplace equation

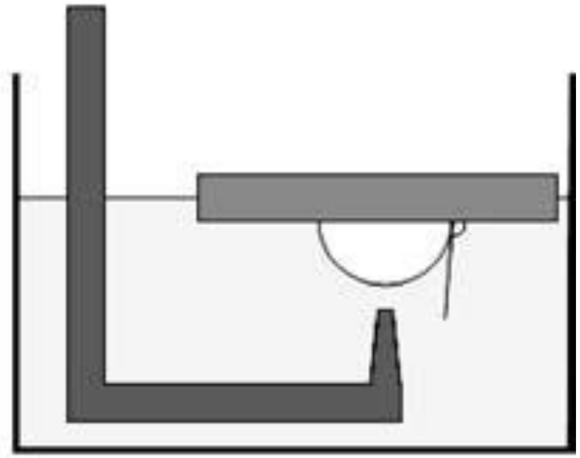
Five ways that the contact angle (θ) can be measured



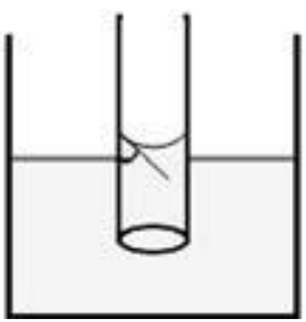
(a) Sessile drop method



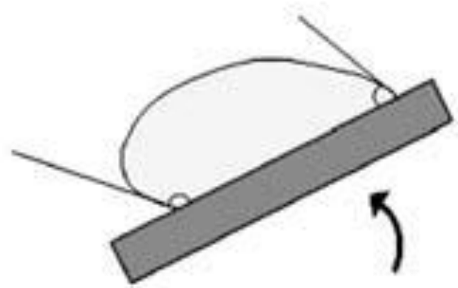
(b) Wilhelmy plate method



(c) Captive air bubble method

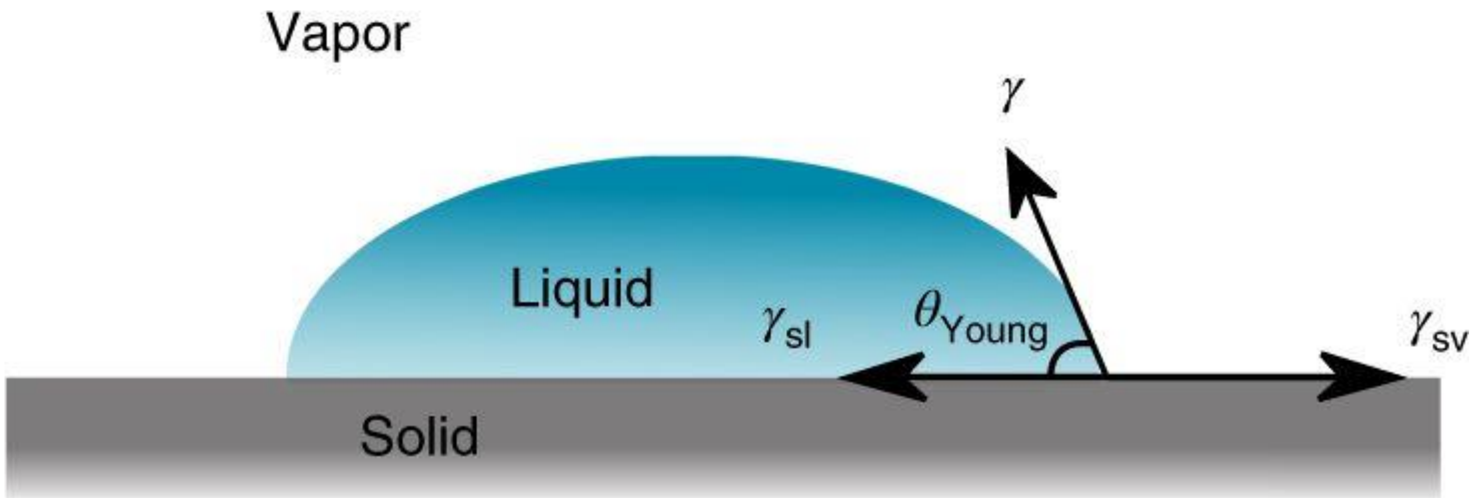


(d) Capillary rise method



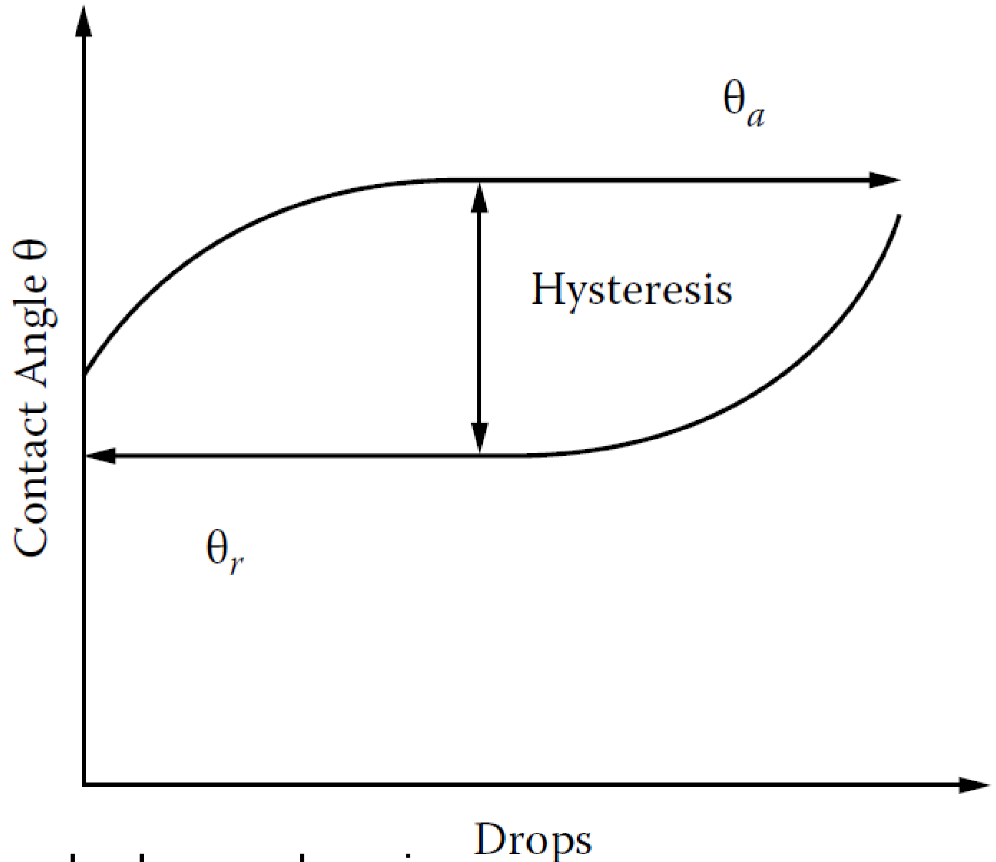
(e) Tilting substrate method

Sessile drop method



- The contact angle is measured by a contact angle goniometer
- An optical system to capture the profile of the liquid droplet on a solid substrate
- The angle formed between the liquid–solid interface and the liquid–vapor interface is the contact angle
- A typical system uses a high-resolution camera and software to capture and analyze the contact angle

Dynamic：



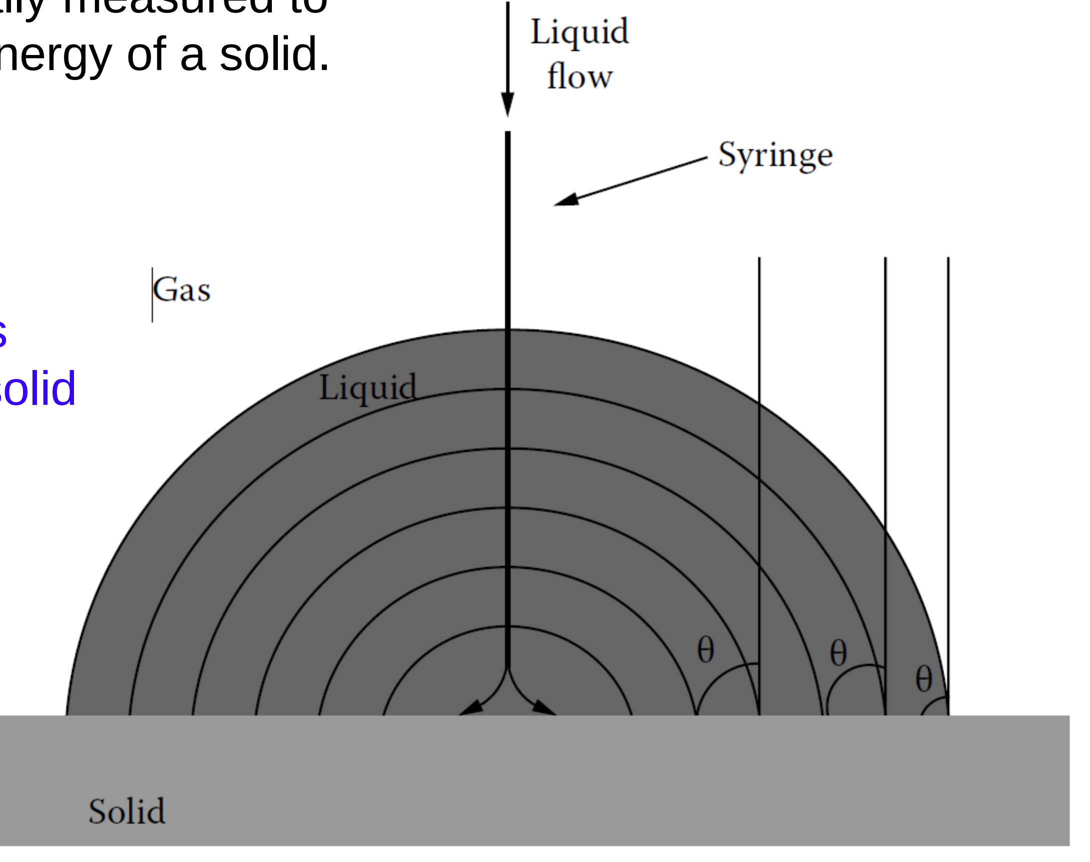
- A plateau is reached upon dropping
- The advancing and retreating angles are usually not equal
- The hysteresis (滞后) can result from sample prehydration, surface roughness, chemical heterogeneity, evaporation, or molecular movement.
- The receding contact angle allows one to measure the degree of hysteresis inherent in the sample surface

Tip: The flow rate should not be too high

Dynamic

Advancing Angle

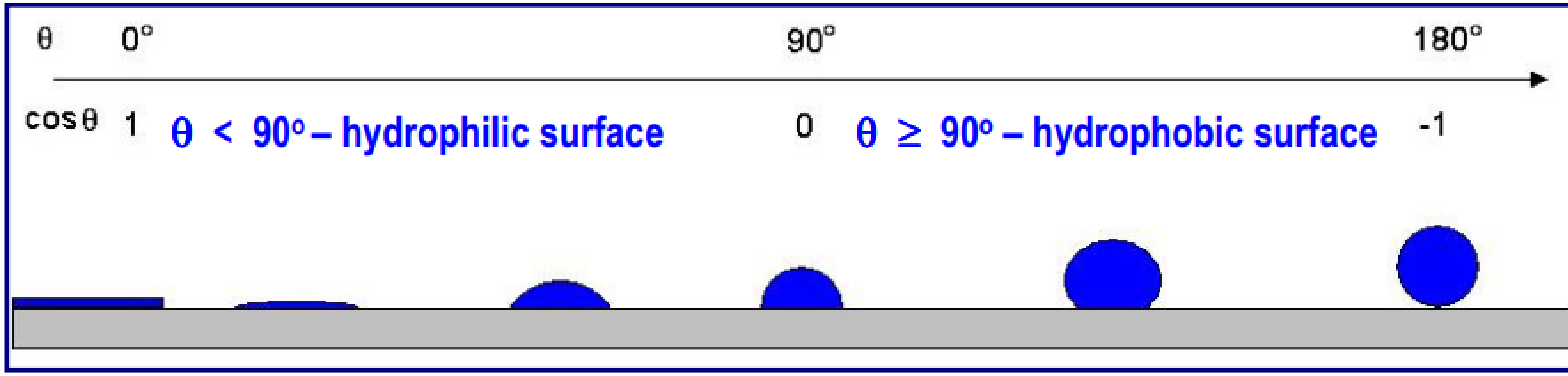
- Advancing angles simulate a fresh surface for the contact angle
- Most reproducible way of measuring contact angles
- Advancing angles are normally measured to determine the surface free energy of a solid.



Receding Angle

- Possible to make statements about the roughness of the solid or chemical inhomogeneties
- Not suitable for calculating surface energies

Note that the contact angle is always denoted by symbol *θ*.



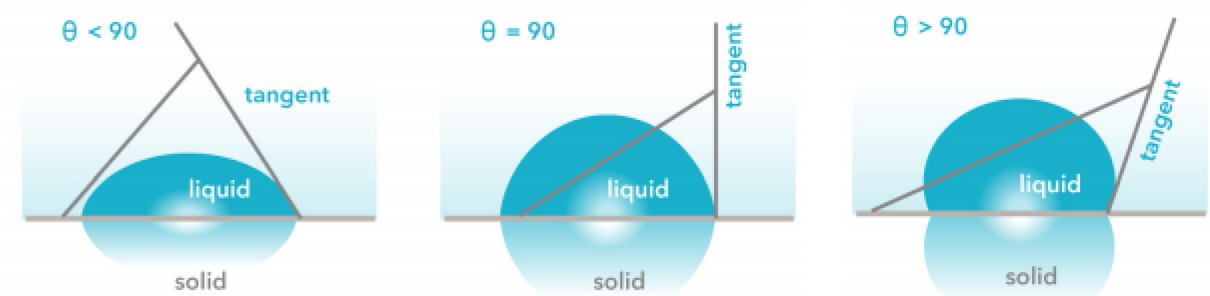


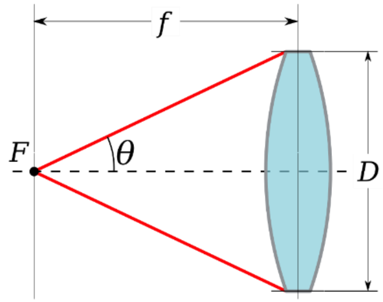
Fig. 13.1. Scheme of contact angles on the hydrophobic and hydrophilic surfaces
http://www.attension.com/?id=1092&cid=

The Optical Microscope

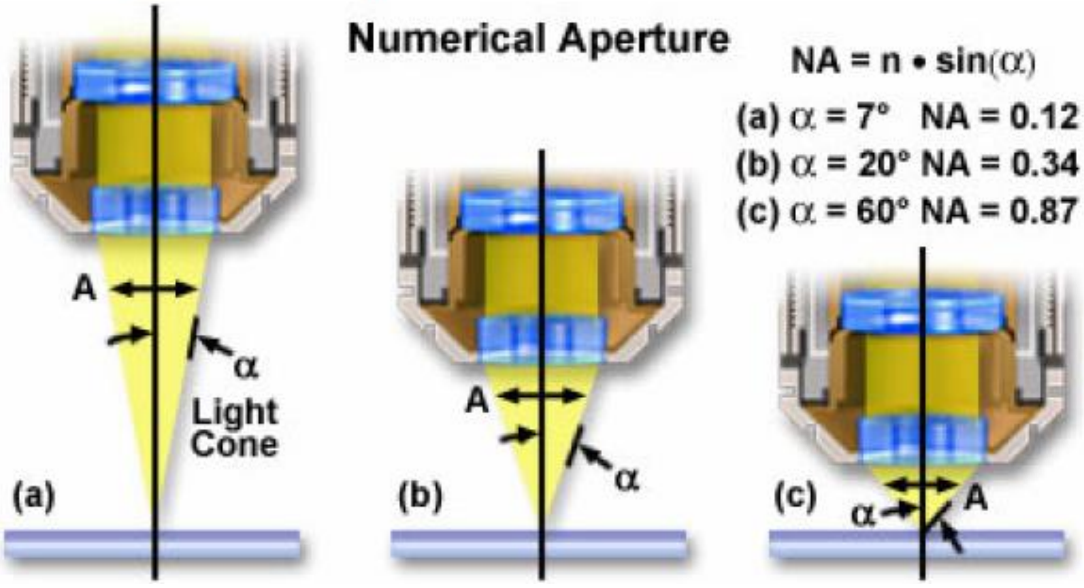
Light Microscopy

- Numerical Aperture, *NA*
  - Ability of the lens to collect light coming off of a specimen

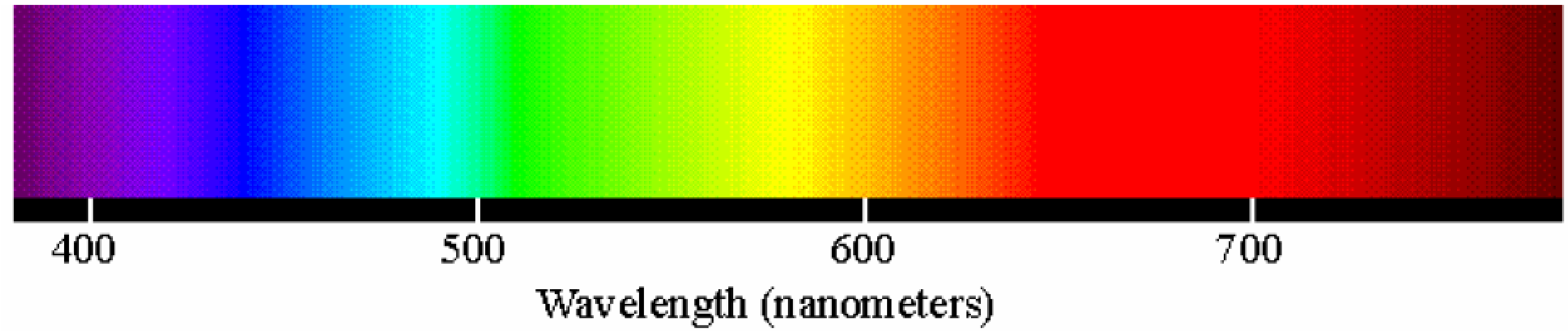
$$NA = n \sin\theta$$



  - n = index of refraction
  - θ =half the angle of acceptance



Light Microscopy



- Resolution, R
  - Minimum distance between two objects which can be distinguished.

$$R = 0.61 \frac{\lambda}{NA}$$

  - NA= numerical aperture λ= wavelength

Scanning Electron Microscopy

Scanning Electron Microscopy

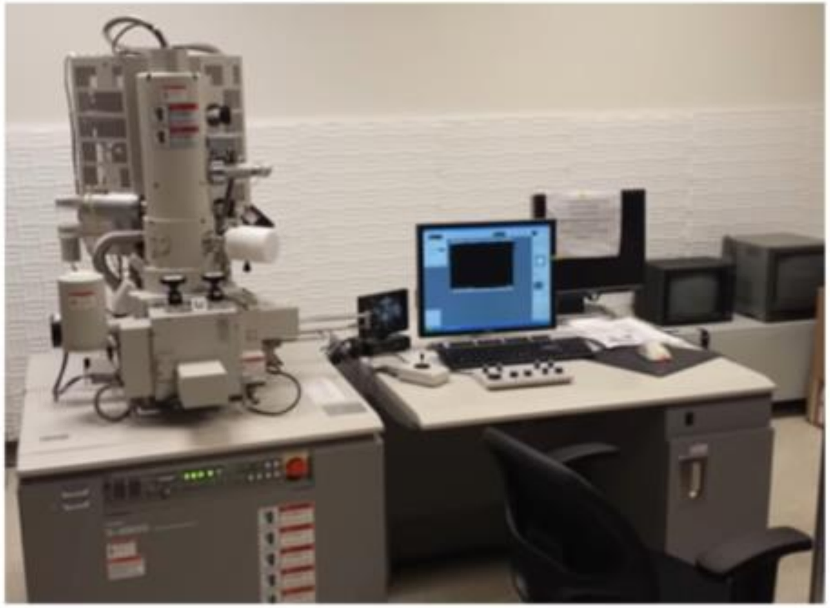
4.1 Scanning Electron Microscopy Principles

4.2 Electron Beam–Specimen Interaction

4.3 SEM Operating Parameters

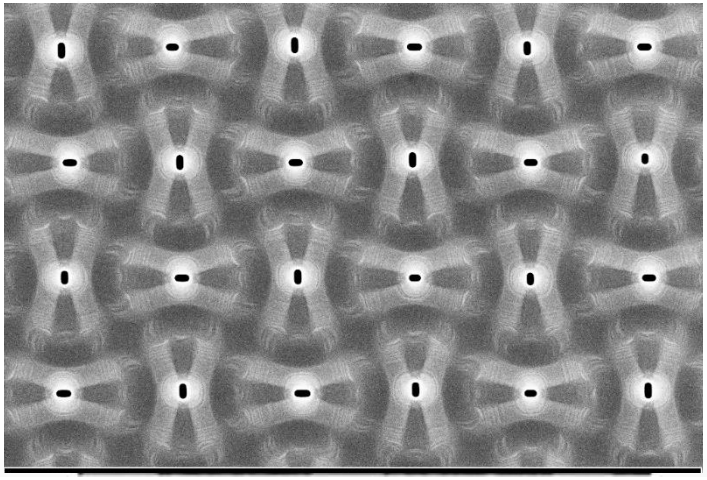
4.4 Applications

What is an Scanning Electron Microscope?



MRL Hitachi S4800

- A Scanning Electron Microscope (SEM) is an instrument for observing and analyzing the surface microstructure of a bulk sample using a finely focused beam of electrons
- SEMs can achieve 1-4 nm imaging resolution depending on the instrument

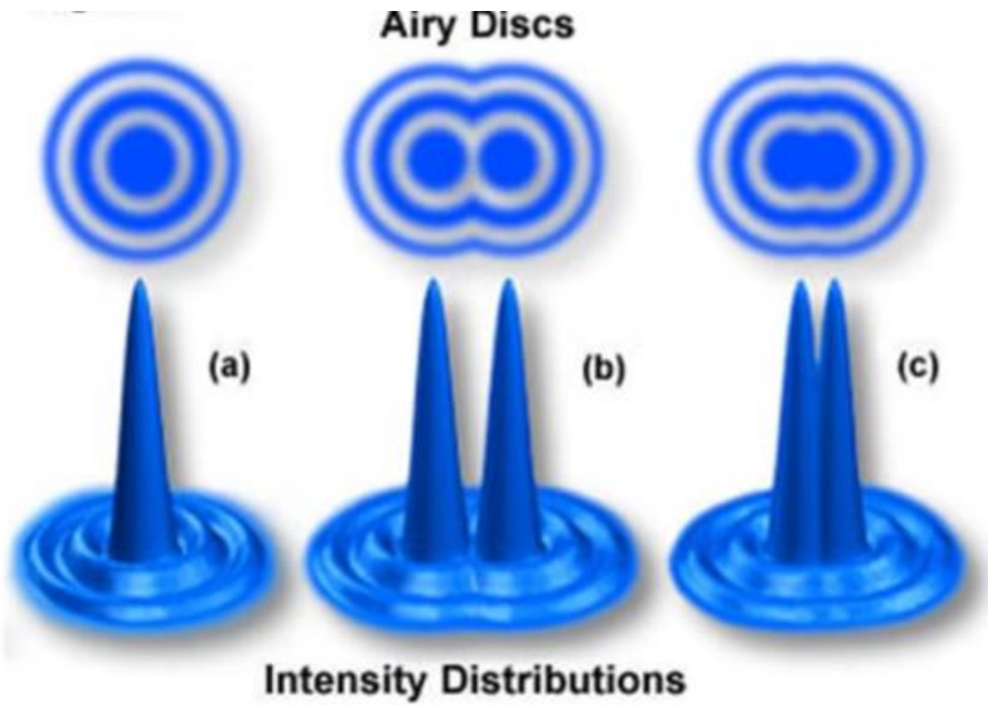


SEM image of Ge on Si from my lab

Primary Applications

- **Surface topography / morphology**
- **Composition analysis**
- Crystallography
- Optical / Electronic properties
- and more …

Resolution-what is it?



- Ability to tell the difference between two close objects vs. one big object
- Electron beam is not a point → Gaussian with nanoscale width
  - Diffraction causes the 'Airy' disc
  - Rayleigh Criterion: Two points are resolvable if the maximum of the first disc coincides with the minimum of the second

瑞利准则

What are the Advantages of Electrons?

Resolution limits (diffraction)

Abbe equation: $$d = \frac{0.61\lambda}{n \sin \alpha}$$

d – resolution
λ – wavelength of particle
n – refractive index relative to free space
α – semi-angle in radians

Microscopy	λ (nm)	nsinα	d (nm)	Mag.*
Light	400	1	**200**	1000
Electron	0.3	0.015	**0.3**	500000

This is the best (approximate) possible resolution. In practice the resolution will be worse…

*Practical mag. = 0.2mm (eye)/d

Resolution

Combined with theory of relativity

$$eU = mc^2 \text{-} m_0 c^2$$ U *is* the accelerated voltage

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\lambda = \frac{h}{mv},$$ From matter wave

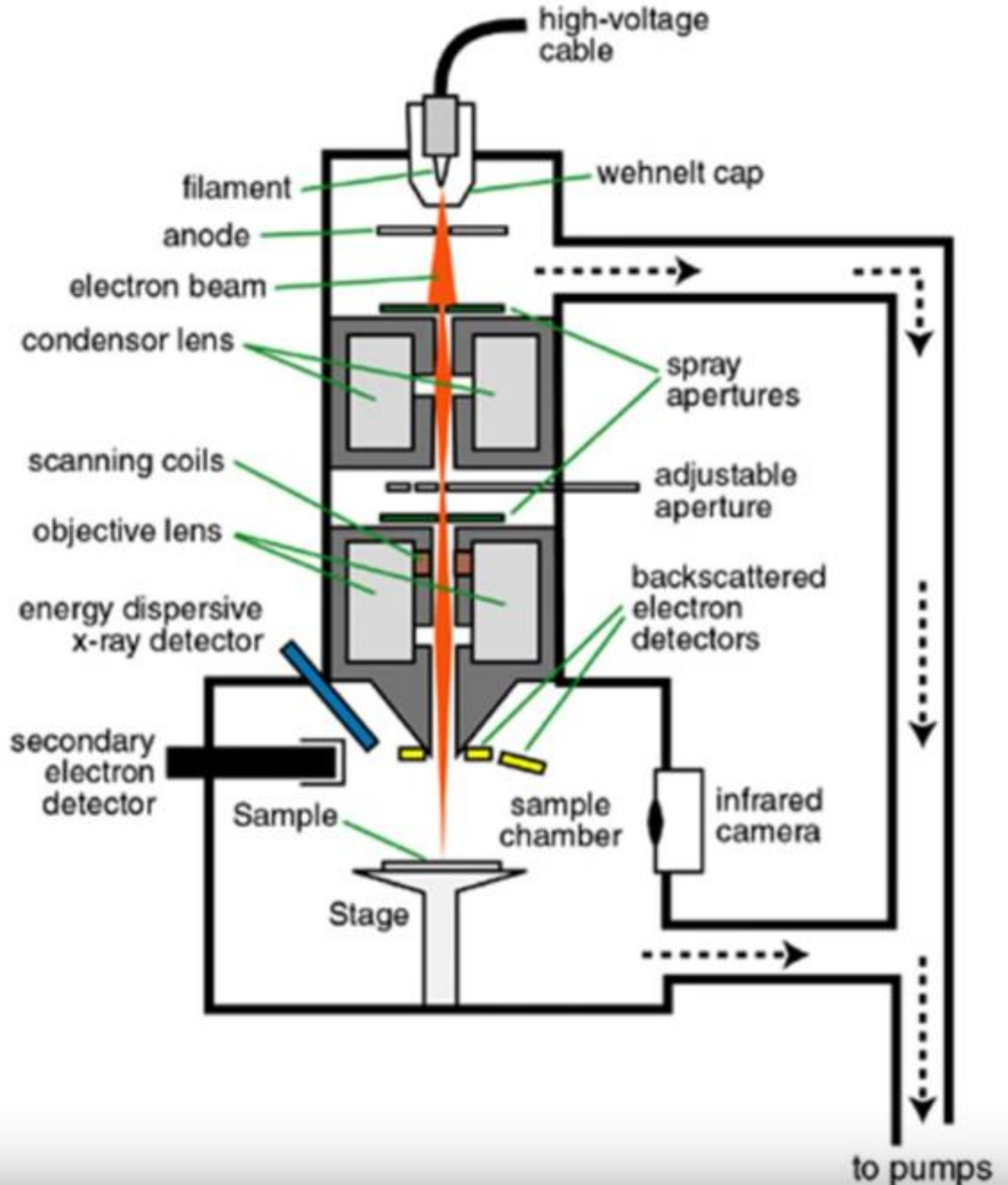
$$\Rightarrow \lambda = \frac{h}{\sqrt{2m_0 eU\left(1+\frac{eU}{2m_0 c^2}\right)}} = \frac{12.25}{\sqrt{U(1+10^{-6}U)}}$$

Scanning Electron Microscopy Principles

1. The Electron Gun
2. The Condenser and Objective Lenses
3. Scanning Coils
4. Specimen Chamber

SEM Component Breakdown

- Vacuum system
- Electron source (gun)
- Optics
  - Lens
  - Aperture
  - Deflectors **(导流板)**
- Scanning system
- Detectors



A General Look at Electron Sources

Purpose:
- Produce electrons
- Roughly shape the beam
- Set the electron energy
- (Accelerating Voltage)

Two source types:
- Thermionic (heat)
- Field emitter (electric field)

Each source has advantages and disadvantages

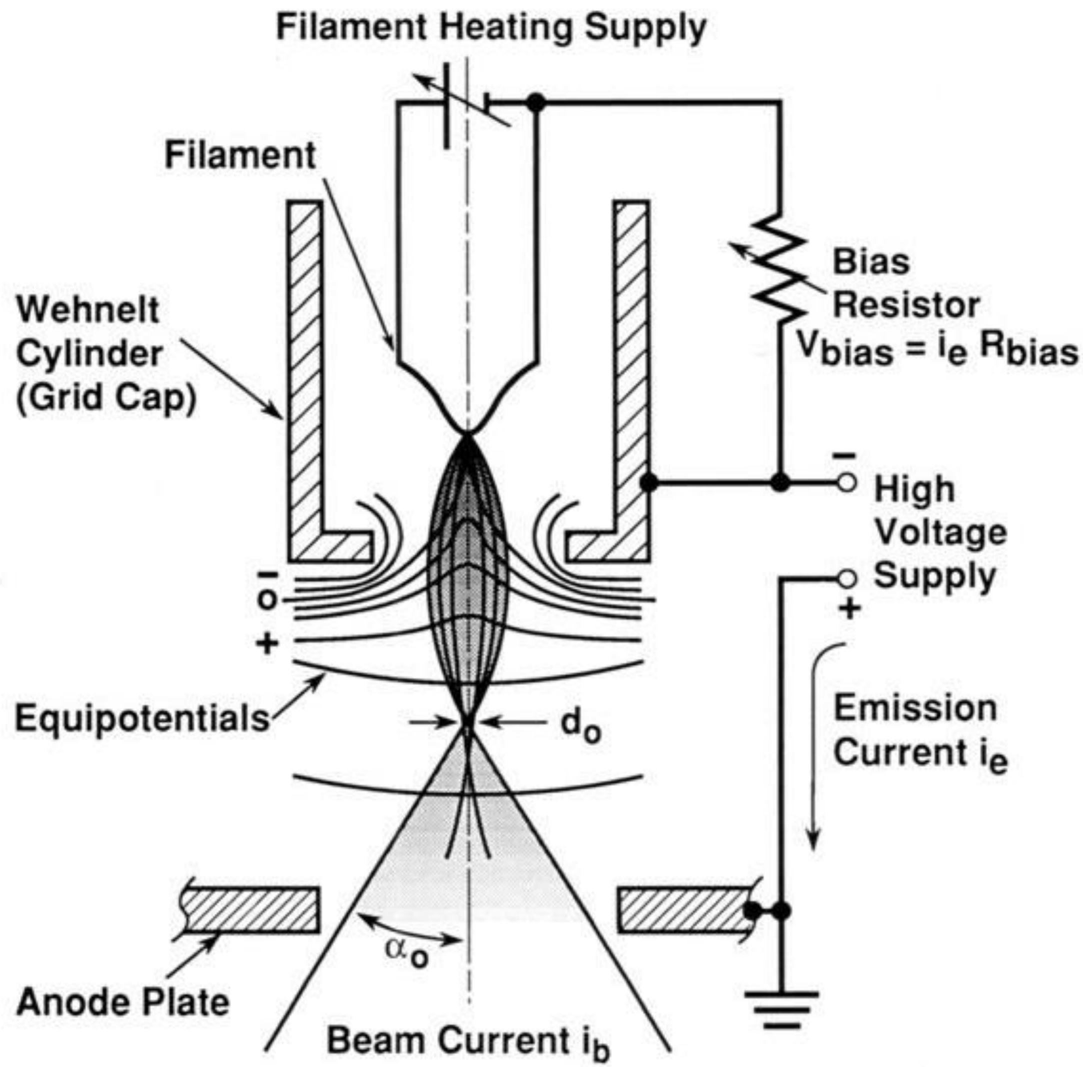
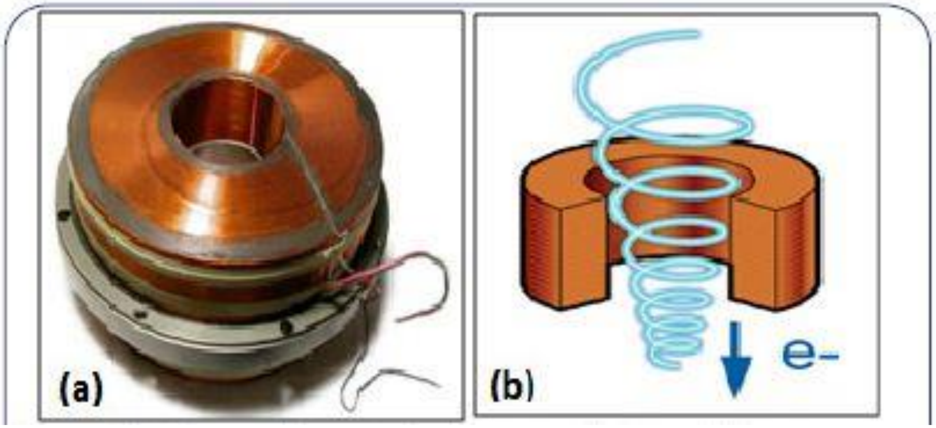


Figure 2.4. Schematic diagram of the conventional self-biased thermionic (triode) electron gun (adapted from Hall, 1966).

Resolution

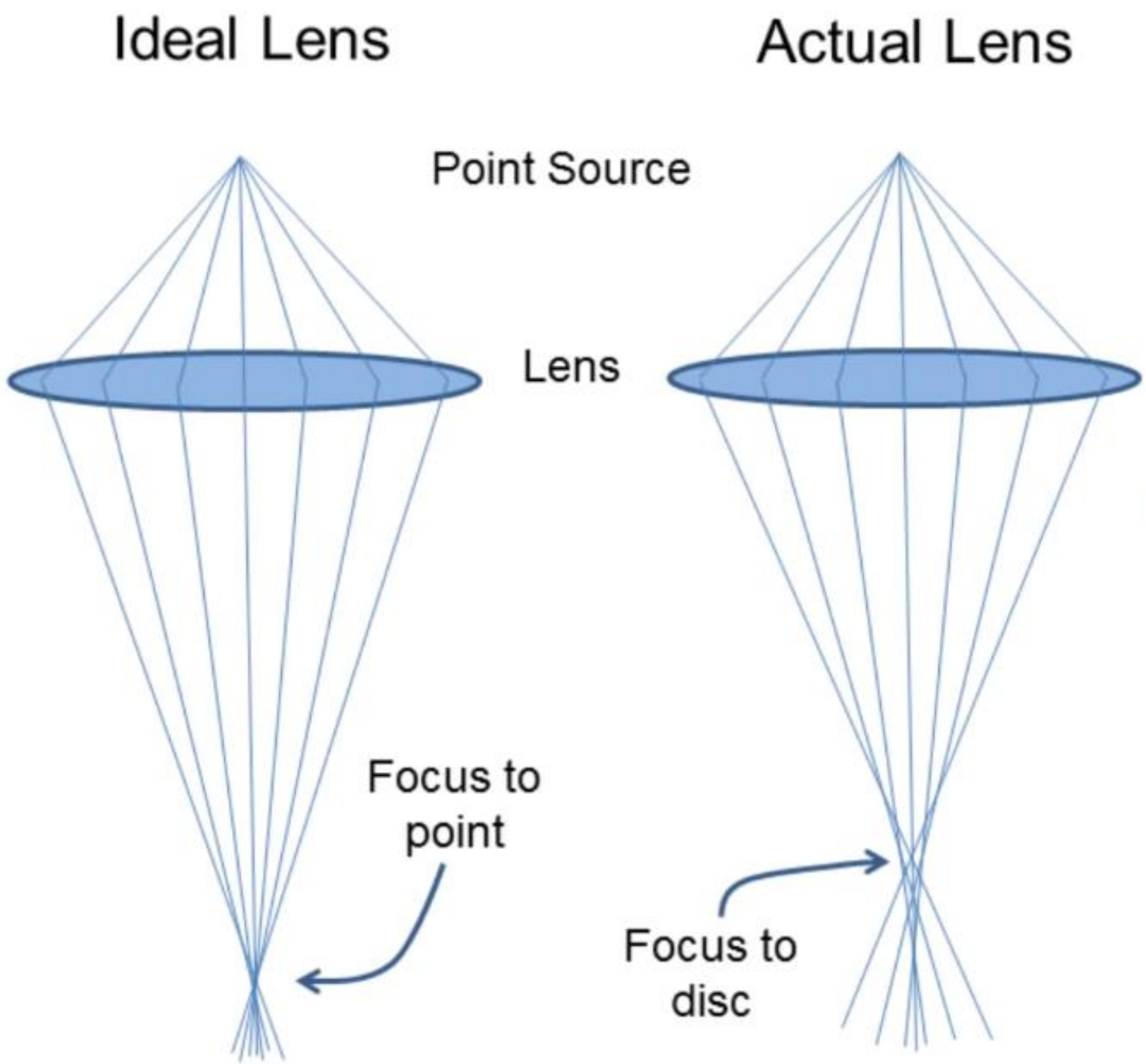
- Obviously, the wavelength is not the main factor affecting the resolution of TEM.
- At present, we can't make electron microscopes with wavelength-limited resolution because we can't make perfect magnetic lenses like optical microscopes.

Magnetic lenses:



- This huge difference is caused by the aberrations of magnetic lenses. Therefore, several other electron microscopes with low acceleration voltage have higher resolution through aberration correction.
- Various aberrations (spherical aberration, chromatic aberration, astigmatism, distortion) of the magnetic lens, especially the influence of the spherical aberration on the numerical aperture of the objective lens, prevent the improvement of the resolution of the electron microscope.

Rays Ideally are Focused to Same Plane…

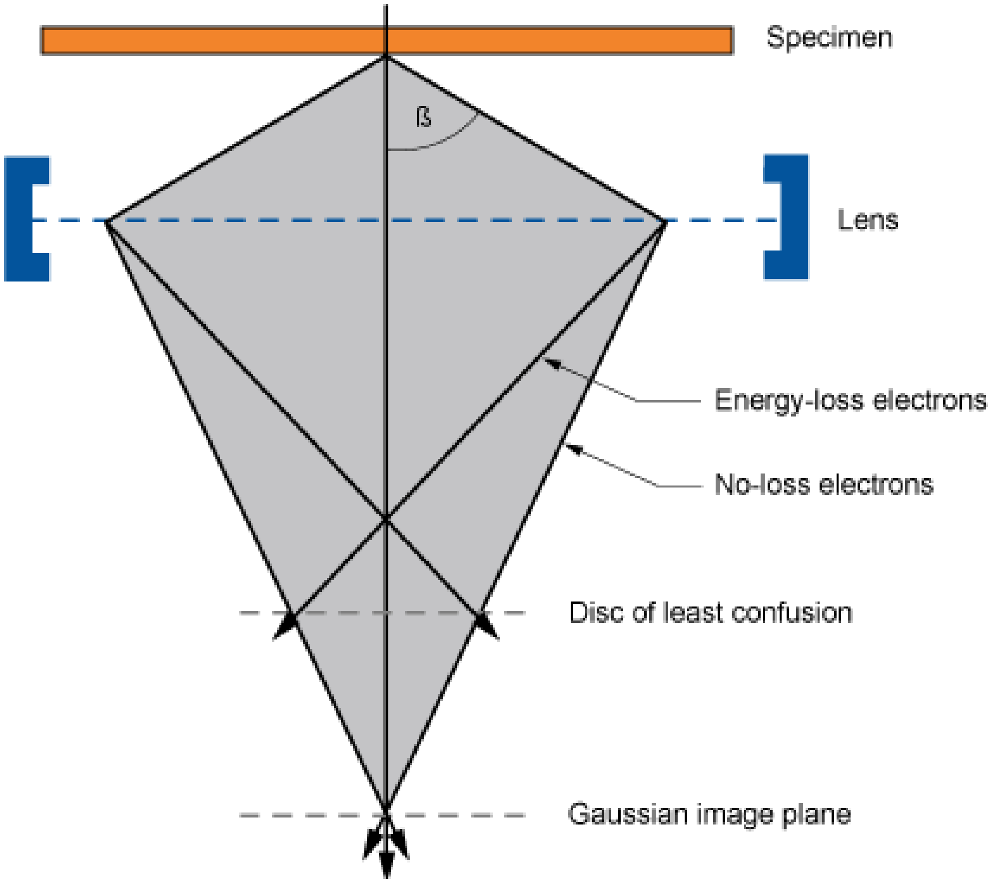


Limiting Parameters

- Spherical Aberration
- Chromatic Aberration
- Aperture diffraction
- Astigmatism

These aberrations reduce resolution!

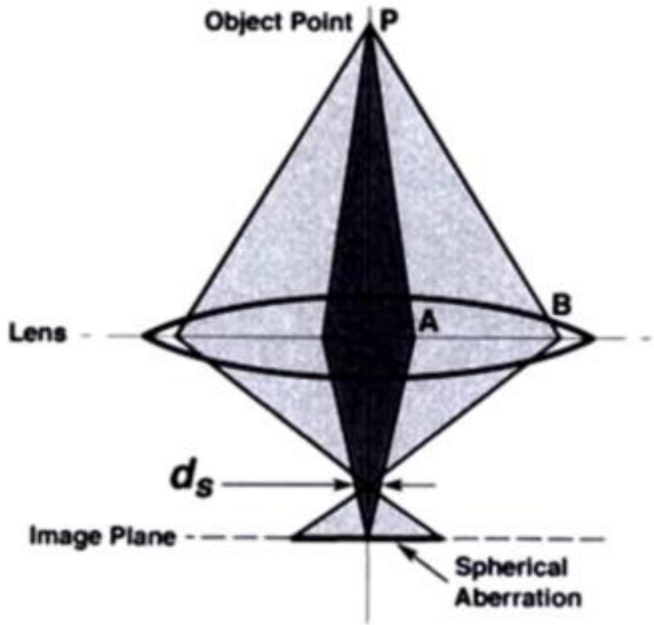
Aberration from Electron Source

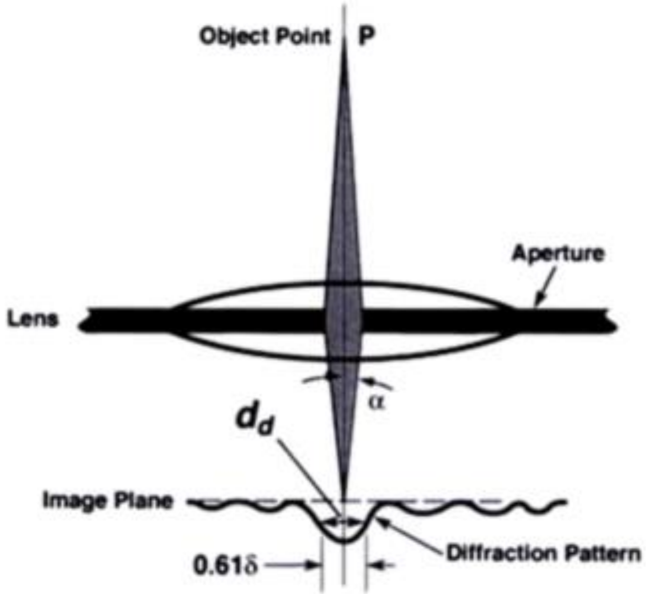


Chromatic Aberration
色差

- Electrons with different energy (velocity) are focused to different points in the image plane
- Increases with source energy spread (heated sources!)
- Decreases with increasing electron energy (E_0)

Aberration from Lenses and Apertures





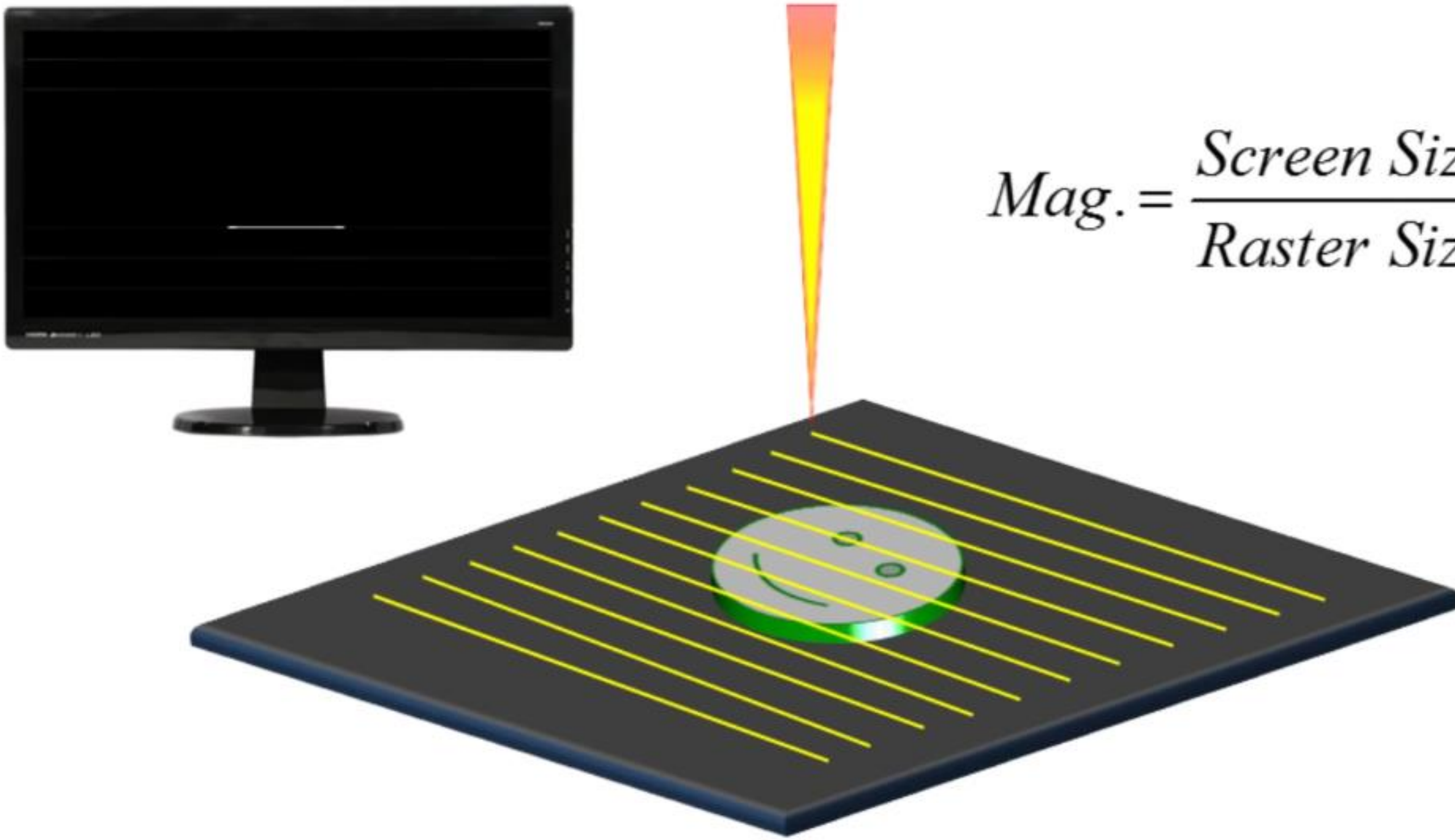
Spherical Aberration (球差)

- Electromagnetic lens (mag-field) are not uniform
  - —Stronger towards outside
  - — Weaker towards center
- Off-axis rays are overfocused
- Solution! Add a limiting aperture

Aperture Diffraction

- Diffraction effects increase with smaller apertures
- Optimum aperture sizes determined by manufacturer

The “Scanning” part of SEM

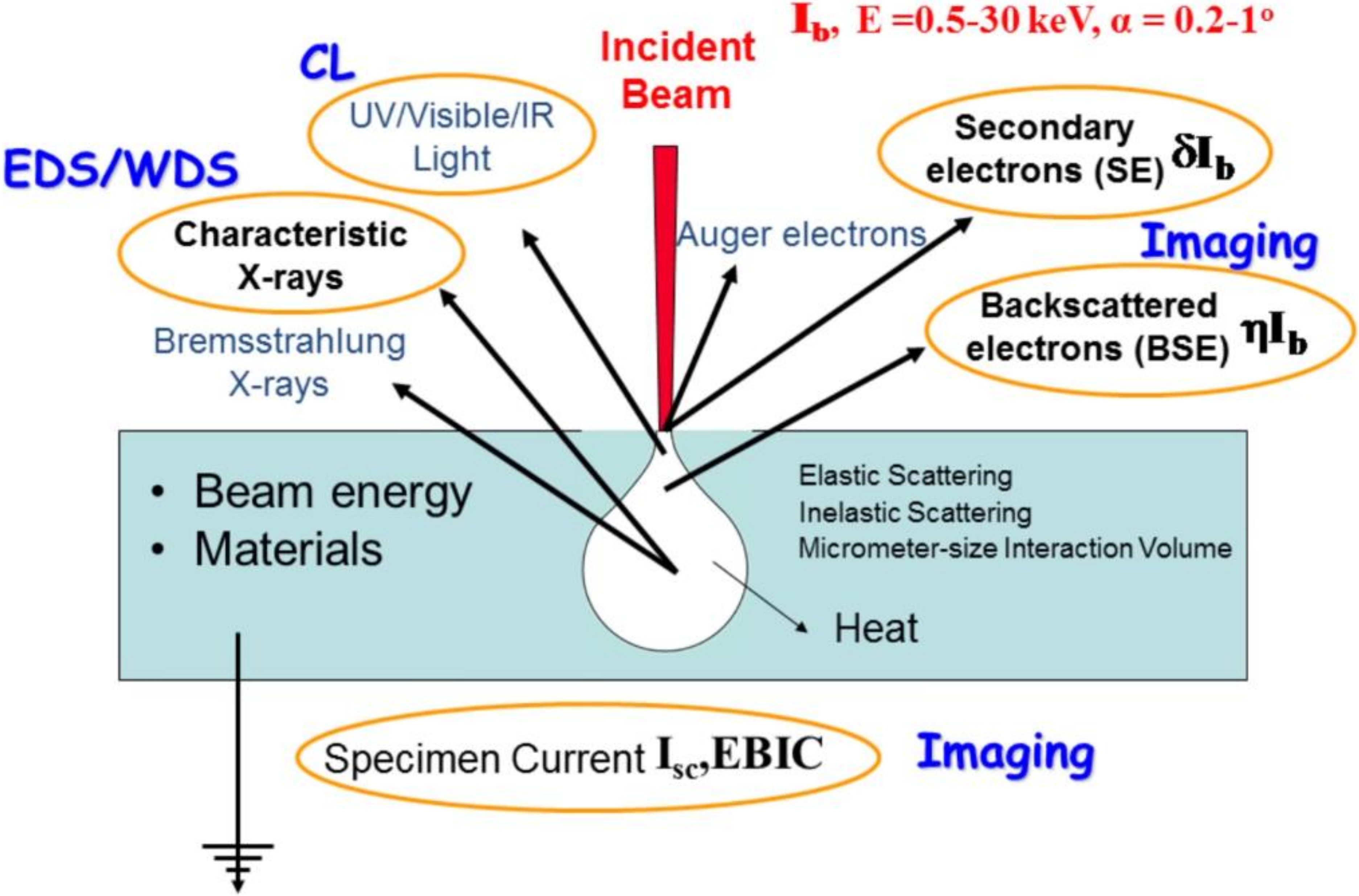


$$Mag. = \frac{Screen\ Size}{Raster\ Size}$$

- Deflection coils are used to steer the beam
- Beam continuously scans across the surface of sample, then flies back to starting point

Electron-Matter Interaction and Detectors

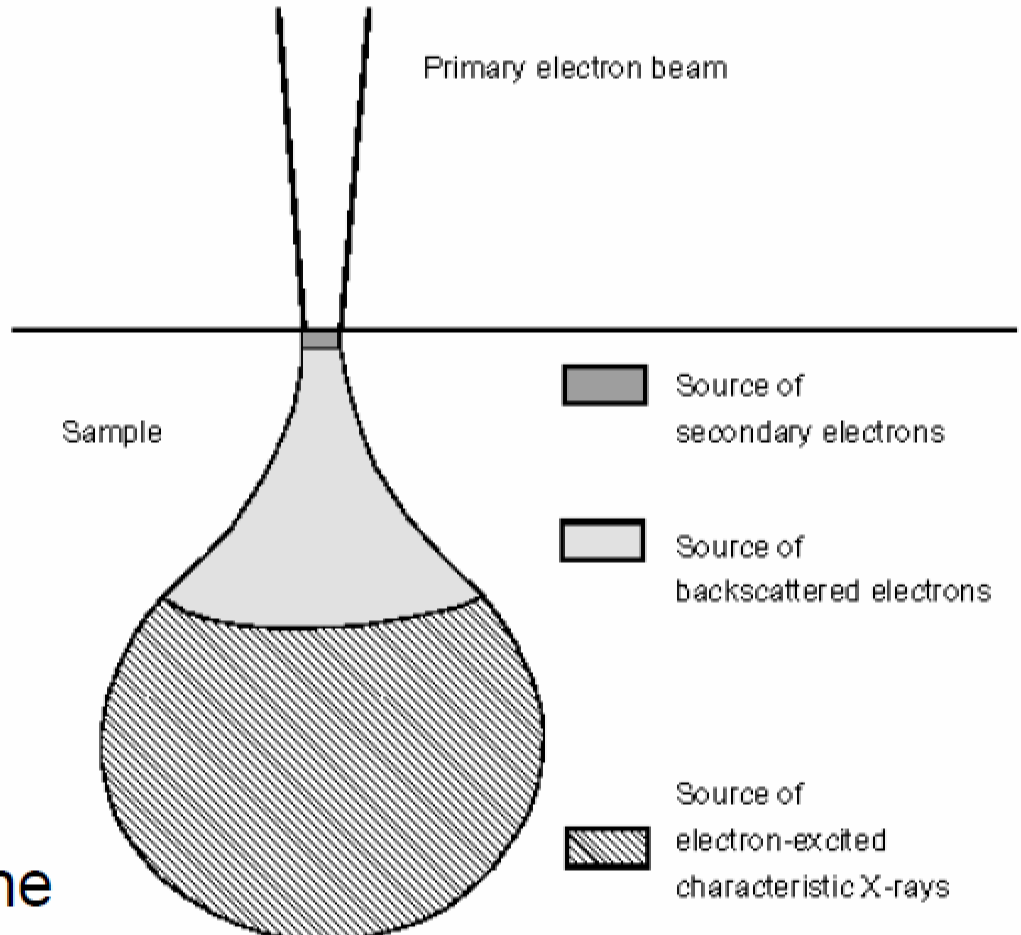
Electron-Matter Interactions



Effective Excitation Volumes

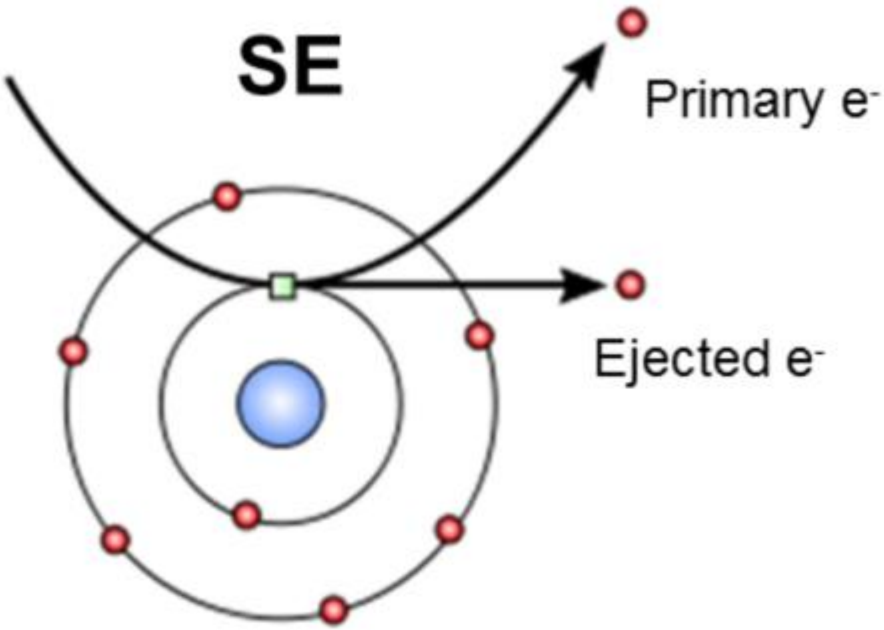
Effective excitation volumes defined by:

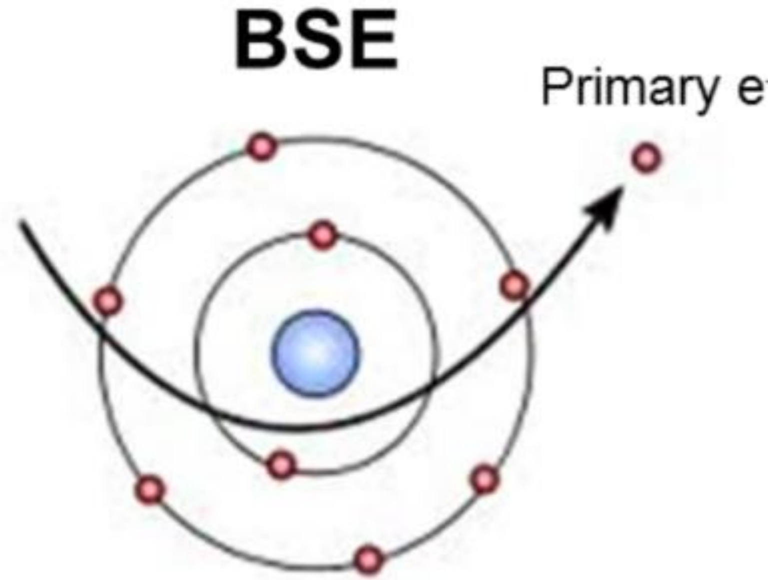
- Excitation Energy
- Atomic Number
- Escape Depth



Escape depth often determines the experimental interaction depth (ie. depth resolution).

Generation of Secondary and Backscatter Electrons

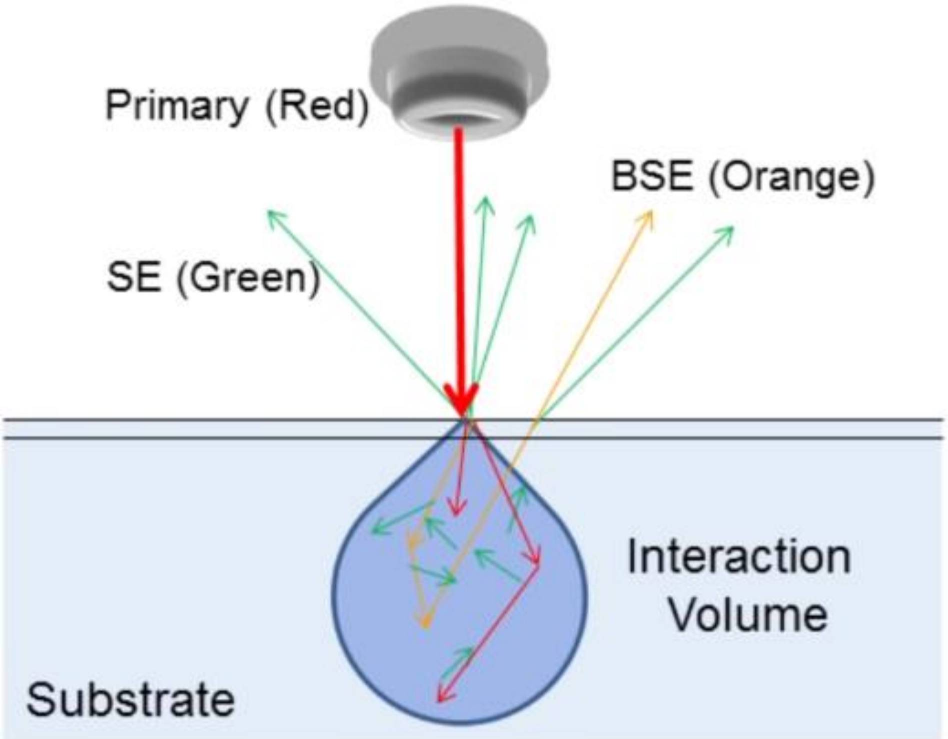


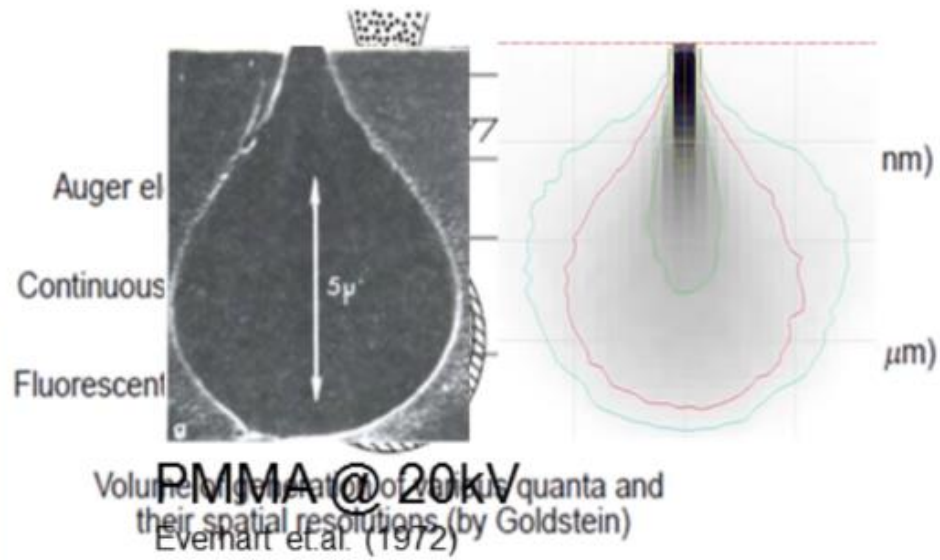


Secondary electrons are low energy electrons (< 50eV) ejected from the specimen atoms by the energetic primary beam

Backscattered electrons are primary beam electrons scattered back out of the sample. BSE have a range of energies (up to the primary energy)

Interaction Volume of Primary Electrons





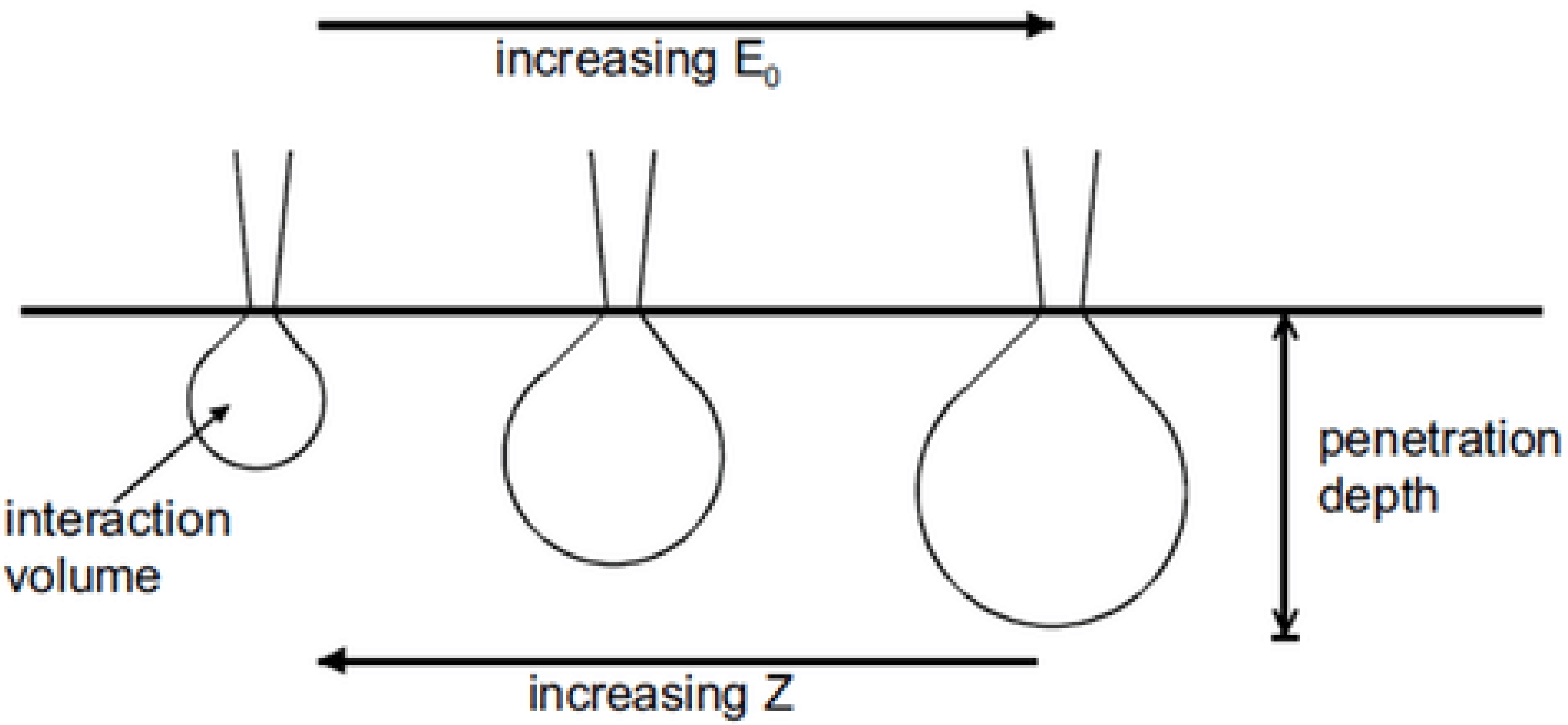
Type	Energy	Range	Sensitivity
SE	<50eV	Few nm **(5-10)**	Surface topography
BSE	<PE	Partial volume	Material (Ave. Z)
X-ray	<PE	Full volume+	Element

Higher primary energy → Larger interaction volume

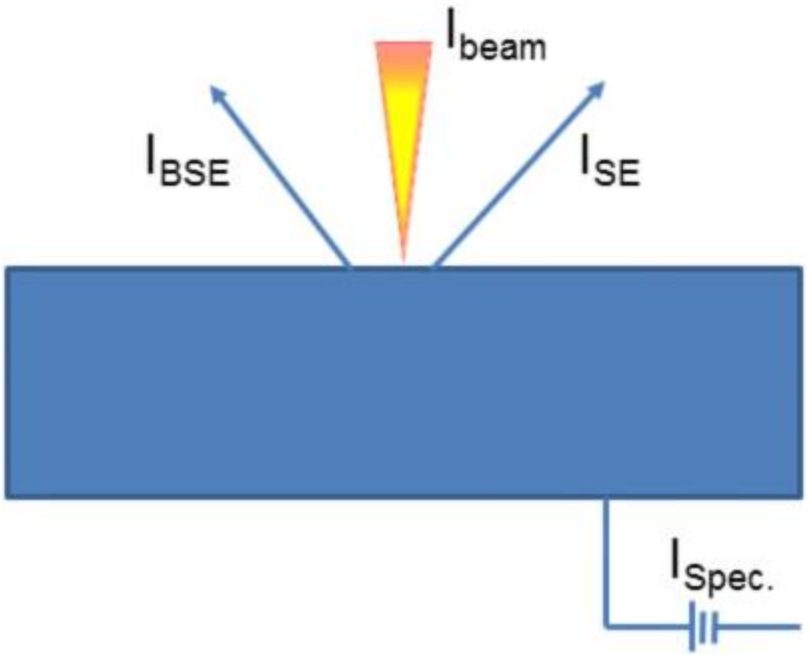
Keep the interaction volume in mind when analyzing samples!

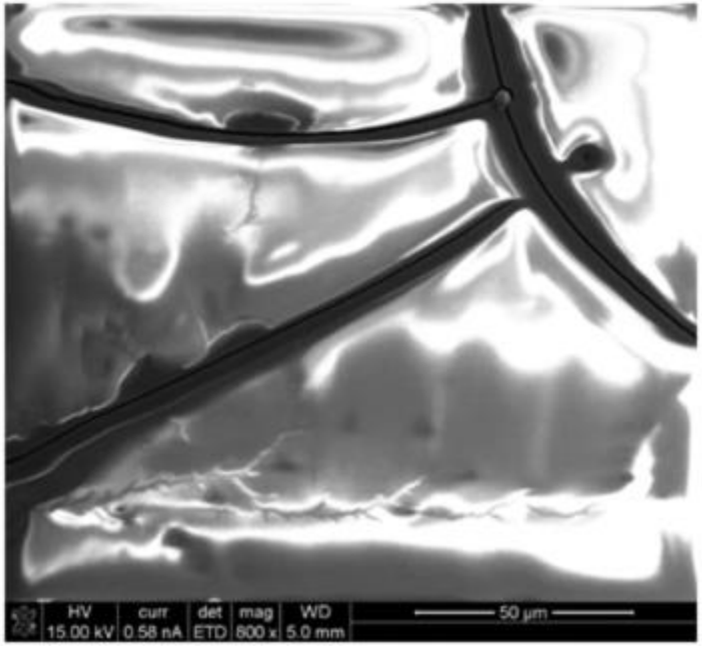
Interaction Volume of Primary Electrons

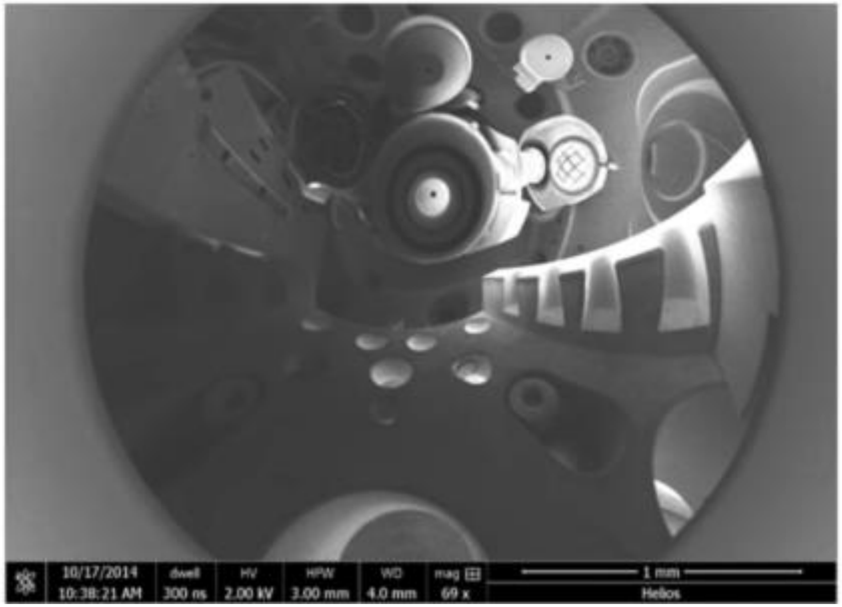
- Interaction volume increased with E
- Interaction volume decreased with Z



Electrons Inject Charge-Where do they go?

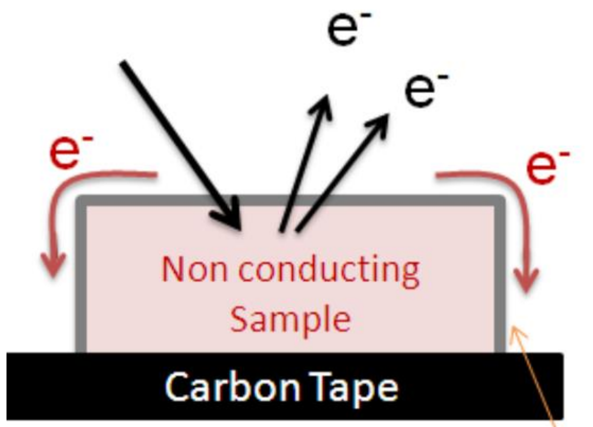




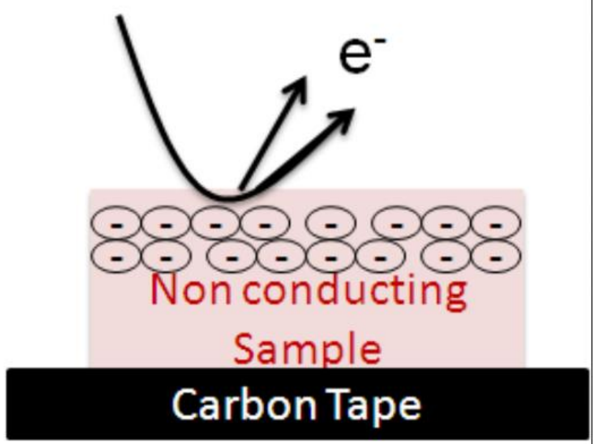


- If the sample is conducting there is no charge buildup
- If the sample is insulating the electron beam will charge the sample surface – image will be distorted!

Non-conductive Specimens-Coating and Beam



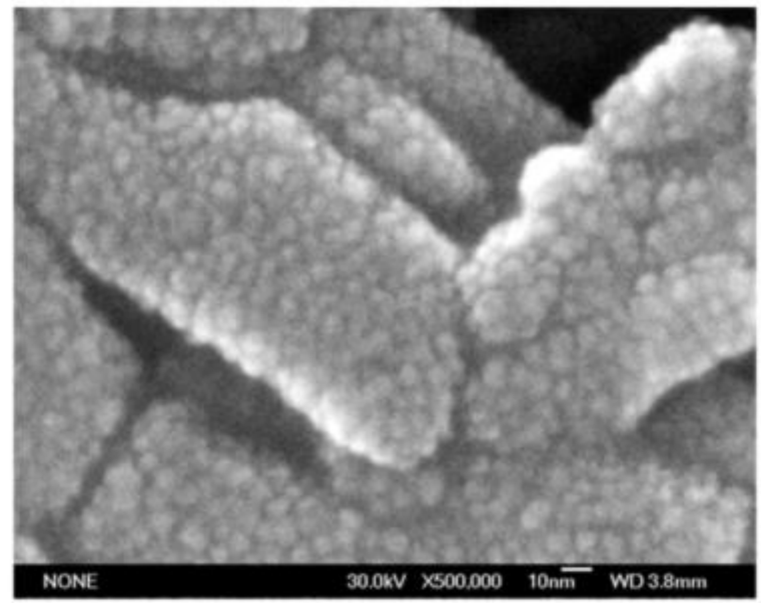
(a) Fully Coating



(b) No Coating

The traditional solution

- coat with few nm of conductor, (e.g. Au/Pd or C) to make sample conductive

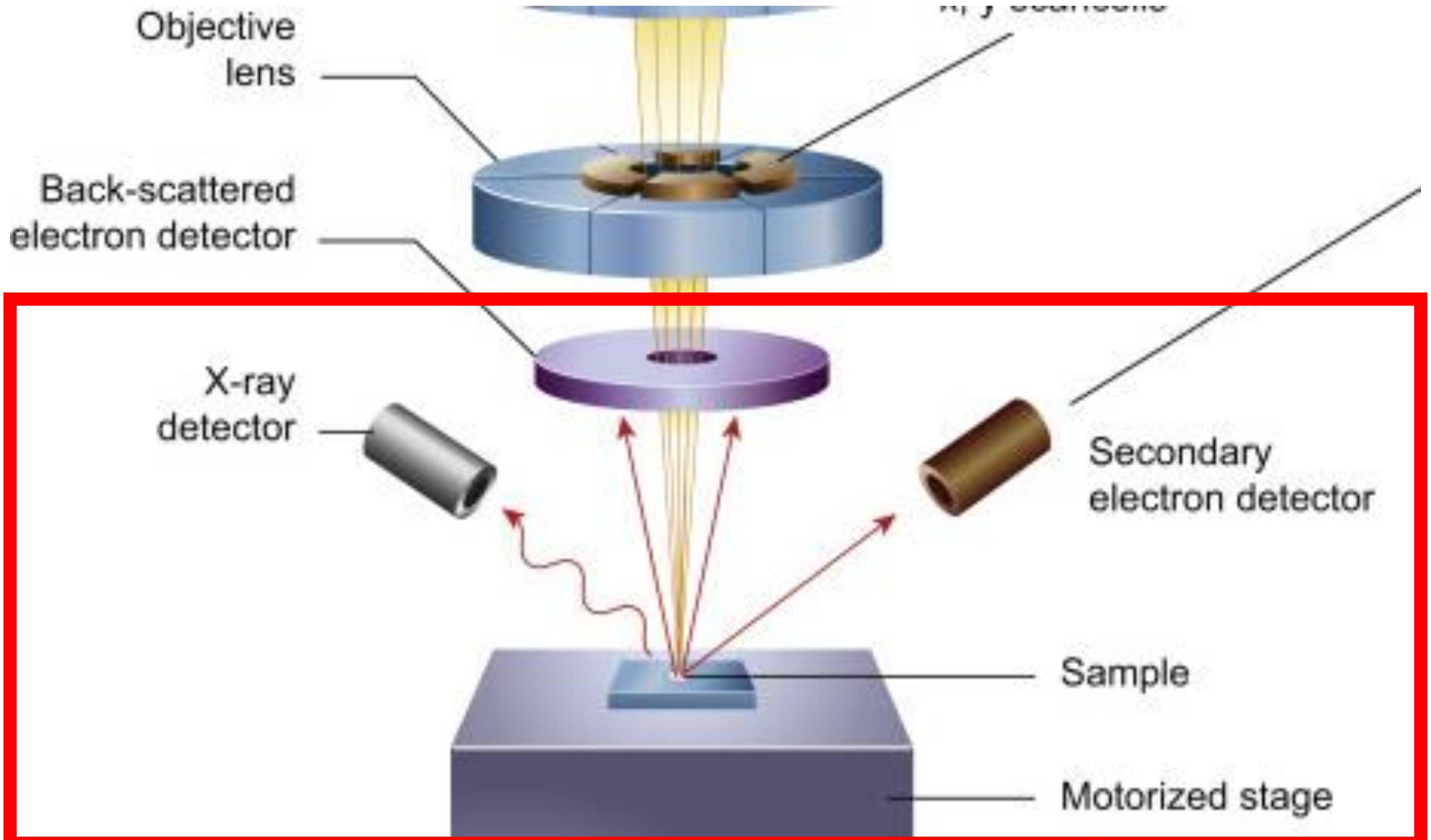


Sputtered Au-Pd on Magnetic Tape

From *Scanning Electron Microscopy and X-Ray Microanalysis*, Joseph I. Goldstein et al. Plenum Press

Specimen Chamber

- Vacuum in the chamber
- Sample should be solid and dry



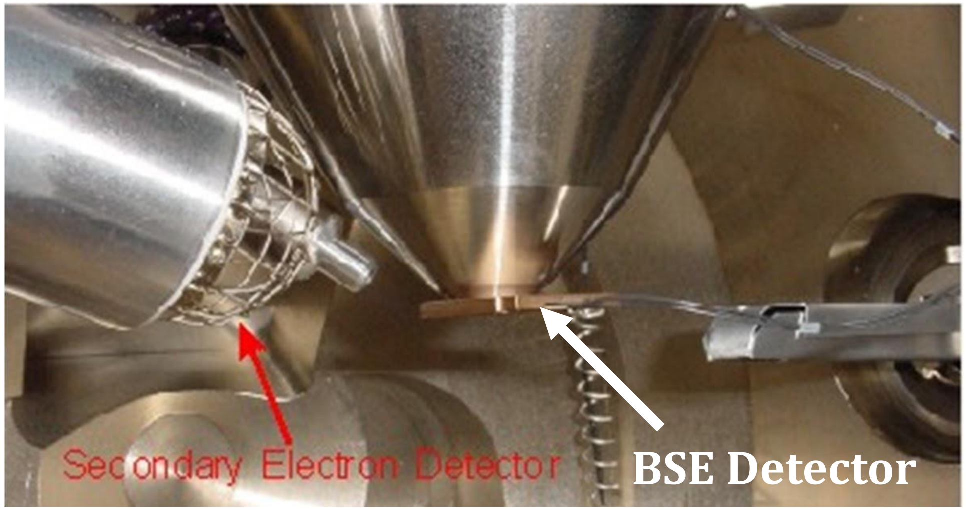
Remarks for Secondary Electrons

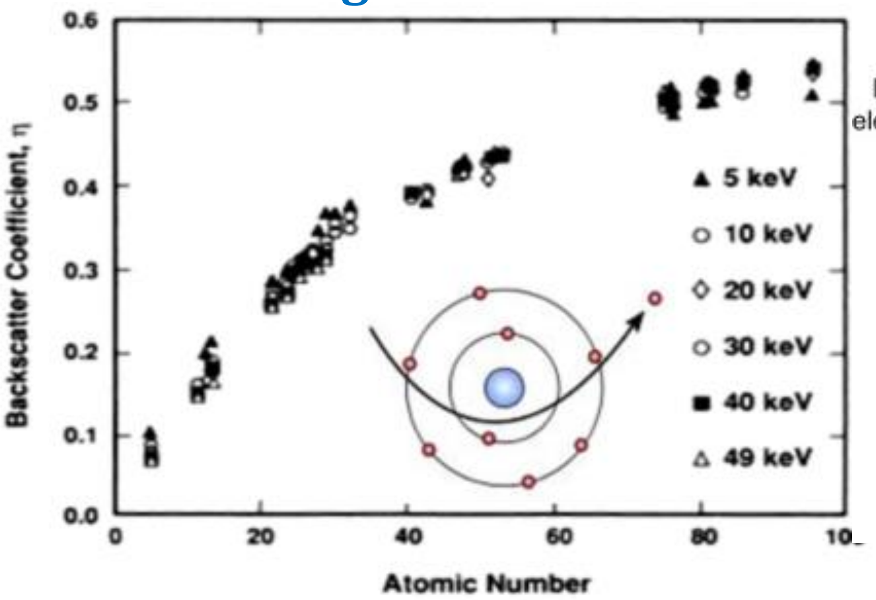
Some of the properties of secondary electrons

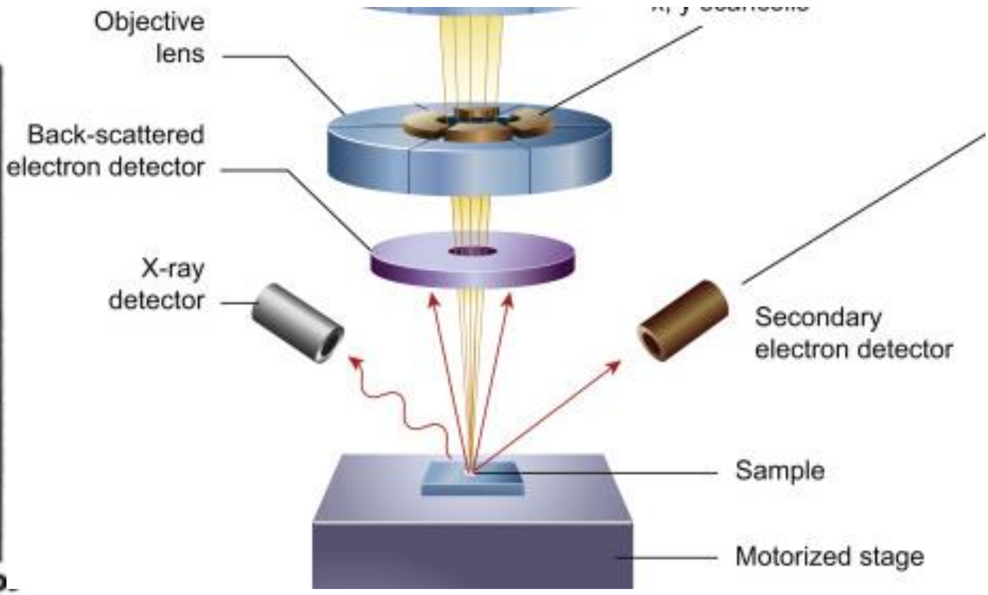
- Secondary electrons are highly sensitive to topographical features of the specimen surface because they depend on the surface orientation with respect to the electron beam.
- There is no significant dependence between the atomic number of the specimen and the intensity of the secondary electron signal.
- The production of secondary electrons increases with the tilt angle of the sample.

Composition Contrast with BSE Detector

- Scattering increase with sample Z
- Quadrant detector placed underneath pole-piece
- Compositional mode imaging most useful on multi-phase samples
- Sensitivity can be as low as 0.1 average Z difference

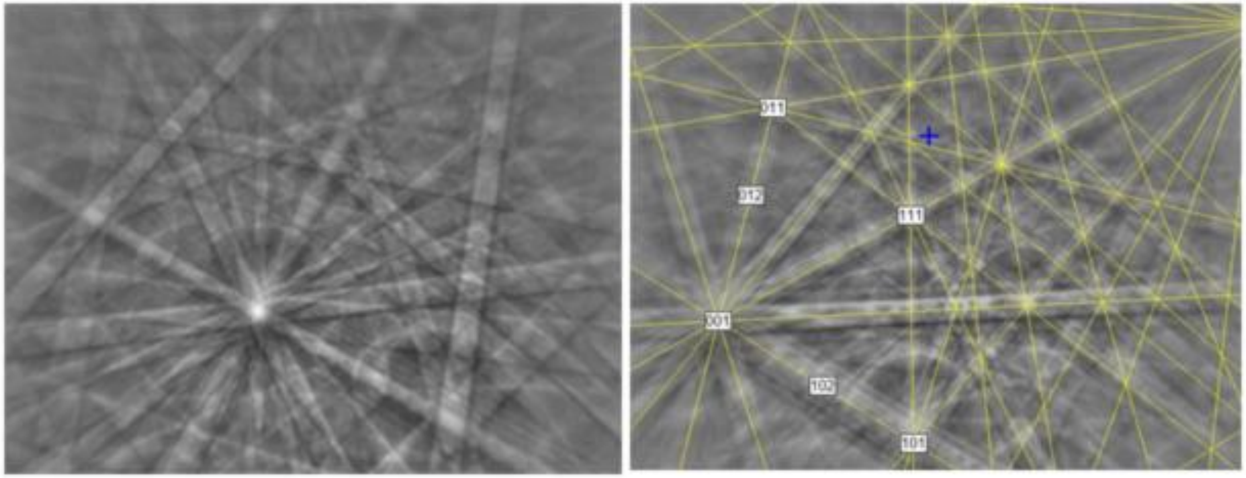






EBSD is a SEM–based microstructural-crystallographic characterization technique commonly used in the study of **crystalline or polycrystalline materials**. The technique can provide information about the **structure, crystal orientation, phase, or strain** in the material.

EBSD of AlN



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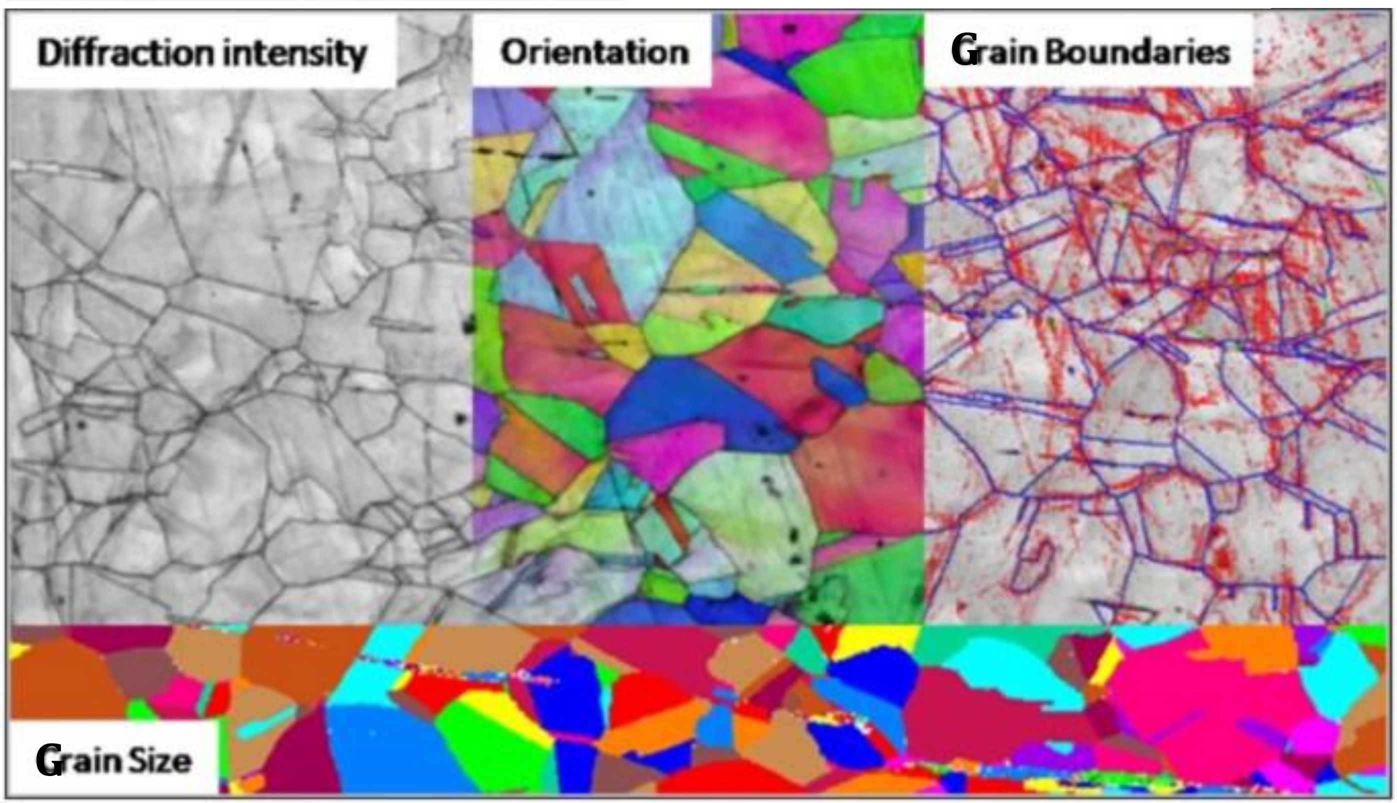
012
111
001
102
101



Diffraction intensity
Orientation
Grain Boundaries
Grain Size

Remarks for Backscattered Electrons

Some of the properties of backscattered electrons

- The signal intensity increases with the increasing of the Z of the sample. Elements with higher Z appear brighter on the SEM screen.

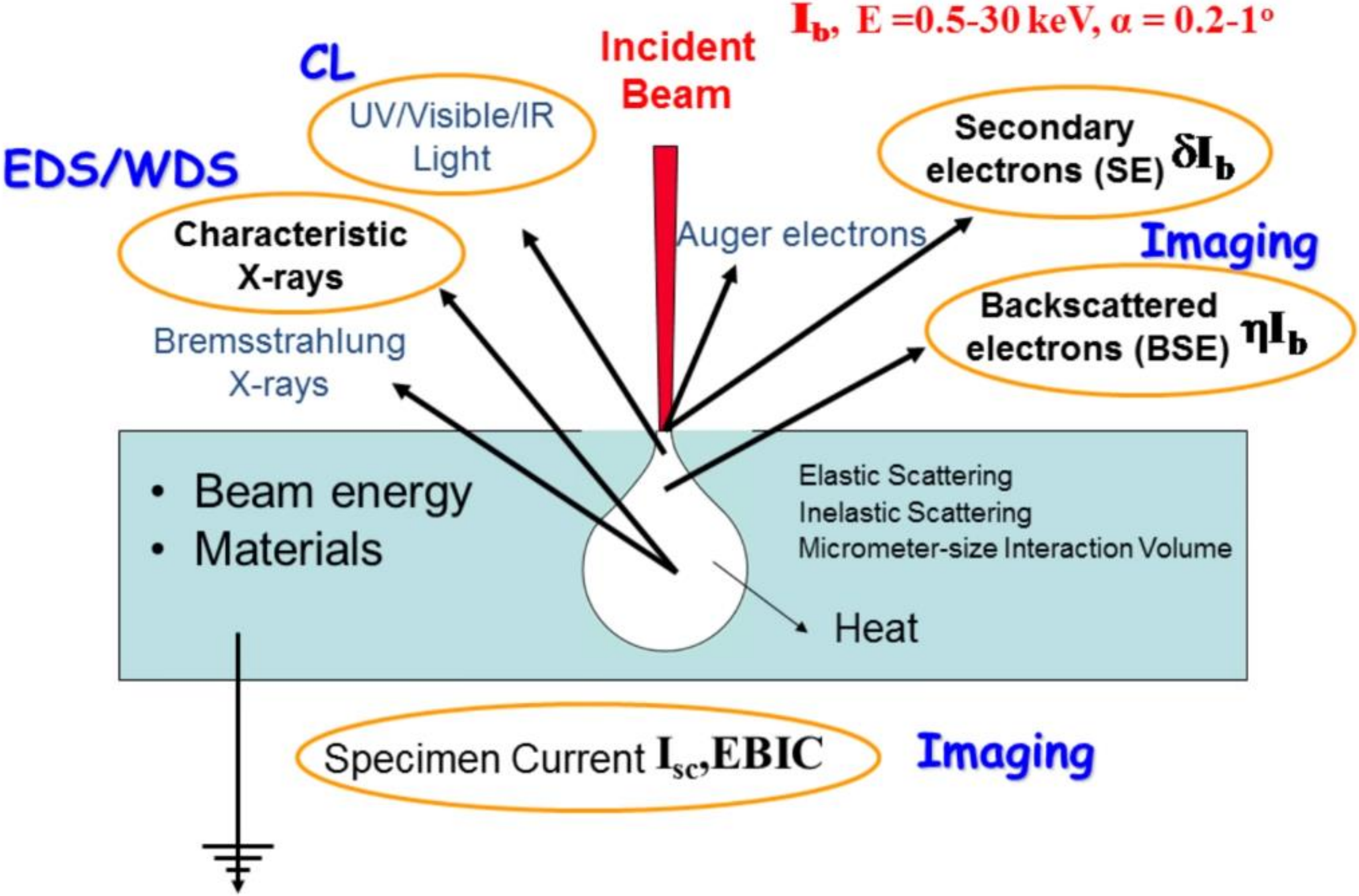
- No significant changes in the backscattered emissions with the increase in acceleration voltage.

- Backscattered electron emissions from the specimen increase with the tilt angle.

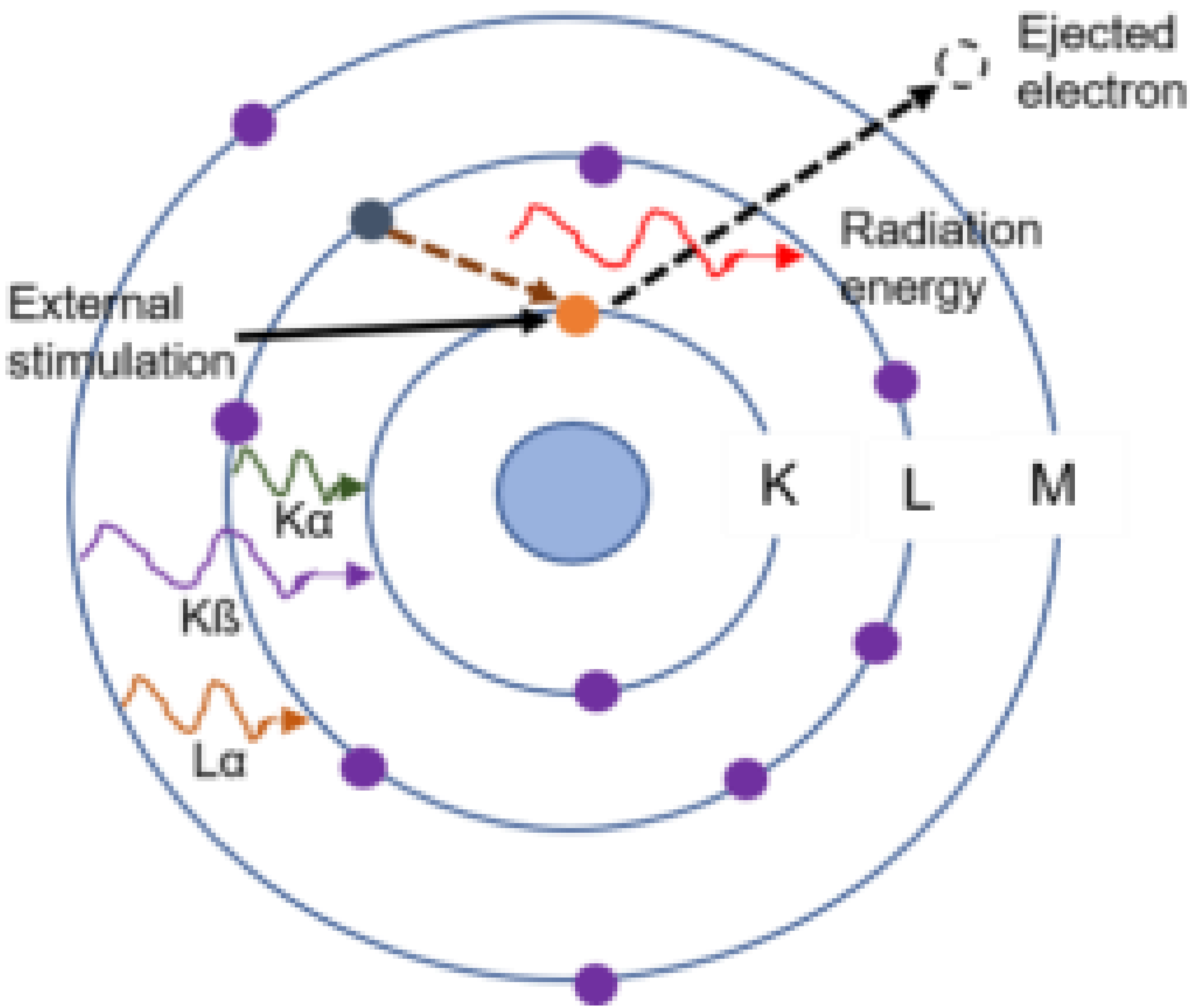
- BSE are mostly affected by subsurface features of the specimen structure, and they are not used for topography analysis of the sample surface. They can be used for imaging purposes but without fine resolution at high magnifications.

Energy-dispersive X-ray spectroscopy (EDS)

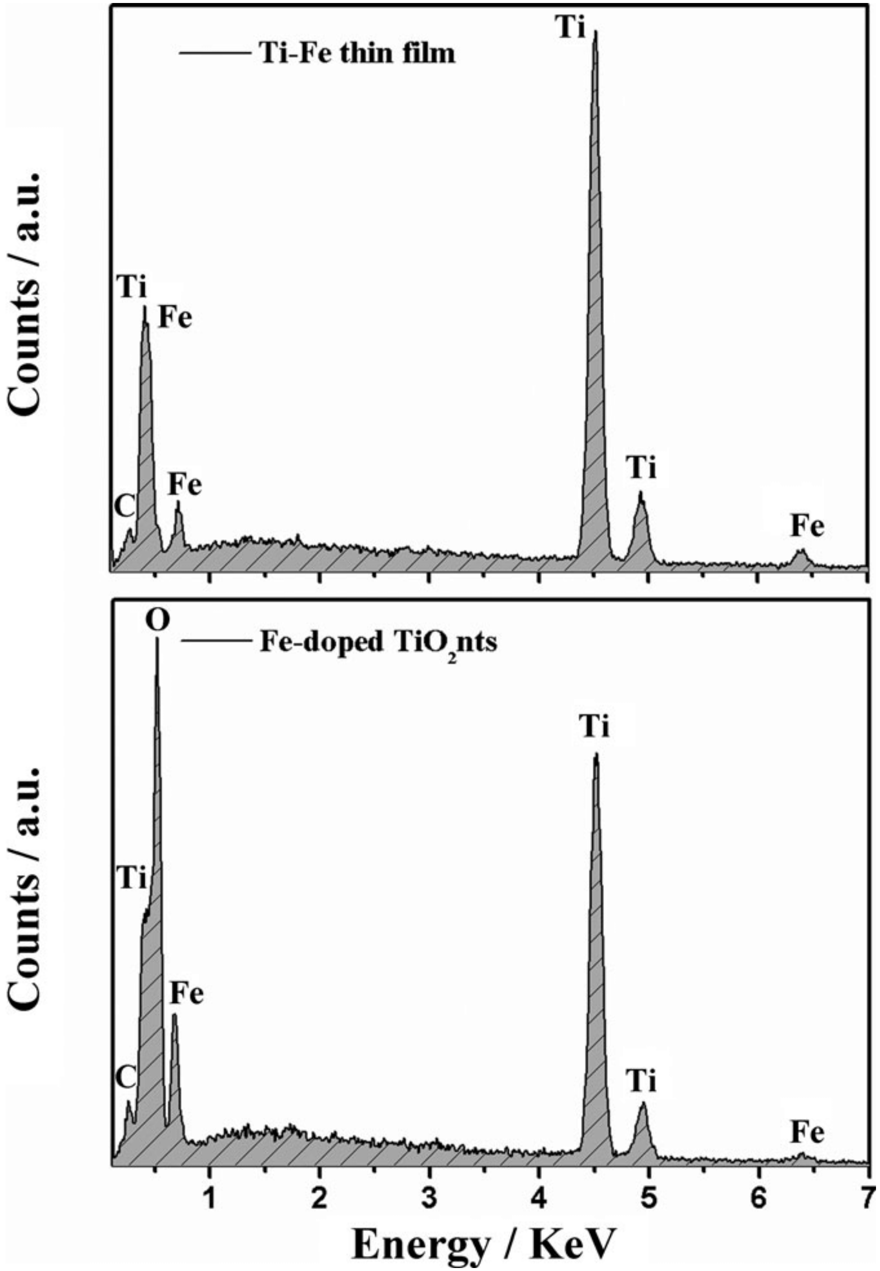
Electron-Matter Interactions



X-ray Generation

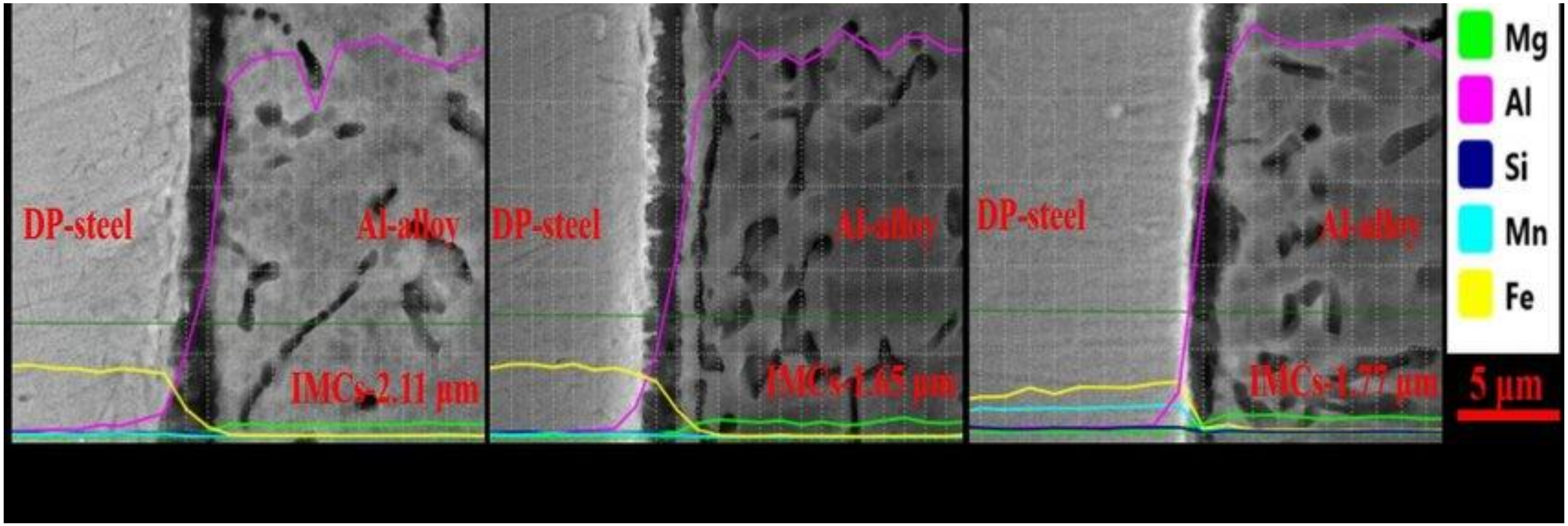


EDS-Microanalysis in the SEM

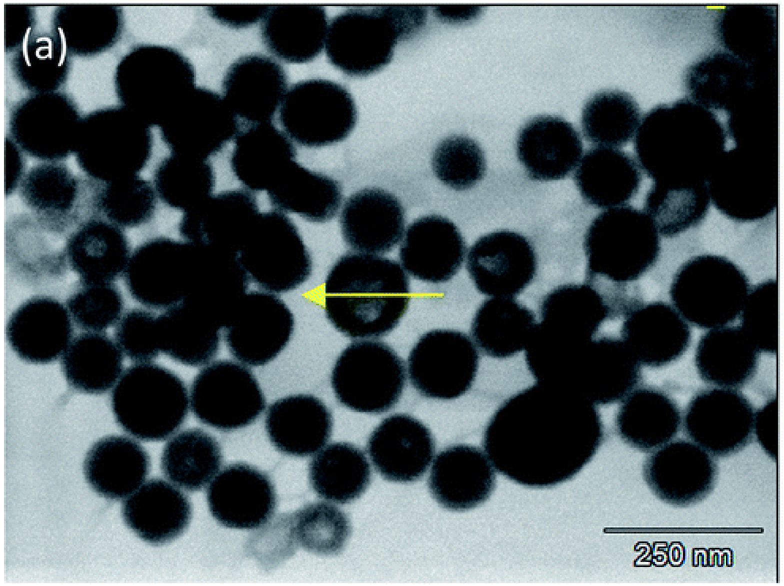


- Qualitative elemental analysis
  - From Beryllium up on periodic table
  - Sensitivities to ~0.1 wt.%
  - Quantitative analysis (many requirements / Limitations)
- Point, line and mapping scans, full spectrum imaging possible
- Analysis from interaction volume
- ~130 eV Mn K-alpha resolution typical for Si(Li) detector

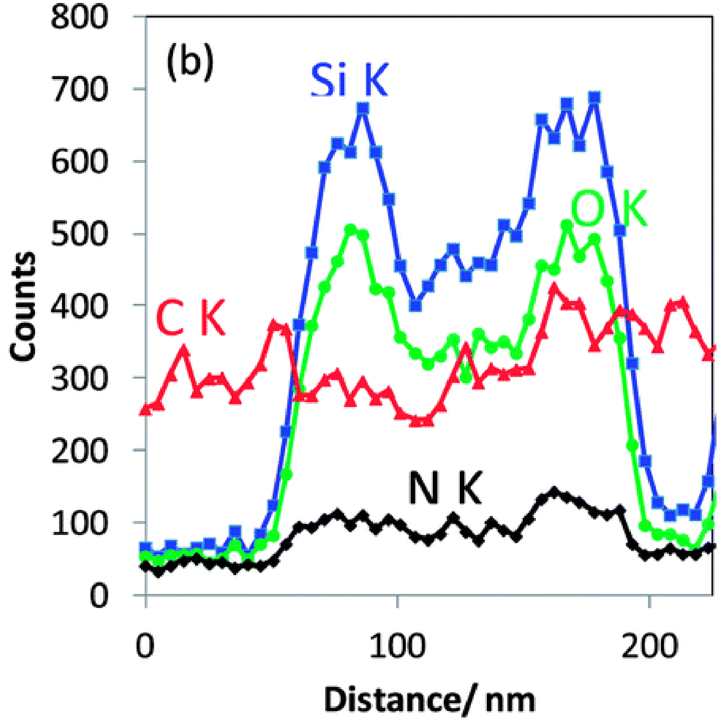
EDS mapping



DP-steel
Al-alloy
IMCs-2.11 μm
DP-steel
Al-alloy
IMCs-1.65 μm
DP-steel
Al-alloy
IMCs-1.77 μm
Mg
Al
Si
Mn
Fe
5 μm

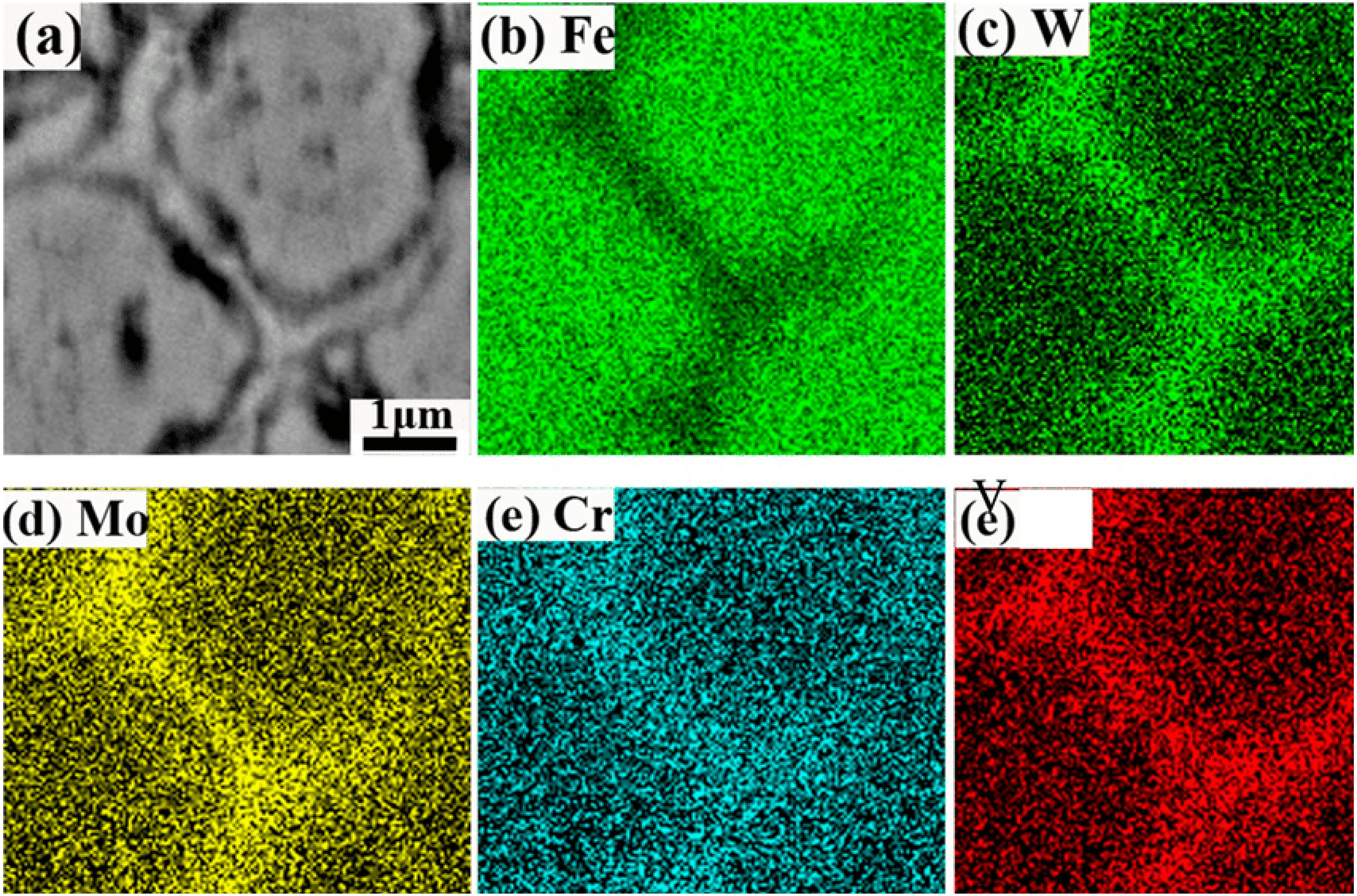


(a)
250 nm



(b)
Si K
O K
C K
N K
Counts
Distance/ nm
0
100
200
300
400
500
600
700
800

EDS mapping



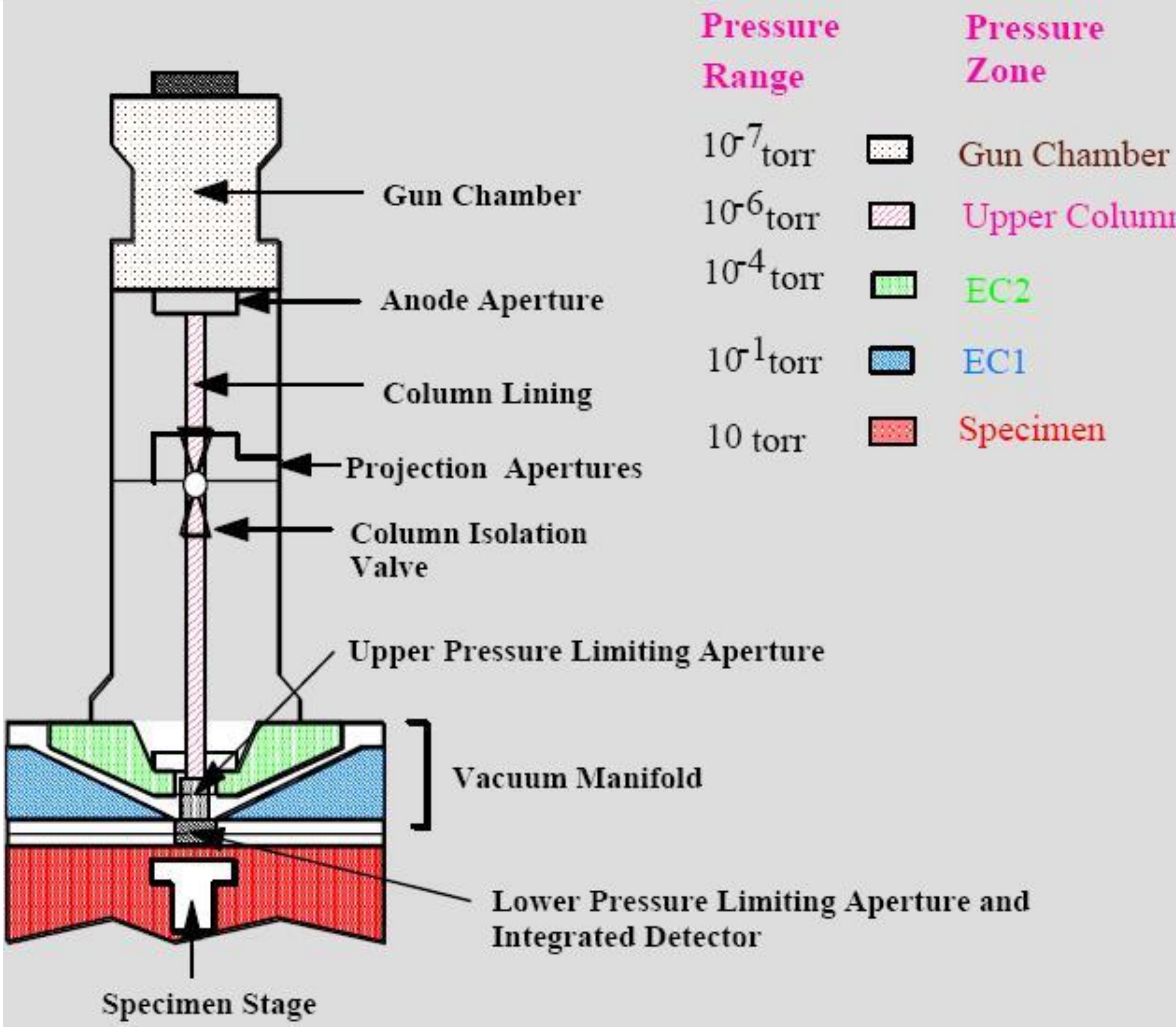
- To see element distribution

Where does SEM Fit? Technique Comparisons…

- SEM is generally thought of as a versatile system or a jack-of –all-trades (全能的)
- Excellent resolution over wide magnification range (but ultimately not as good as surface probe techniques such as STM, AFM, TEM)
- Element identification-excellent choice for “what” and “spatially where” (EDS, WDS)
- Composition-with EDS/WDS, not accurate without standards, not accurate for low Z (<Na) elements (RBS, XPS, Auger may be more appropriate)
- Surface orientation or texture-can be performed with EBSD, does not give phase! (XRD an excellent choice here)

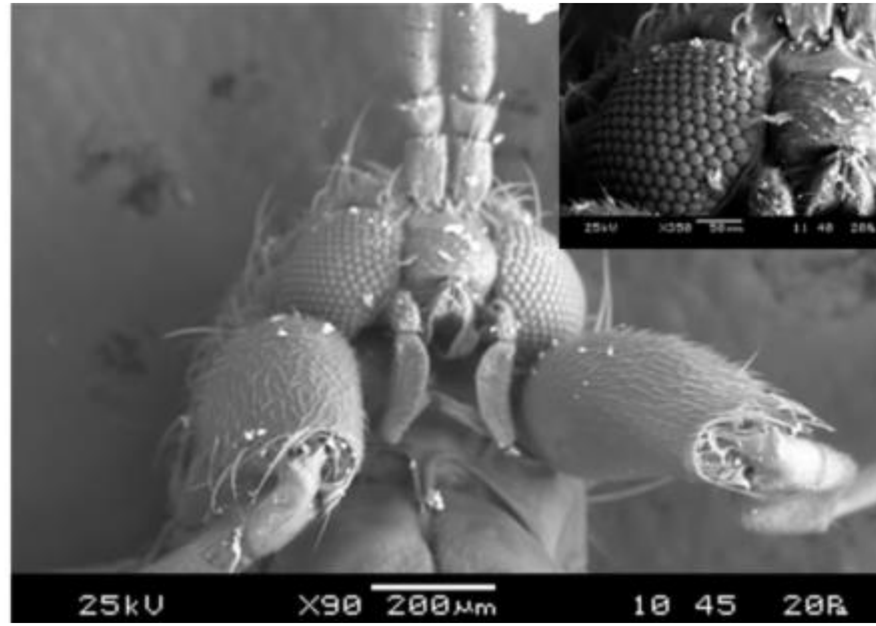
Environmental Scanning Electron Microscopy (ESEM)

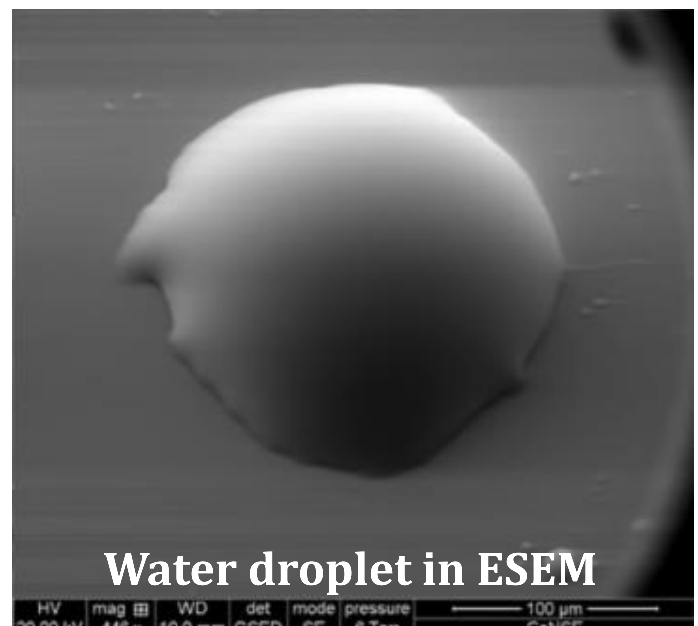
ESEM-schematic



Variable Pressure (Environmental) SEM

- VP: Introduces gas (air, water) into specimen chamber
- Primary electrons ionize the gas
  - Ions are attracted to surface charge
  - Surface is dynamically compensated
  - Good for samples that cannot be coated
  - < 2 torr pressure
- ESEM: Uses cold/heating stages and high pressure
  - Up to 20 torr
  - Can create 100% relative humidity condition
  - Best for observing hydration sensitive samples in native state

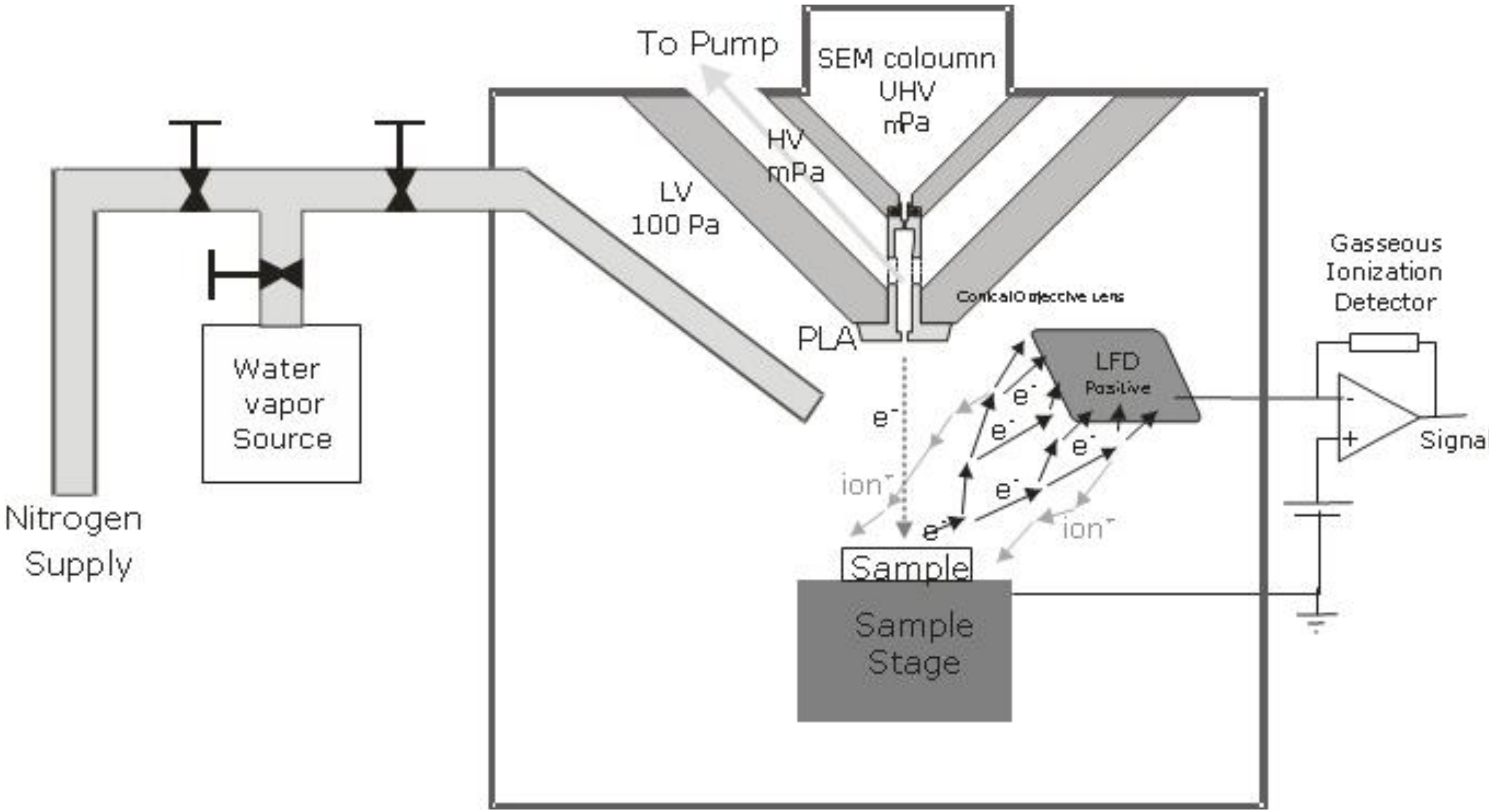




Water droplet in ESEM

Electron beam in ESEM

- Electron beam reduces the intensity, but not the spot size
- Signals are similar to standard SEM



Advantages and Disadvantages of ESEM

Advantages

- Liquid-phase is possible
- Electrically non-conductive specimens do not require the preparation techniques
- The gas itself is used as a detection medium producing novel imaging possibilities
- Gas/liquid/solid interactions can be studied dynamically in situ and in real time, or recorded for post processing (temperature, pressure, humidity)

Disadvantages

- Electron beam remains usable in the gaseous environment.
- The presence of gas may yield unwanted effects in certain applications

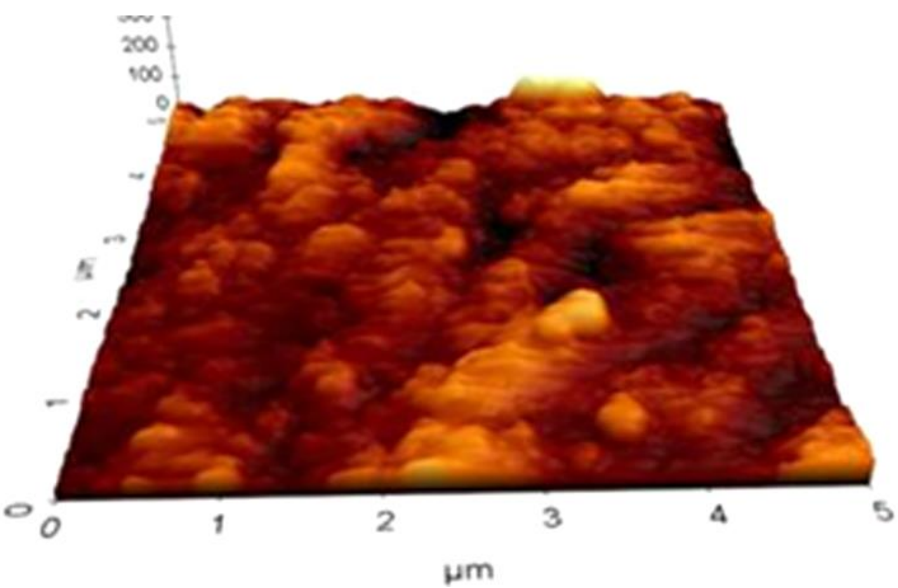
Scanning Probe Microscopy (SPM)

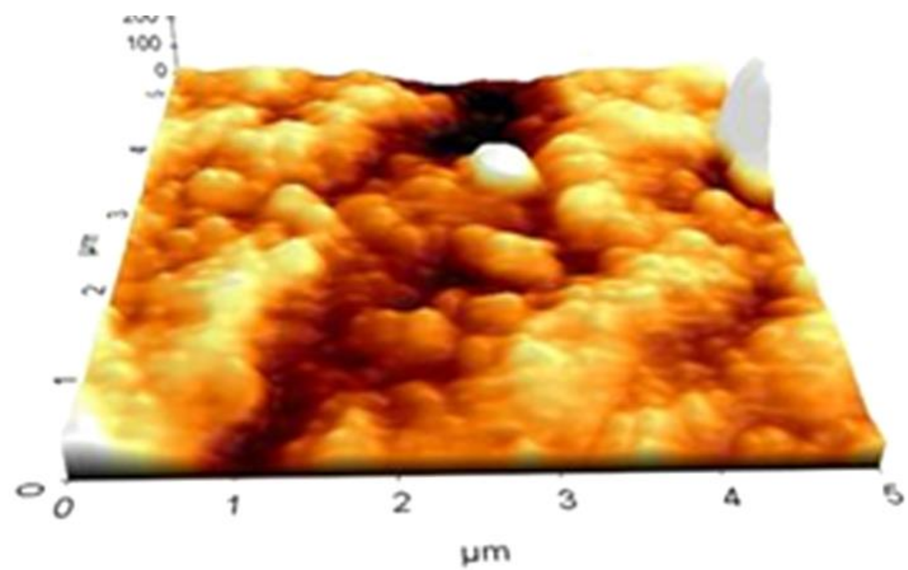
"Seeing" at the nanoscale

- Atomic Force Microscopy (AFM)
- Scanning Tunneling Microscopy (STM)

What does AFM do?

AFM provides a 3D profile of the surface on a nanoscale, by measuring forces between a sharp probe (<10nm) and surface at very short distance (0.2-10nm) probe-sample separation". The probe is supported on a flexible cantilever(悬臂). The AFM tip "gently" touches the surface and records the small force between the probe and the surface."



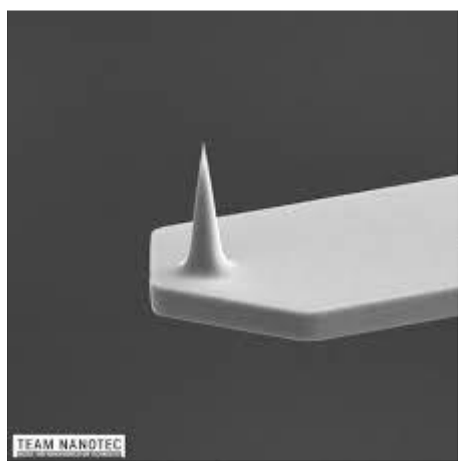


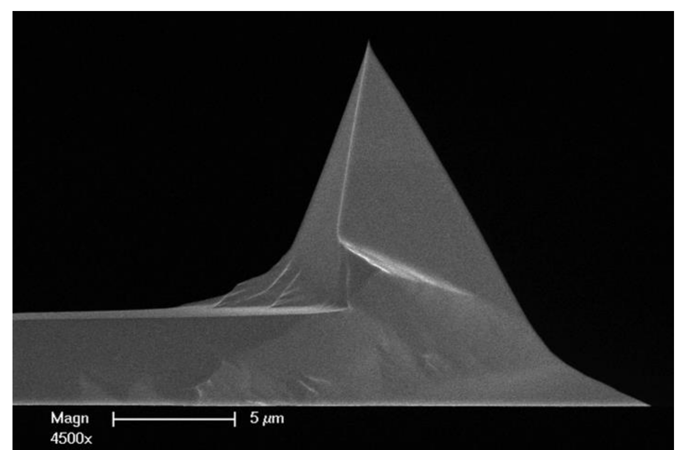
How does it measure the forces?

- The probe is placed on the end of a cantilever (which one can think of as a spring).
- The amount of force between the probe and sample is dependent on the spring constant (stiffness (刚度) of the cantilever, typical value are 0.1– 1N/m) and the distance between the probe and the sample surface.
- This force can be described using Hooke's Law

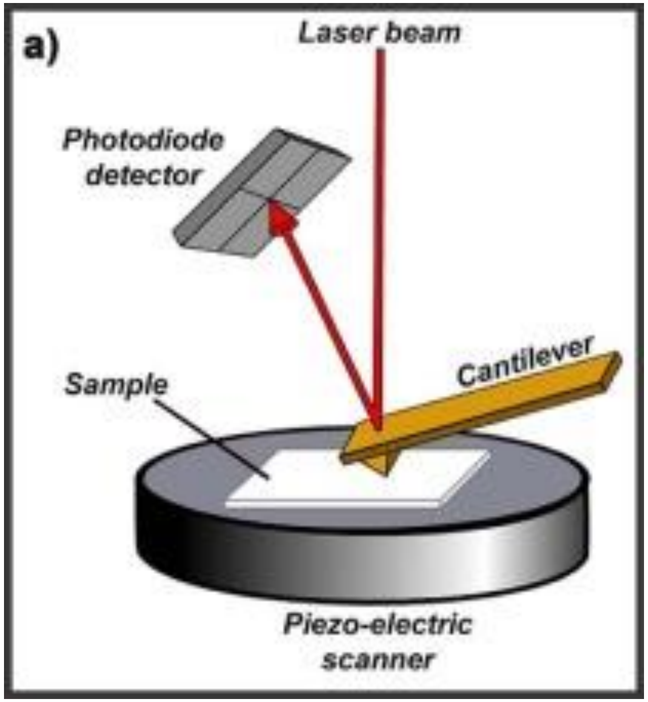
$$F=-k\bullet x$$

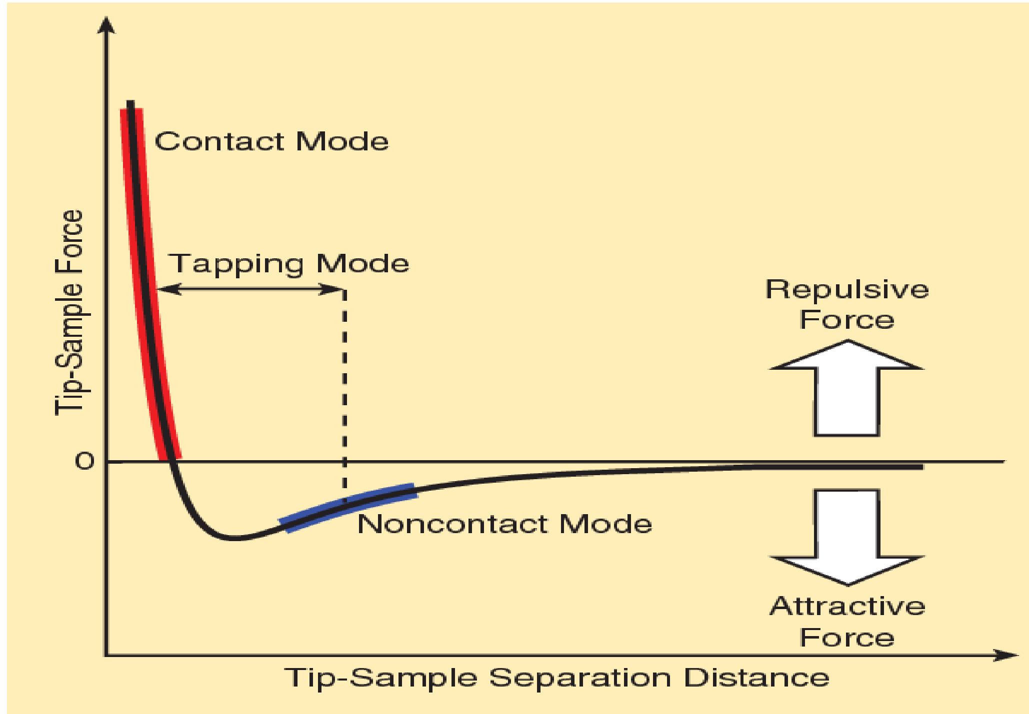
Typical forces are 10^{-6}-10^{-9} N





Types of forces measured





“The dominant interactions at short probe-sample distances in the AFM are Van der Waals interactions. However, long-range interactions (i.e. capillary, electrostatic, magnetic) are significant further away from the surface. These are important in other SPM methods of analysis. During contact with the sample, the probe predominately experience *repulsive* Van der Waals forces (**contact mode**) to the tip deflection described previously. As the tip moves further away from the surface *attractive* Van der Waals forces are dominant (**non-contact mode.**)”

Modes of Operation

There are 3 primary imaging modes in AFM:

(1) Contact AFM

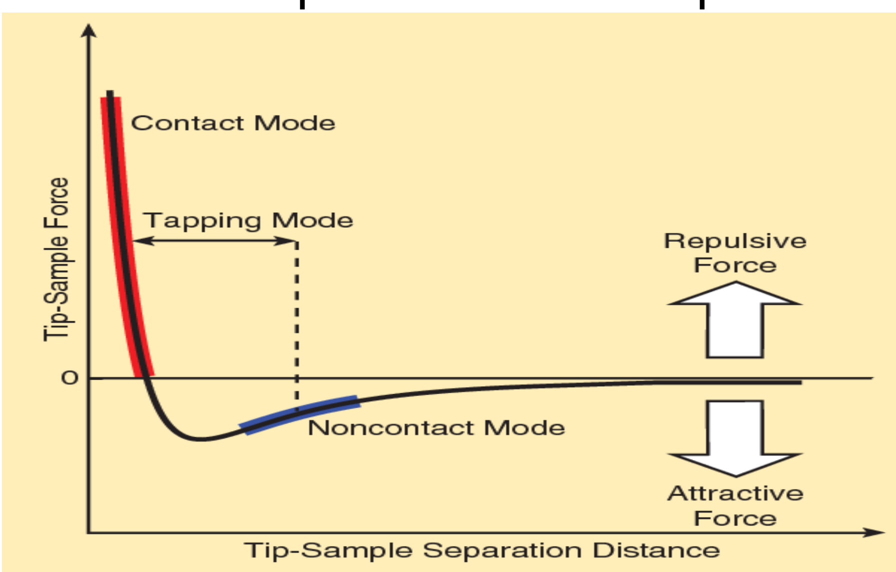
<0.5nm probe-surface separation, the forces are repulsive

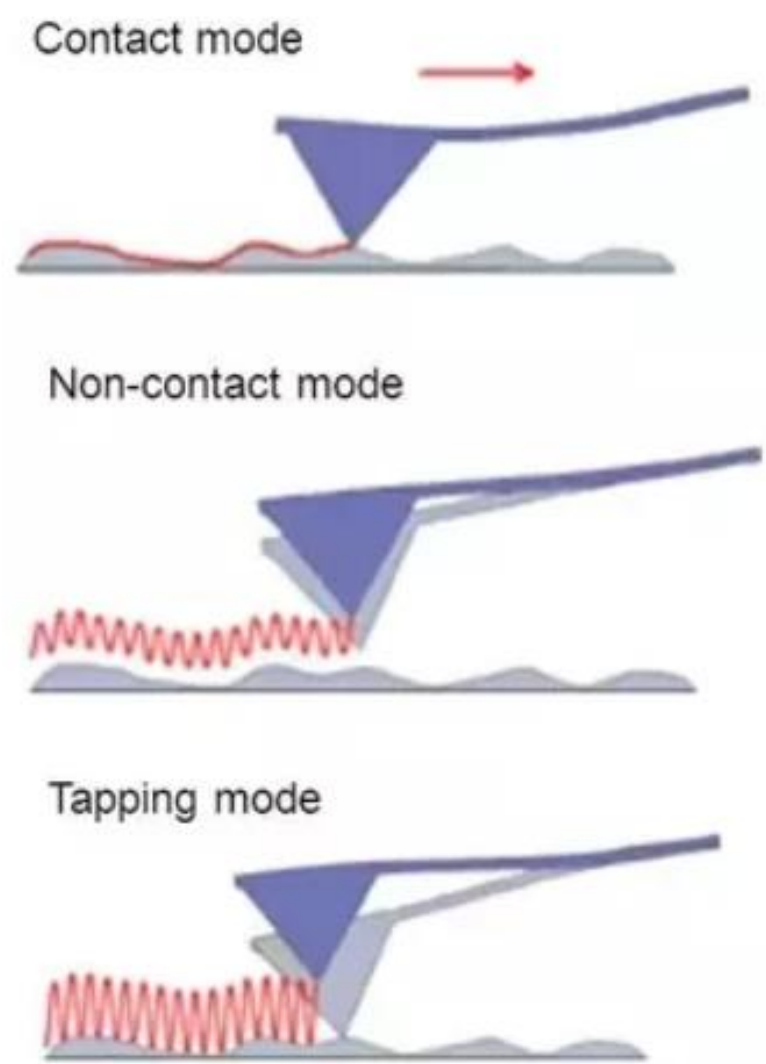
(2) Non-contact AFM

0.5-10 nm probe-surface separation, the forces are attractive

(3) Intermittent contact (tapping mode AFM)

0.5-2 nm probe-surface separation

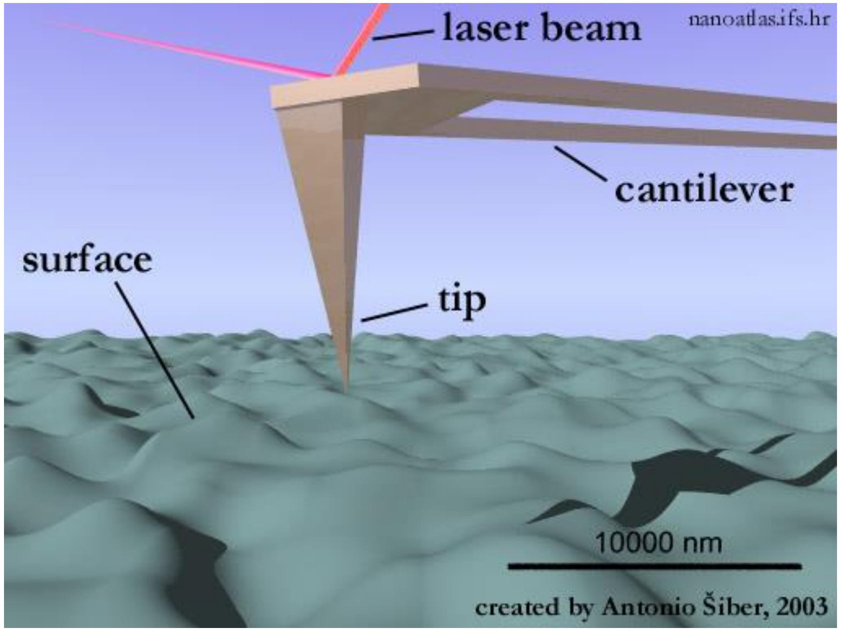




The AFM cantilever can be used to measure both attractive force mode and repulsive forces.

Contact Modes AFM

Contact Modes AFM: (repulsive Van der Waals) When the spring constant of cantilever is less than surface, the cantilever bends. The force on the tip is repulsive. By maintaining a constant cantilever deflection (using the feedback loops) the force between the probe and the sample remains constant and an image of the surface is obtained.

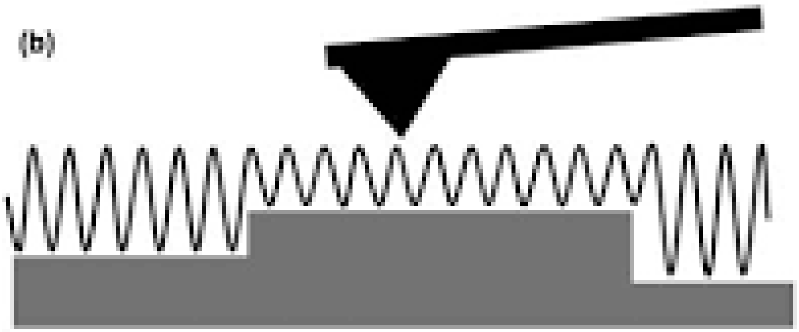


- **Advantages: fast scanning, good for rough samples, used in friction analysis.**
- **Disadvantages: at time force can damage/deform soft samples (however imaging in liquids often resolves this issue)**

✓ In ambient conditions there is also a capillary force exerted by the thin water layer present (2-50 nm thick).

Tapping Mode

Intermittent Mode (Tapping): The imaging is similar to contact. However, in this mode the cantilever is oscillated at its resonant frequency, with oscillation amplitudes of 20-100 nm. The probe lightly "taps" on the sample surface during scanning, contacting the surface at the bottom of its swing. By maintaining a constant oscillation amplitude a constant tip-sample interaction is maintained and an image of the surface is obtained.



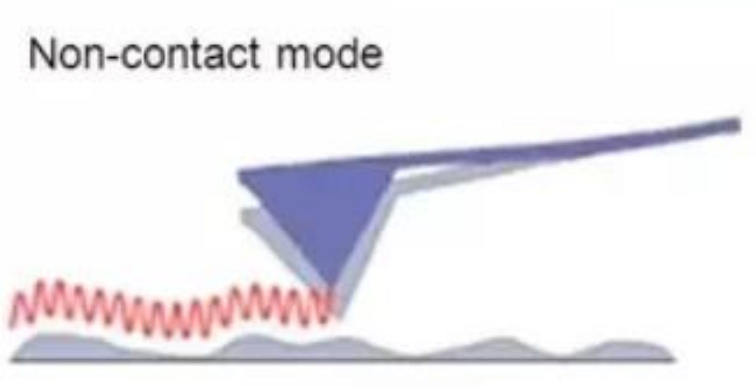
Tapping Mode

- **Advantages: allows high resolution of samples that easily damaged and/or loosely held to a surface; good for biological samples**

- **Disadvantages: more challenging to image in liquids, slower scan speeds needed.**

Non-contact Mode

Non-contact Mode: (attractive Van der Waals) The probe does not contact the sample surface, but oscillates above the adsorbed fluid layer on the surface during scanning. (Note: all samples unless in a controlled UHV or environmental chamber have some liquid adsorbed on the surface). Using a feedback loop to monitor changes in the amplitude due to attractive VdW forces the surfaces topography can be measured.



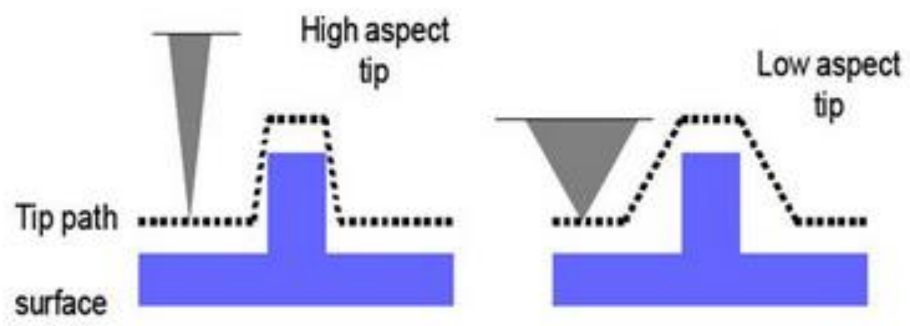
0.5-10 nm probe-surface separation, the forces are attractive

- **Advantages: VERY low force exerted on the samples (10-12N), extended probe lifetime.**

- **Disadvantages: generally lower resolution; contaminant layer on surface can interfere with oscillation; usually need ultra high vacuum (UHV) to have best imaging.**

Limitations of AFM

The AFM can be used to study a wide variety of samples (i.e. plastic, metals, glasses, semiconductors, and biological samples such as the walls of cells and bacteria). Unlike STM or scanning electron microscopy it does not require a conductive sample. **However, there are limitations in achieving atomic resolution. The physical probe used in AFM imaging is not ideally sharp. As a consequence, an AFM imaging does not reflect the true sample topography, but rather represents the interaction of the probe with the sample surface.** This is called tip convolution. **This does not often influence the height of a feature but the lateral resolution."**



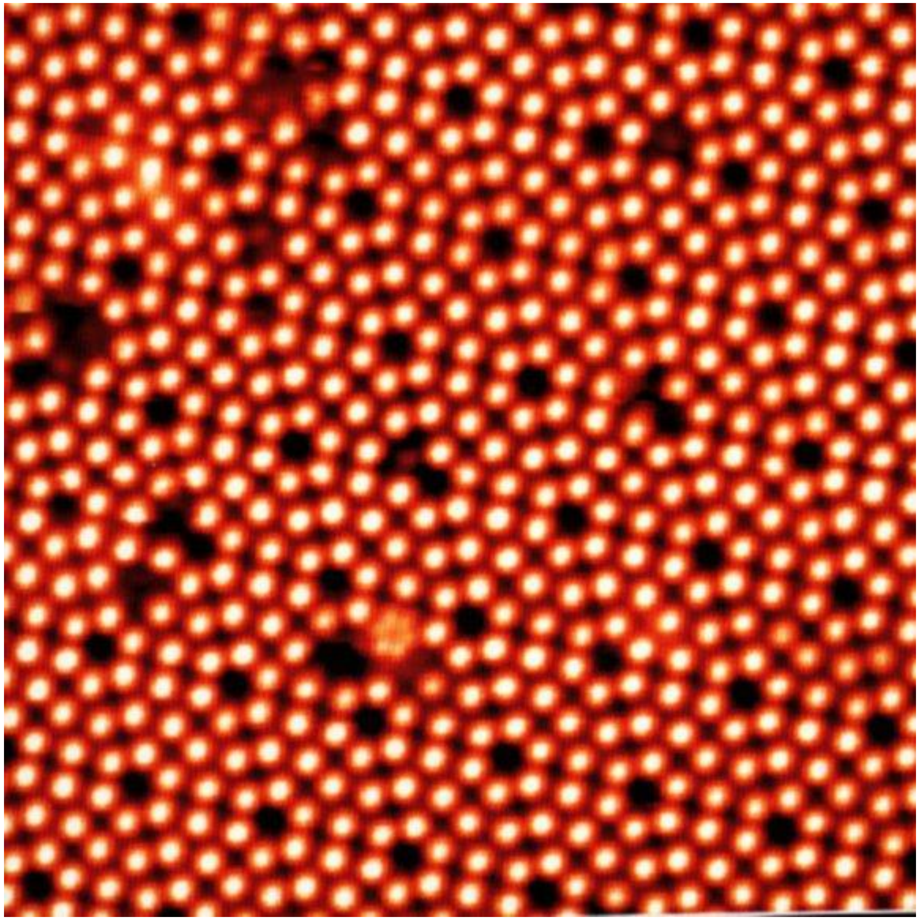
Ideally a probe (tip) with a high aspect ratio will give the best resolution. The radius of curvature of the probe leads to tip convolution. This does not often influence the height of a feature but the lateral resolution.

AFM vs SEM

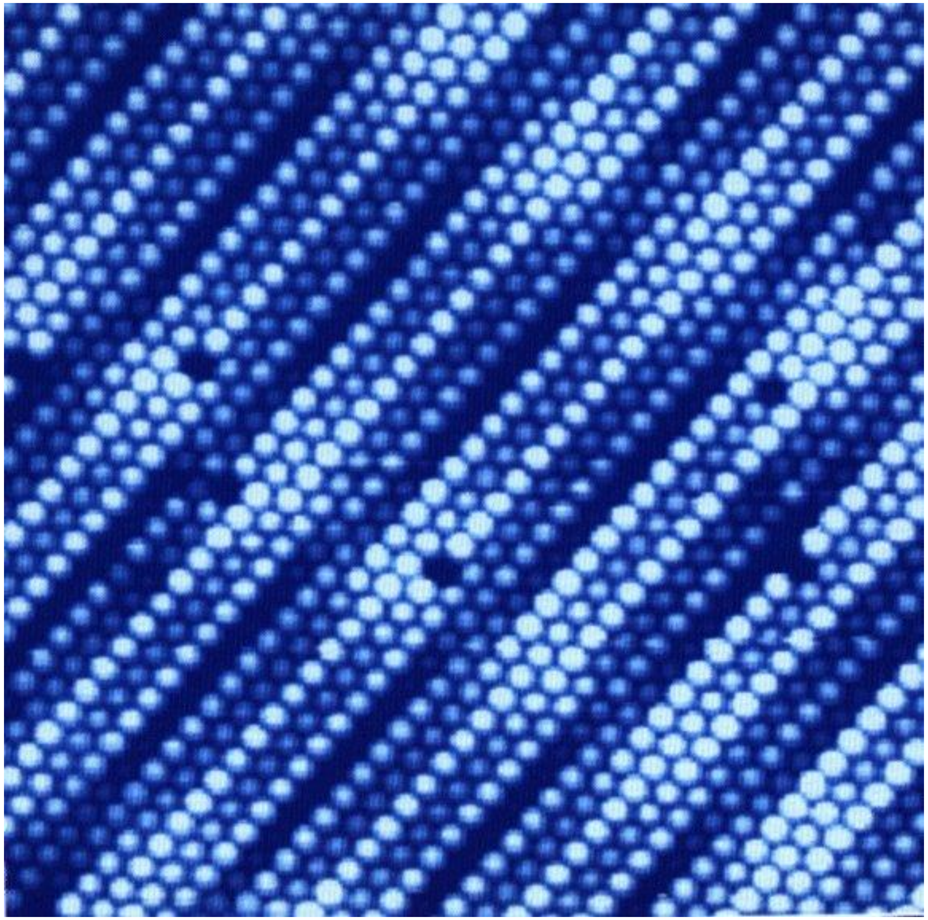
	AFM	SEM
Real 3D image of the studied surface with x, y, z coordinates for each pixel	√	X
Dielectric samples	√	X
Speed	X	√
Samples that can not be exposed to vacuum	√	X
Chemical composition	X	√

Scanning Tunneling Microscopy

STM Images of Point Defects



Clean **{111} Si** surface in ultra high vacuum conditions

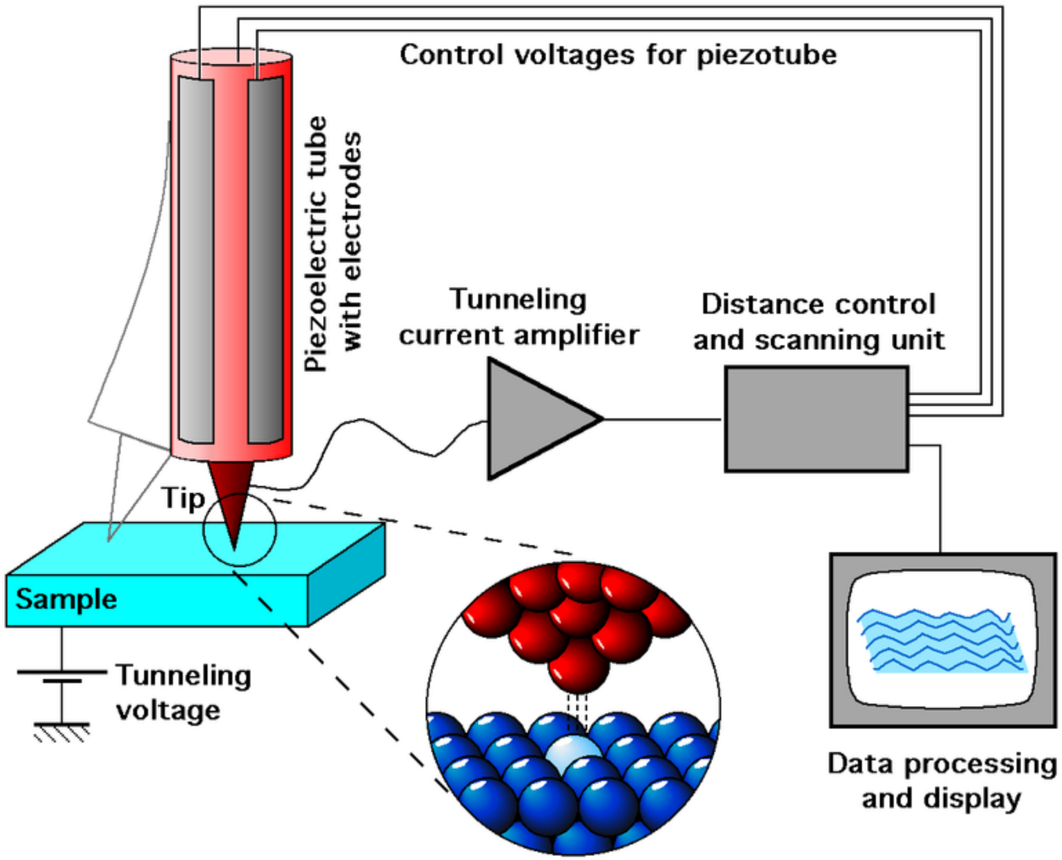


A STM image of a Pt surface. Vacancies are clearly visible

Source: https://www.tf.unikiel.de/matwis/amat/def_en/kap_4/illustr/g4_2_4.html

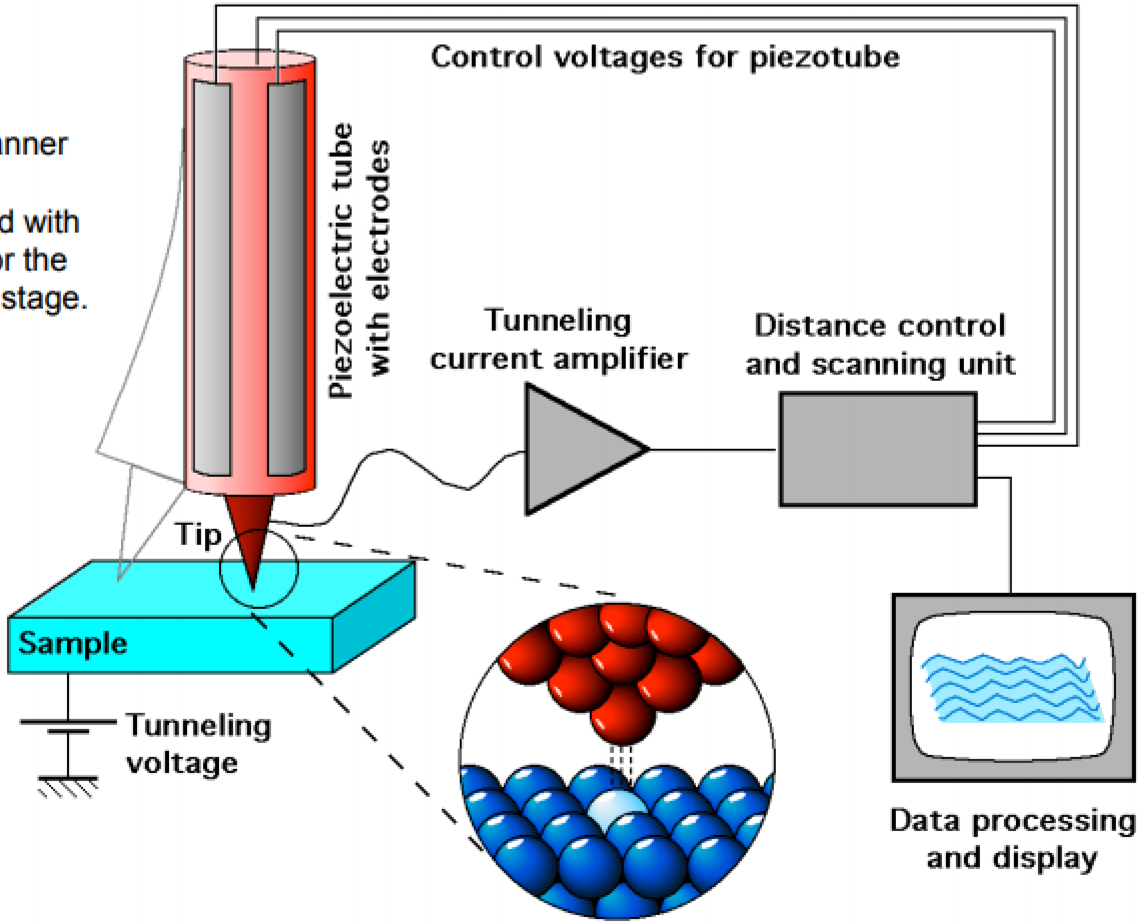
STM

- Similar to AFM, a sharp tip is raster scanned over a sample surface at very short separations (~ 1nm)
- In STM, the signal monitored for the feedback loop is the quantum mechanical tunneling current
- In the most common STM image mode, a piezoelectric scanner moves the tip up and down as it is scanned across the sample to maintain a constant value of this current



Basic components of STM:

The scanner can be mounted with the tip or the sample stage.

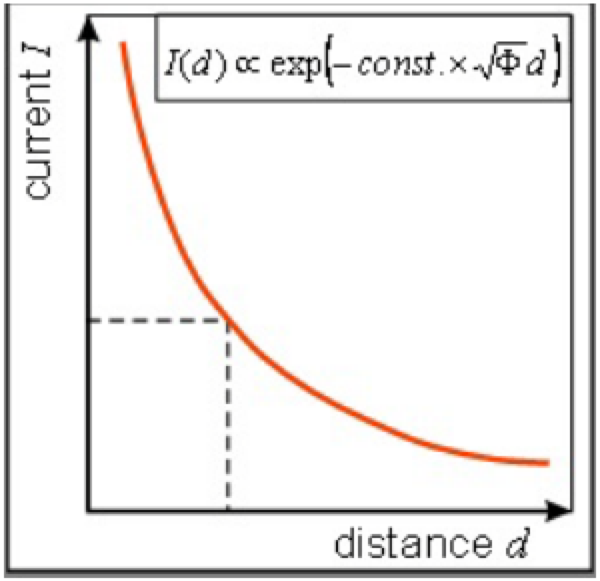


<u>Five basic components:</u>

1. Metal tip,
2. Piezoelectric scanner,
3. Current amplifier (nA),
4. Bipotentiostat (bias),
5. Feedback loop (current).

- Tunneling current from tip to sample or vice-versa depending on bias;
- Current is exponentially dependent on distance;
- Raster scanning gives 2D image;
- Feedback is normally based on constant current, thus measuring the height on surface.

Tunneling Current



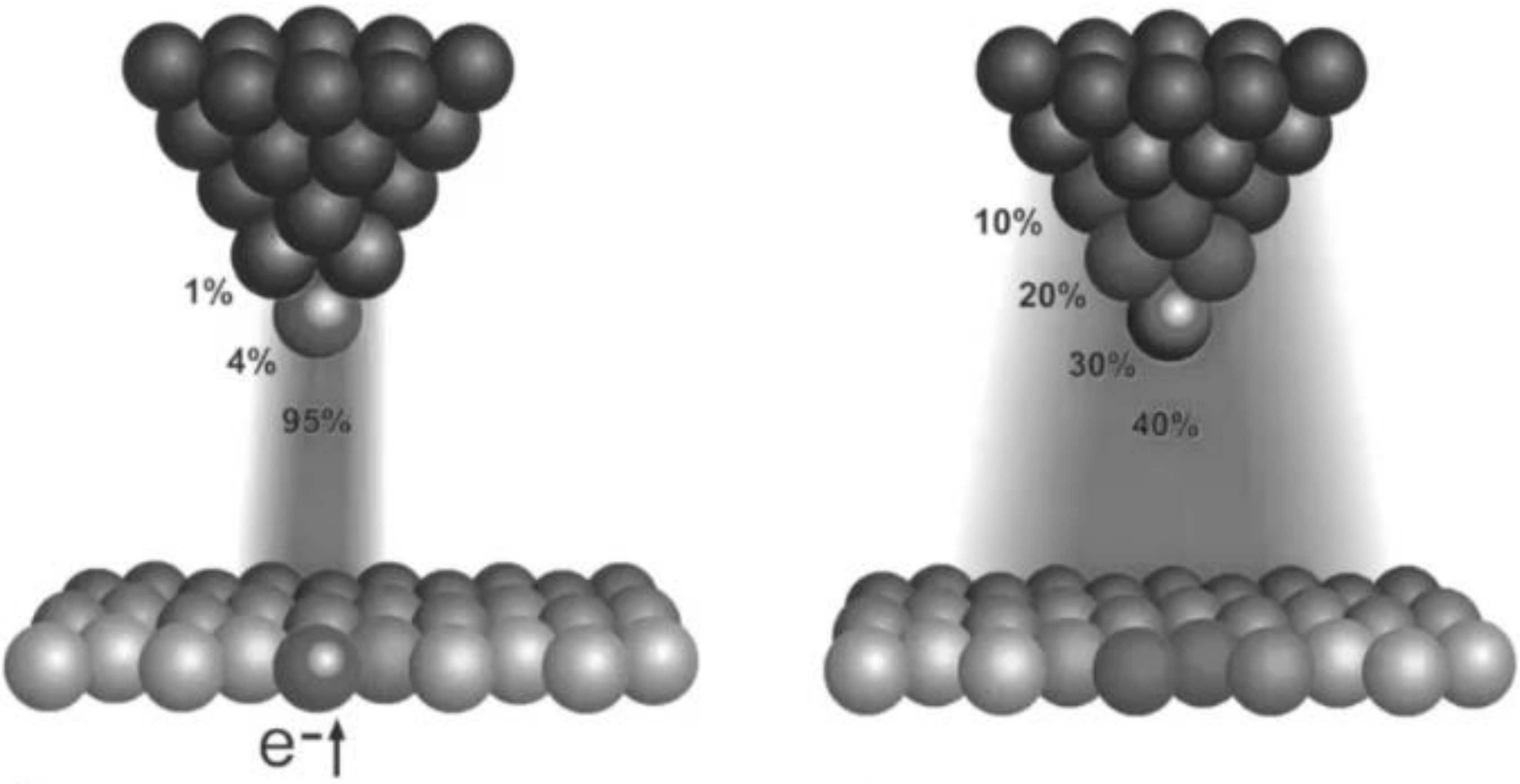
$$I(d) = constant \cdot V \cdot e^{\left(-2\frac{\sqrt{2m\Phi}}{\hbar}d\right)}$$

Φ: the work function (energy barrier),
e: the electron charge,
m: the electron mass,
h: the Planck's constant,
V: applied voltage,
d: tip-sample distance.

Next page: Φ of common metals.

- A thin metal tip is brought in close proximity of the sample surface. At a distance of only a few Å, the overlap of tip and sample electron wavefunctions is large enough for an electron tunneling to occur.
- When an electrical voltage *V* is applied between sample and tip, this tunneling phenomenon results in a net electrical current, the 'tunneling current'. This current depends on the tip-surface distance *d*, on the voltage *V*, and on the height of the barrier Φ:
- This (approximate) equation shows that the tunneling current obeys Ohm's law, i.e. the current *I* is proportional to the voltage *V*.
- The current depends exponentially on the distance *d*.
- For a typical value of the work function Φ of 4 eV for a metal, the tunneling current reduces by a factor **10** for every **0.1** nm increase in *d*. This means that over a typical atomic diameter of e.g. **0.3 nm**, the tunneling current changes by a factor **1000**! *This is what makes the STM so sensitive.*
- The tunneling current depends so strongly on the distance that it is dominated by the contribution flowing between the last atom of the tip and the nearest atom in the specimen --- single-atom imaging!

Interatomic action, STM (left) and AFM (right)



An STM gives true atomic resolution. Because the dependence of the tunneling current on the tip-to-sample separation is exponential, only the closest atom on a good STM tip interacts with the closest atom on the sample. For AFM, the dependence of cantilever deflection on tip-to-sample separation is weaker. The result is that several atoms on the tip interact simultaneously with several atoms on the sample

STM Tips

- STM tip should be conducting (metals, like Pt);
- STM plays with the very top (outermost) atom at the tip and the nearest atom on sample; so the whole tip is not necessarily very sharp in shape, different from the case of AFM, where spatial “contact” is necessary and crucial for feedback.

Comparison of AFM and Other imaging techniques

- AFM vs. STM
  - resolution of STM is better than that of AFM
  - STM is applicable only to conducting samples
  - AFM is applicable to both conductors and insulators
- AFM vs. SEM
  - AFM provides extraordinary topographic contrast direct height measurements and unobscured views of surface features
  - No (conductive) coating is necessary for AFM samples
- AFM vs. TEM
  - 3D AFM image obtained without expensive sample preparation
  - More complete information than two dimensional profiles from TEM

Portland State University

Nanoelectronics & Packaging Lab

AFM vs STM

- Resolution of STM is better than AFM because of the exponential dependence of the tunneling current on distance.

- STM is generally applicable only to conducting samples while AFM is applied to both conductors and insulators.

- AFM offers the advantage that the writing voltage and tip-to-substrate spacing can be controlled independently, whereas with STM the two parameters are integrally linked.

Summary

STM

- real space imaging
- high lateral and vertical resolution
- STS probe electronic properties
- sensitive to noise
- image quality depends on tip conditions
- not true topographic imaging
- only for conductive materials

AFM and others

- apply to nonconducting materials : biomolecules, ceramic
- real topographic imaging
- probe various physical properties: magnetic, electrostatic, hydrophobicity, friction, elastic modulus, etc.
- can manipulate molecules and fabricate nanostructures
- low lateral resolution ~50A
- contact mode can damage the sample
- image distortion due to the presence of water
- difficulties in chemical identification

X-RAY PHOTOELECTRON SPECTROSCOPY (XPS)

Introduction to xps

- What is XPS? General Theory

- How can we identify elements and compounds?

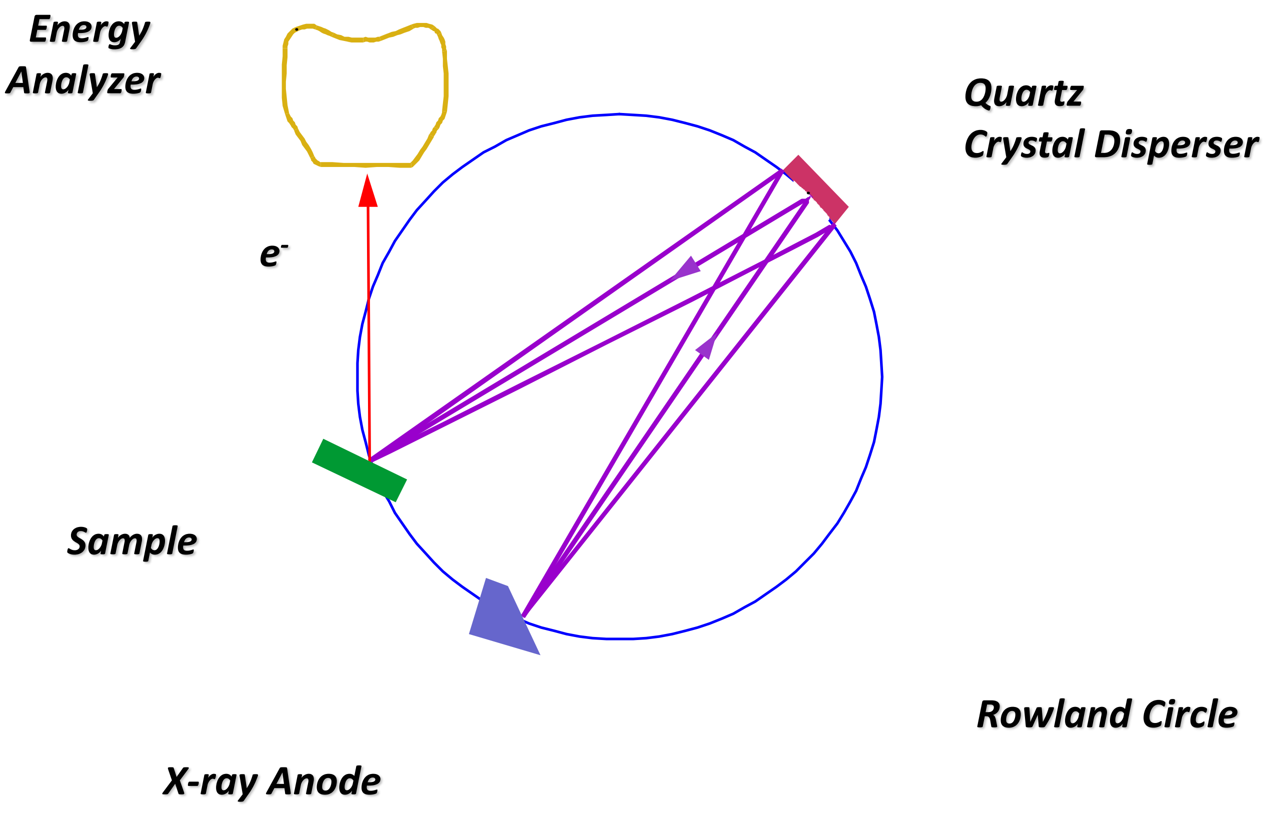
- What is the instrumentation of XPS?

WHAT IS XPS?

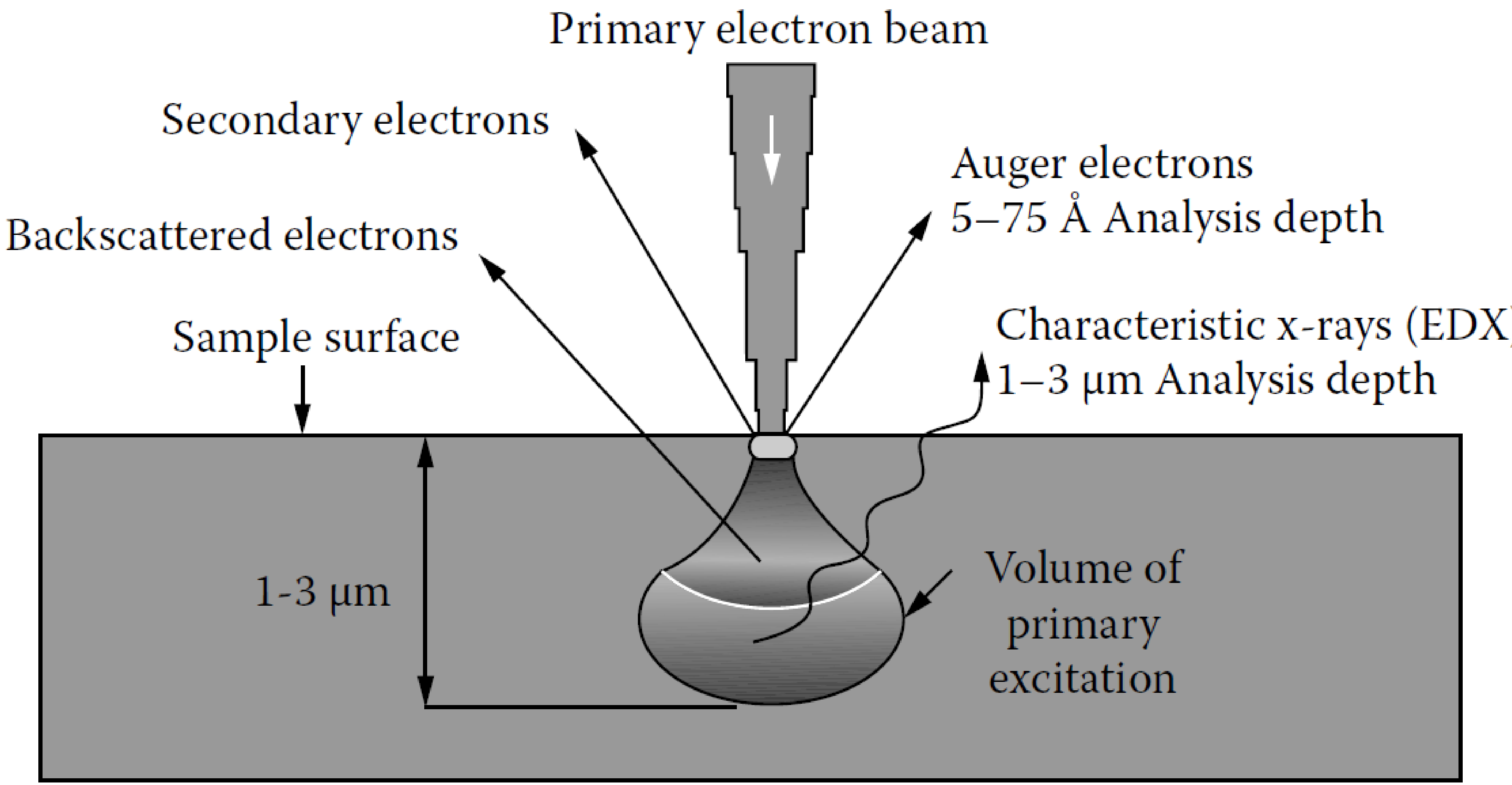
- *X-ray Photoelectron Spectroscopy (XPS) is also known as Electron Spectroscopy for Chemical Analysis (ESCA) is a widely used technique to investigate the chemical composition of surfaces.*

  - Measure the elemental composition, chemical stoichiometry, chemical state, and electronic state of the elements that exist in a material

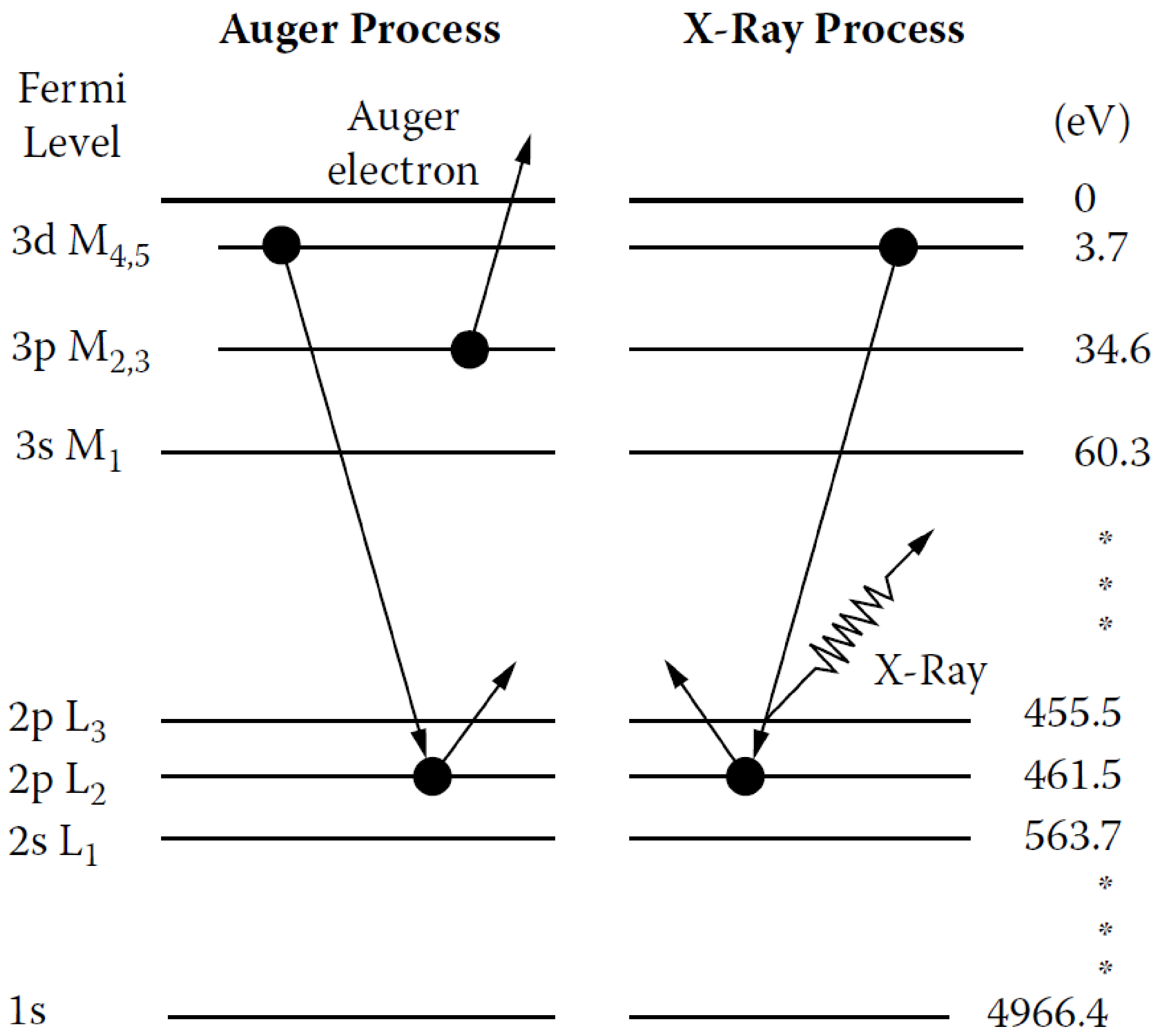
Schematic of X-ray Monochromator



Generation of electron beam

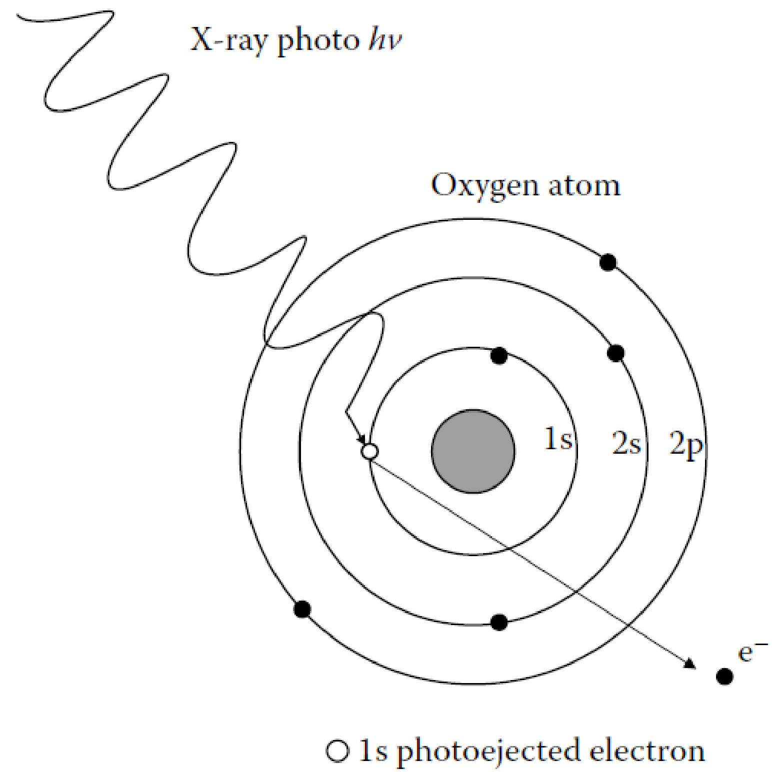


Auger and x-ray processes after ionization



Photoionization

Photoionization is the physical process in which an incident x-ray photon with energy of hv ejects one or more electrons from an atom, ion, or molecule.



- Not every photon that encounters an atom or ion will photoionize it. The probability of photoionization is related to the photoionization cross section, which depends on the energy of the photon and the target element being considered.

- For photon energies below the ionization threshold, the photoionization cross section is zero. Above the threshold, the cross section decreases as E^{-3} (photon energy)

The process of photoionization can be written as:

$$A + h\nu \rightarrow A^{+} + e^{-}.$$

The left-hand side is an atom of no charge (neutral), and a photon of energy *hv*; the right-hand side is an ionized atom (now one positive charge) plus an ejected electron (one negative charge).

From the energy viewpoint, the energies before and after the striking of the photon on the nucleus should be equal (conservation of energy):

$$E(\mathrm{A}) + h\nu = E(\mathrm{A}^{+}) + E(\mathrm{e}^{-}).$$

Since the energy of the ejected electron E(e^{-}) is present solely as kinetic energy (E_K)

$$E_K = h\nu - \underline{[E(A^{+}) - E(A)]}.$$

This term is actually the difference in energy between the ionized and neutral atoms. In fact, that is what we call the binding energy (E_B) of the electron.

Therefore, above Equation can be rewritten as

$$E_K = h\nu - E_B.$$

E_B is a direct measure of the energy required to remove only the electron concerned from its initial level to the vacuum level.

- E_B is unique for a given atom and a given orbital.
- The kinetic energy E_K of the photoelectron is the energy of the ejected electron after it leaves the "hold" of the atom. E_K varies when a different photon energy (*hv*) of the x-ray is used E_K is therefore not an intrinsic material property

- The E_B of the electron is that which identifies the electron specifically, both in terms of its parent element and atomic energy level.

- As E_K is measurable for a given x-ray source, *hv* is known. Thus it is easy to calculate E_B, from which one can identify the atom (element) that a particular electron had escaped from because no two elements share the same set of electronic binding energies.
- At a given photon energy, any changes in E_B are reflected in E_K, which means that changes in the chemical environment of an atom can be followed by monitoring the changes in the photoelectron energies, leading to the provision of chemical information.

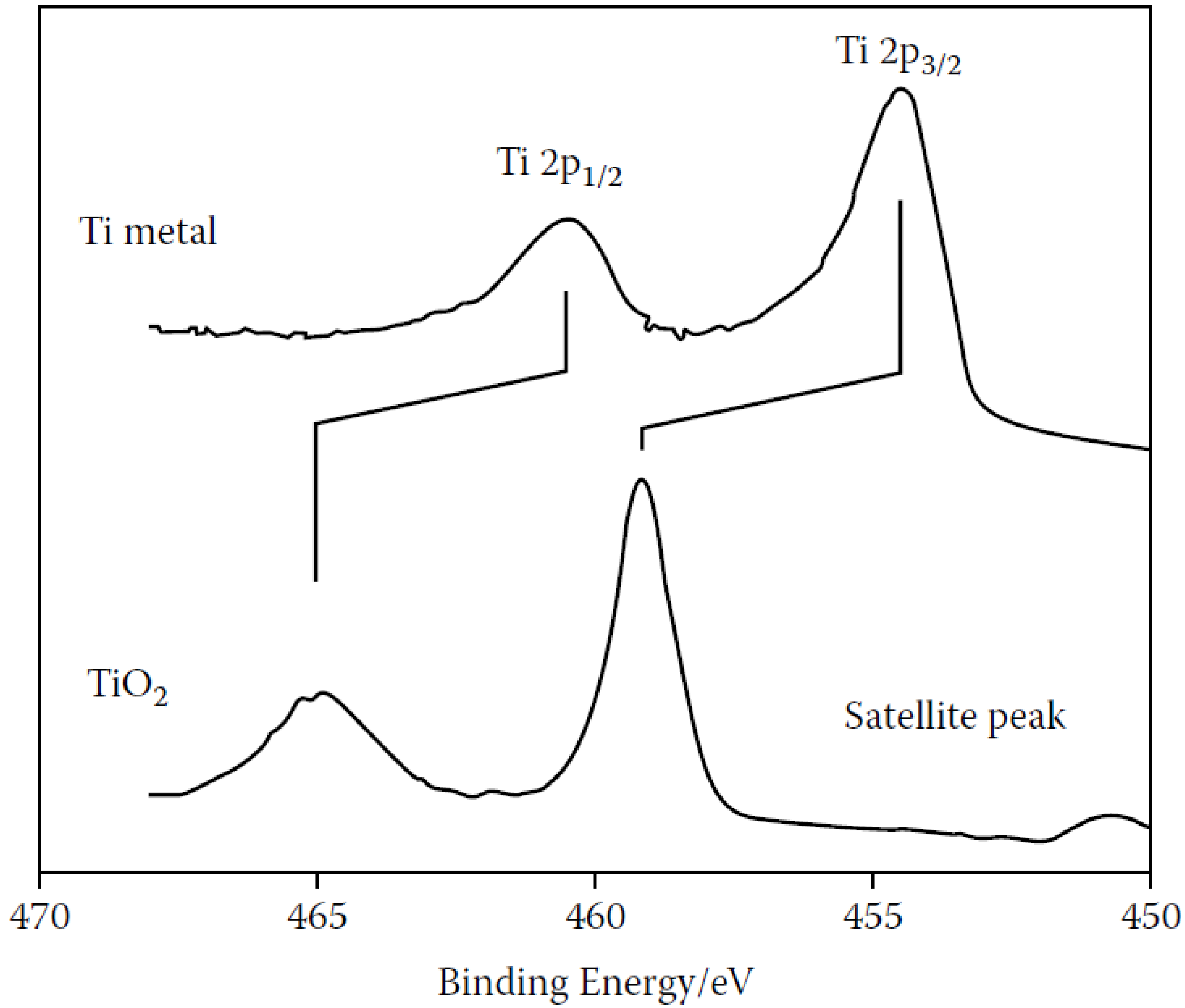
Chemical Shift in XPS

- The initial state is the ground state of the atom before ionization (either by photoionization or primary electron bombardment).

- If the energy of the atom's initial state is changed, for example, by formation of chemical bonds with other atoms or by alteration of the local chemical and physical environment (such as formation of oxide, thus a neutral state becomes a charged state), the energy of electrons in that atom will change.

- The change in energy, is the so-called chemical shift.

Electronic Effects- Spin-Orbit Coupling

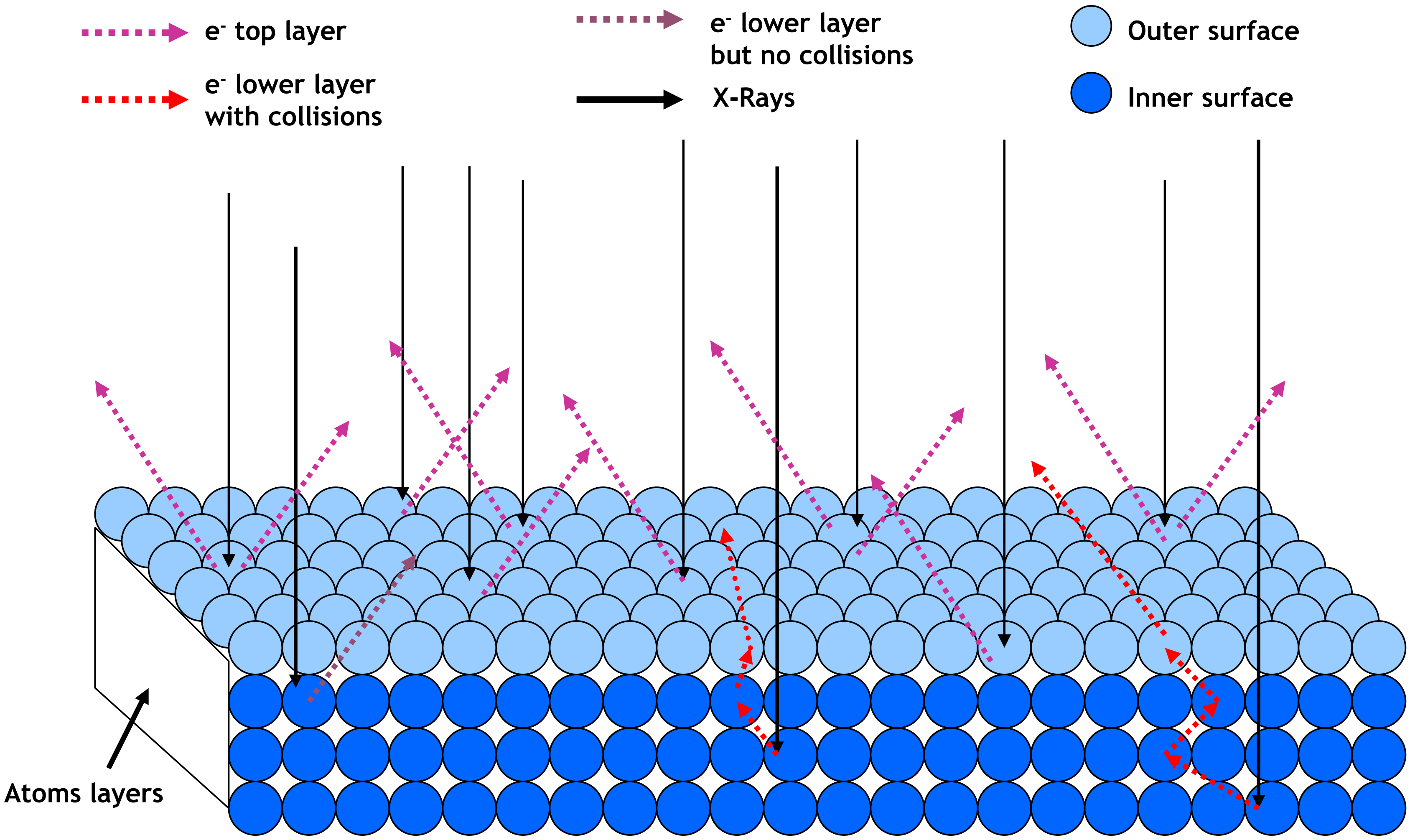


Elemental Shifts

	Binding Energy (eV)		
Element	***$2p_{3/2}$***	***3p***	Δ
Fe	707	53	654
Co	778	60	718
Ni	853	67	786
Cu	933	75	858
Zn	1022	89	933

Electron-nucleus attraction helps us identify the elements

X-Rays on the Surface



Although x-rays can penetrate much deeper into a material, photoelectrons free of inelastic collision are only from the top 1 to 10 nm of the material; thus, XPS is a very surface-sensitive method for chemical analysis.

X-Rays on the Surface

- The X-Rays will penetrate to the core e^- of the atoms in the sample.

- Some e^-s are going to be released without any problem giving the Kinetic Energies (KE) characteristic of their elements.

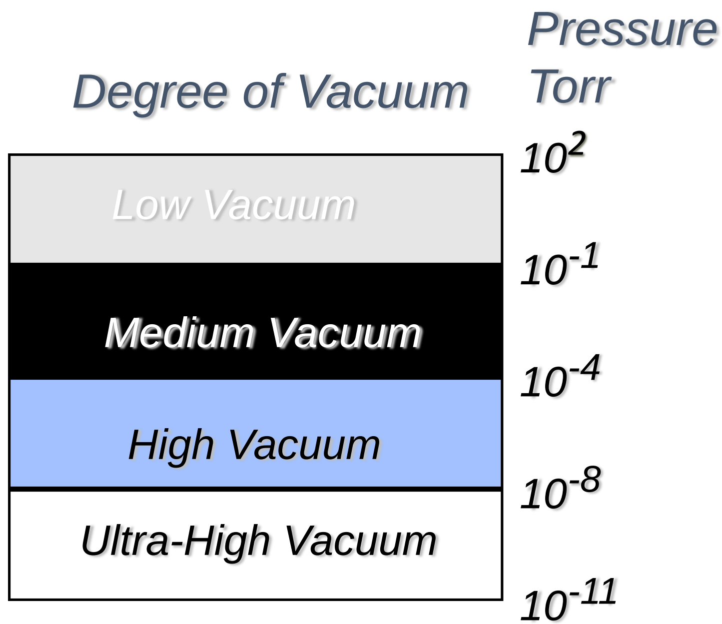
- Other e^-s will come from inner layers and collide with other e^-s of upper layers
  - These e- will be lower in lower energy.
  - They will contribute to the noise signal of the spectrum.

Instrumentation for X-ray Photoelectron Spectroscopy

COMPONENTS OF XPS:

- ❖A source of X-rays
- ❖An ultra high vacuum (UHV)
- ❖An electron energy analyzer
- ❖ magnetic field shielding
- ❖An electron detector system
- ❖A set of stage manipulators

Why UHV for Surface Analysis?



- Remove adsorbed gases from the sample.
- Eliminate adsorption of contaminants on the sample.
- Prevent arcing and high voltage breakdown.
- Increase the mean free path for electrons, ions and photons.

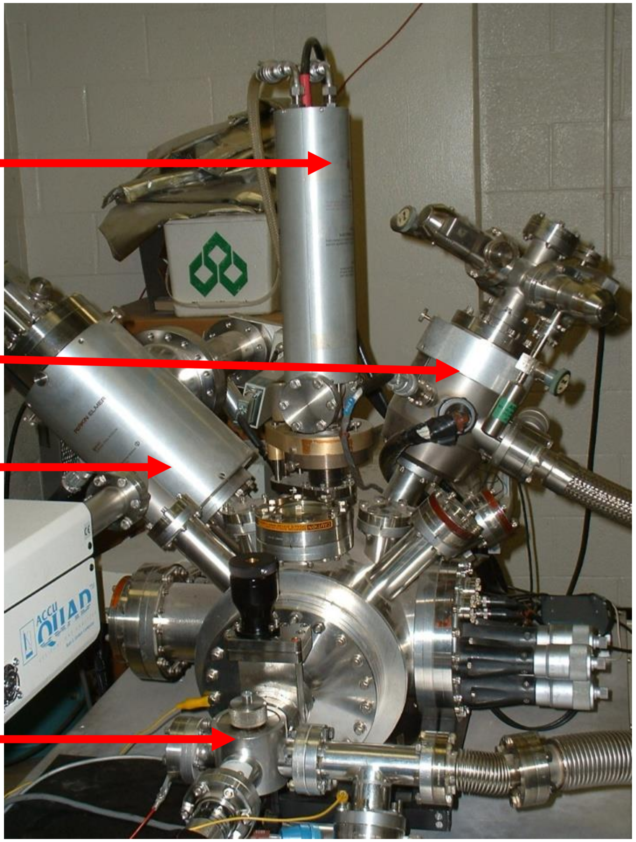
XPS Instrument

X-Ray Source

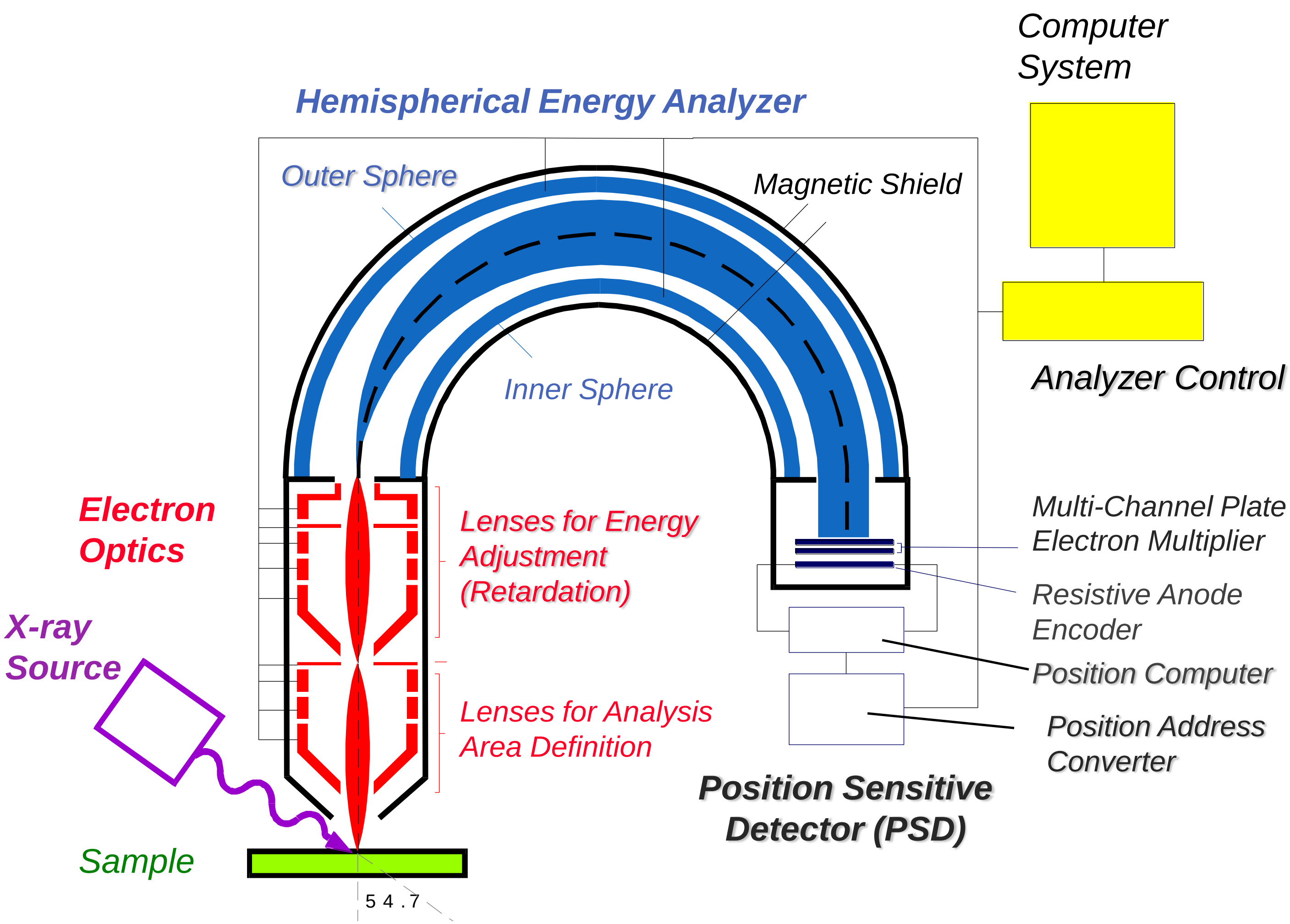
Ion Source

SIMS Analyzer

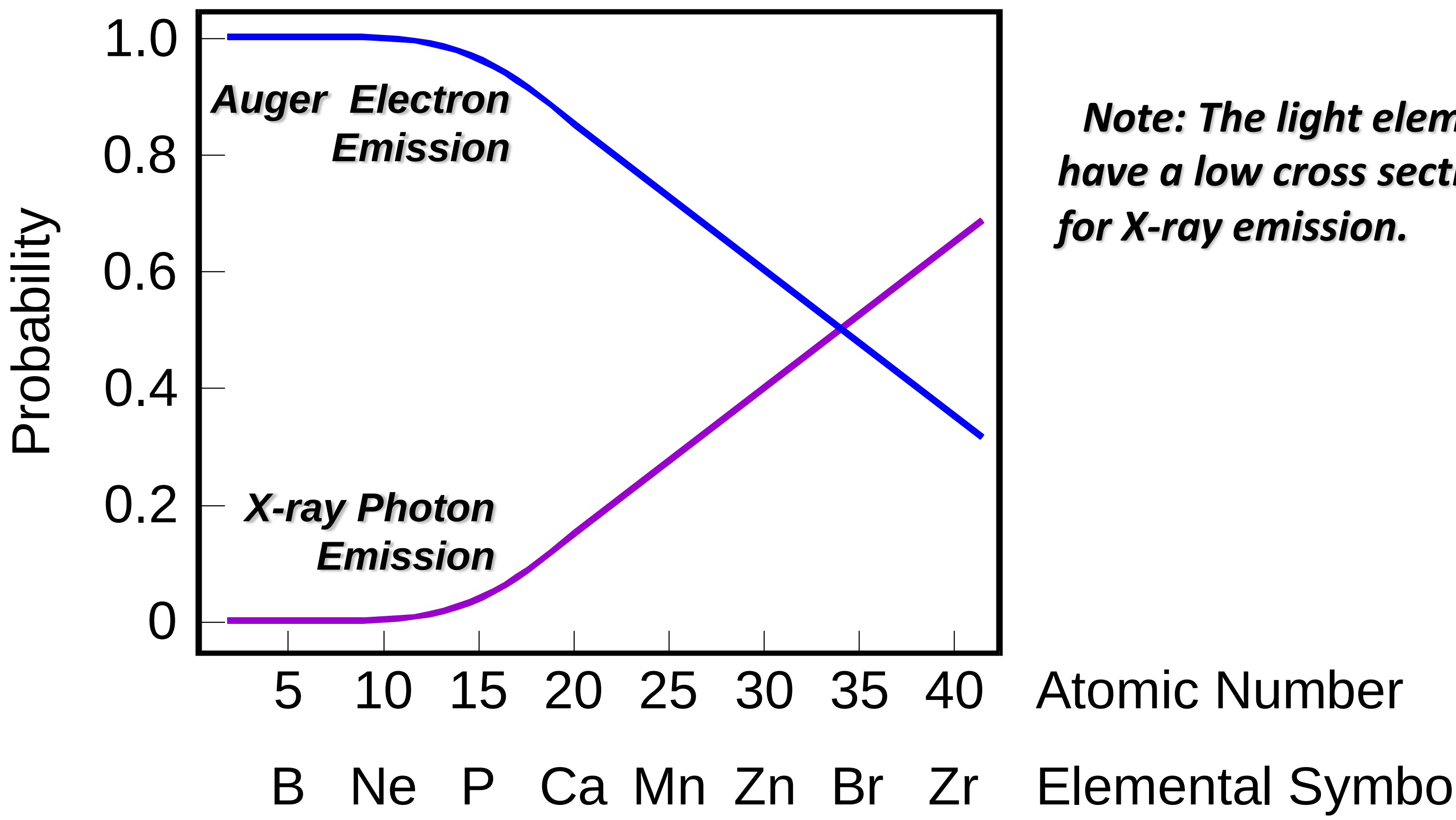
Sample introduction
Chamber



X-ray Photoelectron Spectrometer



Relative Probabilities of Relaxation of a K Shell Core Hole



Note: The light elements have a low cross section for X-ray emission.

How Does XPS Technology Work?

- A monoenergetic x-ray beam emits photoelectrons from the from the surface of the sample.
- The x-ray photons The penetration about a micrometer of the sample
- The XPS spectrum contains information only about the top 10 - 100 Å of the sample.
- Ultrahigh vacuum environment to eliminate excessive surface contamination.
- Cylindrical Mirror Analyzer (CMA) measures the KE of emitted e^-s.
- The spectrum plotted by the computer from the analyzer signal.
- The binding energies can be determined from the peak positions and the elements present in the sample identified.

which materials are analazıed?

- ❖XPS is routinely used to analyze inorganic compounds,metals,semiconductors,polymers, ceramics,etc.

- ❖Organic chemicals are not routinely analyzed by XPS because they are readily degraded by either the energy of the X-rays or the heat from non-monochromatic X-ray sources

ANALYSIS OF XPS

- XPS detects all elements with (Z) >3. It cannot detect H (Z = 1) or He (Z = 2) because the diameter of these orbitals is so small, reducing the catch probability to almost zero

- Dedection unit: ppt and some conditions ppm
- ppt (parts-per-trillion, 10−12)

conclusıon

XPS devices is also named ESCA(Electron Spectroscopy Chemical Analysis)It works base on the photoelectron effect .

XPS is primarly used for chemical analysis as determining thicknesses and emprical formulas of different elements and,binding energies,densities of electronic states.

XPS is used from starting Li(A>=3) because of smaller orbital's radius and analysis of organic compounds cannot be done with XPS because of degredation with radiation.

XPS is used also determination of others like; metals,ceramics,semiconductors,papers,etc.

X-ray Diffraction

The basic method of crystal diffraction

1. X-ray diffraction

X-ray is an electromagnetic wave generated by electrons accelerated by a high voltage striking the "target" material

$$h\gamma_{\max} = eU \qquad h\frac{c}{\lambda_{\min}} = eU$$

$$\lambda_{\min} = \frac{hc}{eU} \approx \frac{1.2\times10^{3}}{U} \ (\mathrm{nm})$$

$$h \approx 6.62\times10^{-34}\,\mathrm{J\cdot s} \qquad c = 3\times10^{8}\,\mathrm{m/s} \qquad e \approx 1.6\times10^{-19}\,\mathrm{C}$$

$\mathbf{U = 10^4\,V}$, $\lambda \sim 0.1\,\mathrm{nm}$ In practice, normally 40 kV is applied

X-rays can only interact with electrons of the sample

The basic method of crystal diffraction

2. Electron diffraction

$$\lambda = \frac{h}{P}, \quad \frac{P^2}{2m} = eU, \quad P = \sqrt{2meU}, \quad \lambda = \frac{h}{\sqrt{2meU}}$$

$$h \approx 6.62 \times 10^{-34}\,\mathrm{J \cdot s} \qquad e \approx 1.6 \times 10^{-19}\,\mathrm{C} \qquad m \approx 9.1 \times 10^{-31}\,\mathrm{kg}$$

$$\lambda \approx \sqrt{\frac{1.5}{U}}\ \text{(nm)} \qquad U = 150\ \mathrm{V}, \lambda \sim 0.1\,\mathrm{nm}$$

Electron waves are scattered by electrons and nuclei, with strong scattering and weak transmission

The basic method of crystal diffraction

3. Neutron diffraction

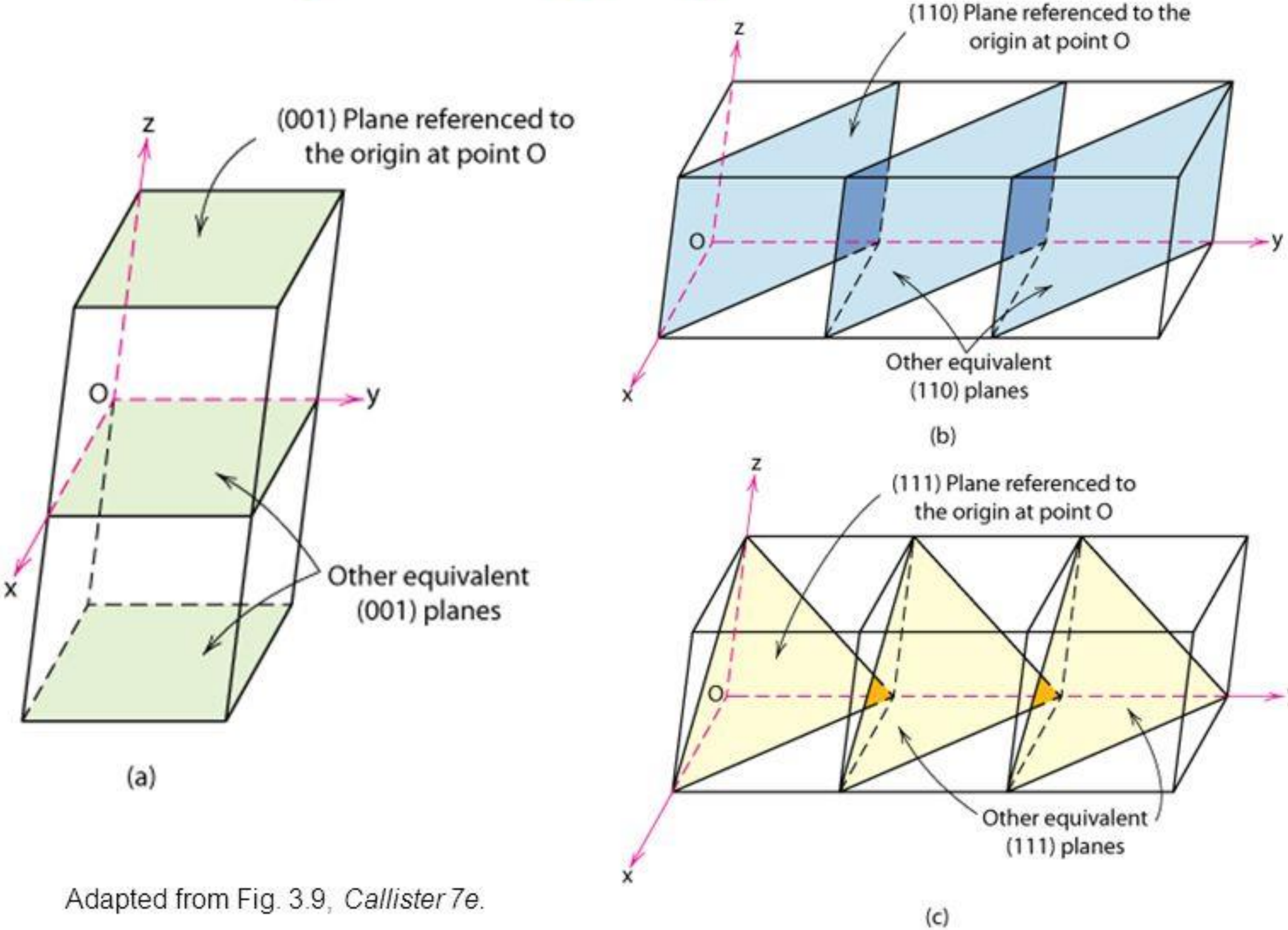
$$m_{中} \approx 2000 m_{电} \qquad \lambda \approx \sqrt{\frac{1.5}{2000\text{U}}} \approx \sqrt{\frac{7.5}{\text{U}}} \times 10^{-2}\,\text{nm}$$

$$\text{U} \sim 0.1\text{V}, \lambda \sim 0.1\text{nm}$$

- Neutrons are mainly scattered by atomic nuclei, and light atoms also have strong scattering of neutrons, so they are often used to determine the position of hydrogen and carbon in the crystal.

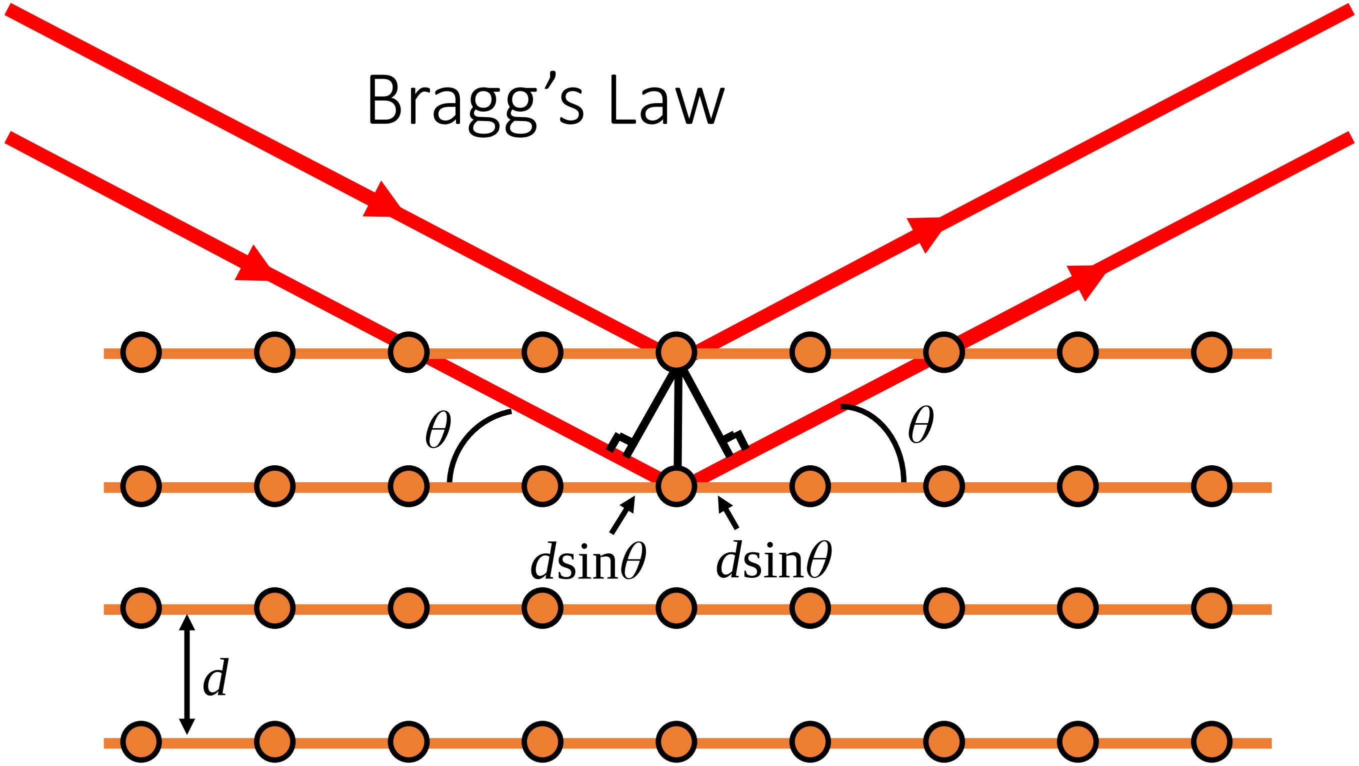
- Neutrons have magnetic moments and are particularly suitable for studying the structure of magnetic substances.

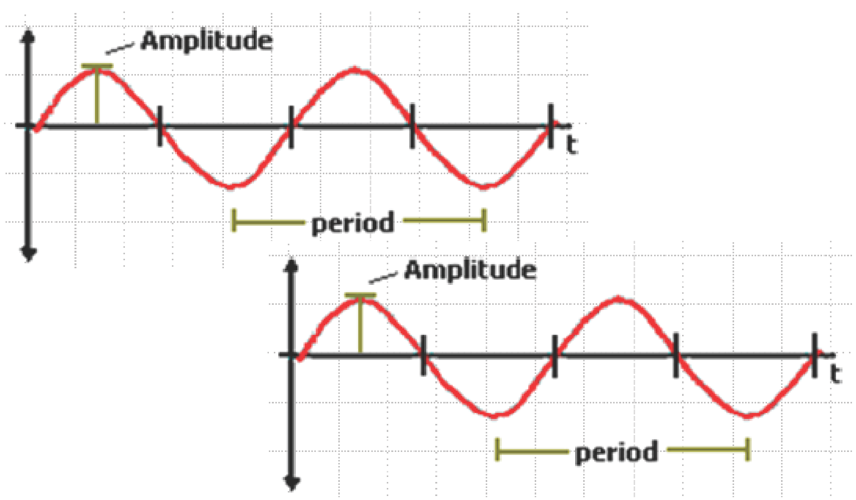
Crystallographic Planes



Adapted from Fig. 3.9, *Callister 7e.*

Bragg's Law





Path length difference = $2d\sin\theta$

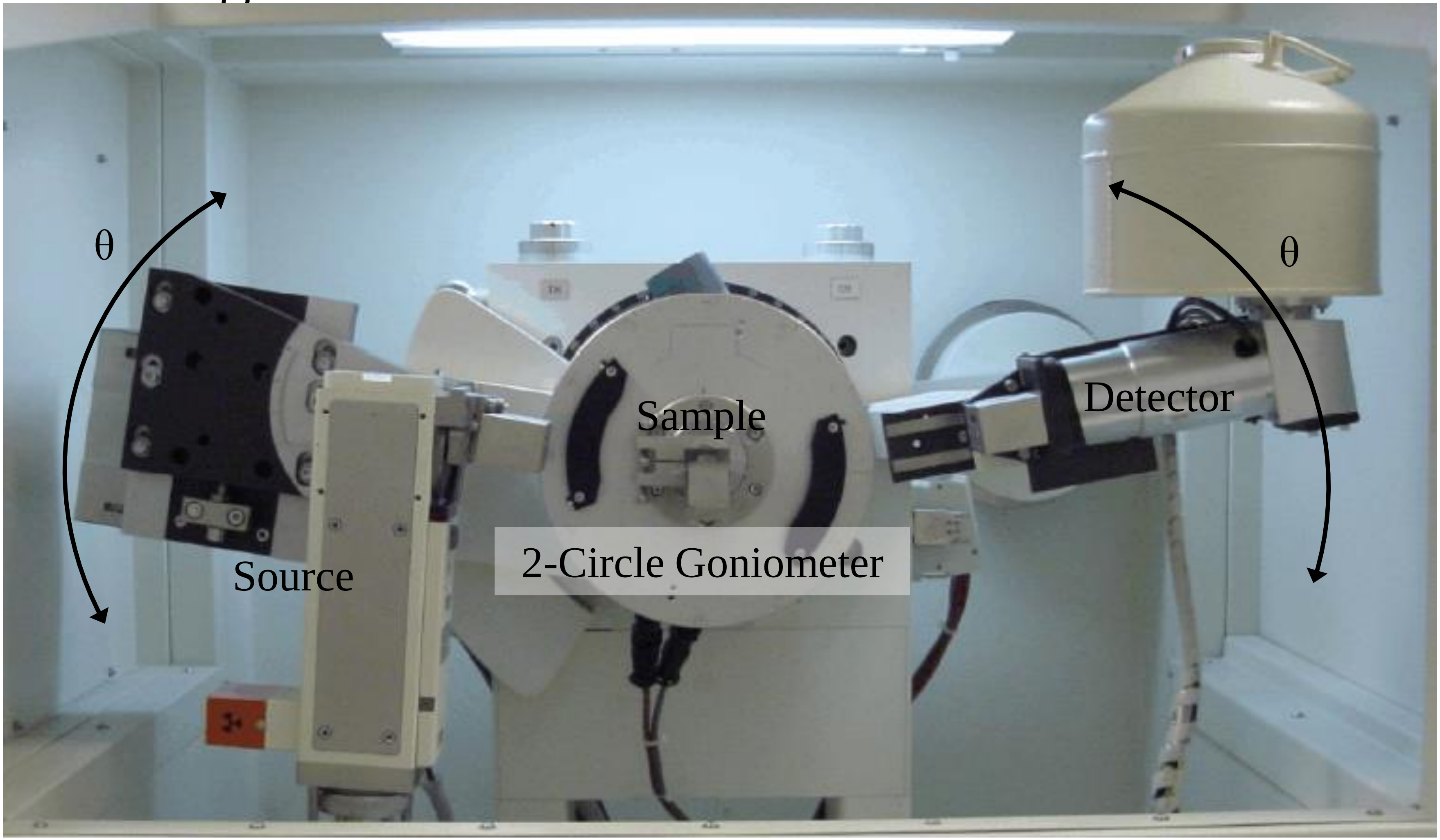
Constructive interference for

$2d\sin\theta = n\lambda$

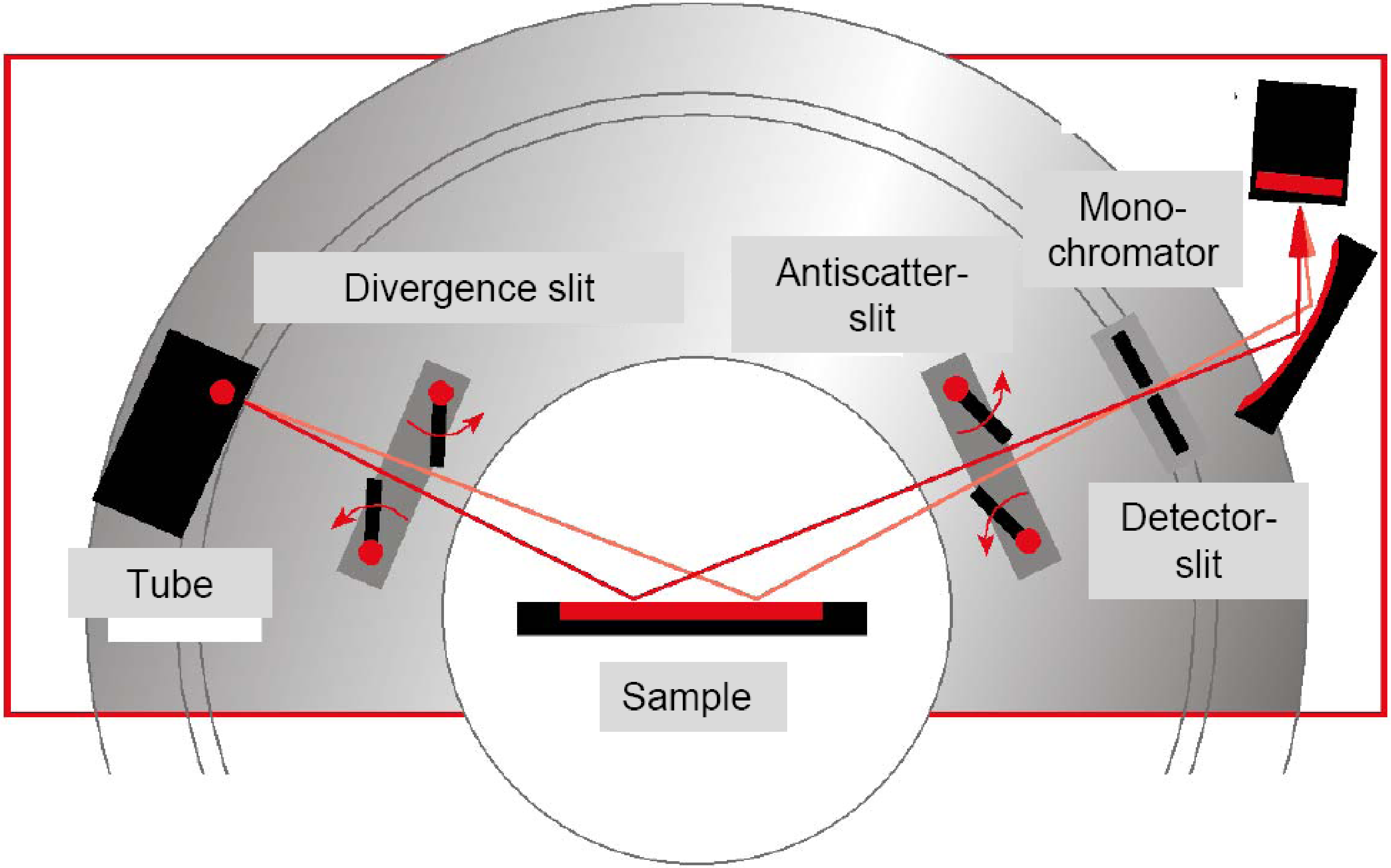
Destructive interference for

$2d\sin\theta = n\lambda/2$

Scintag Powder Diffractometer



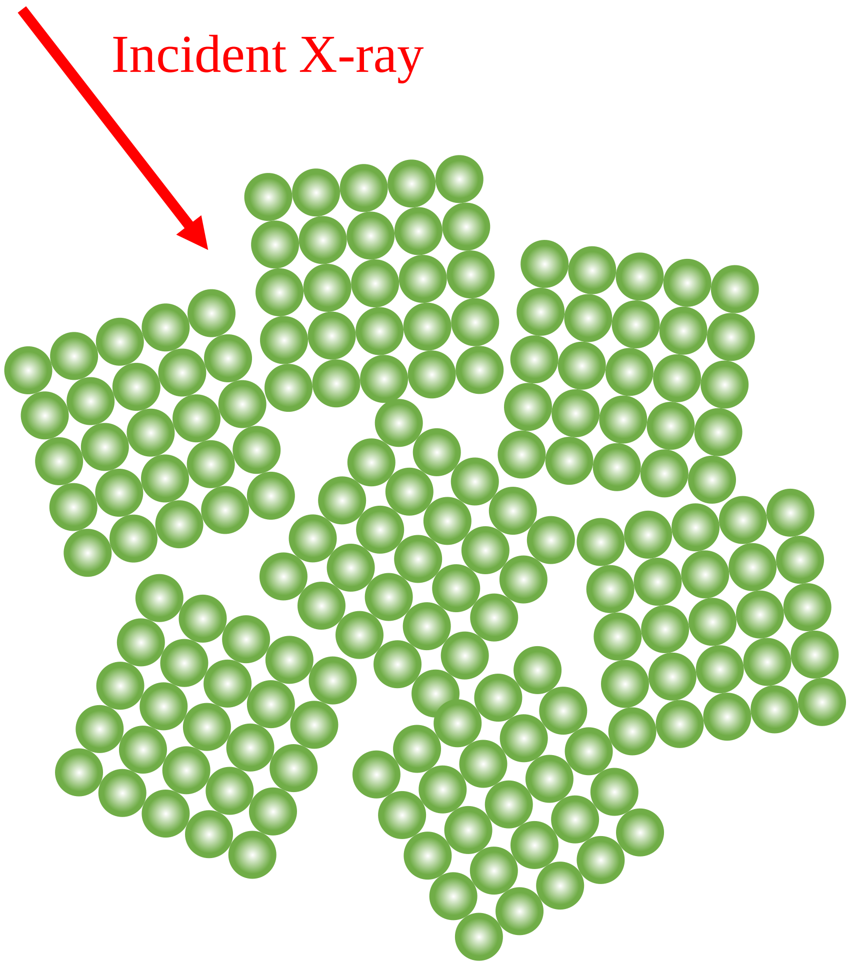
Typical Powder Diffraction System

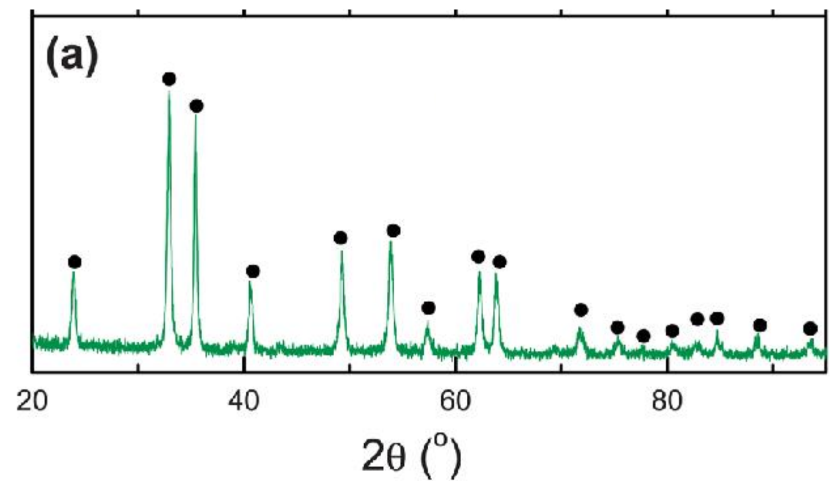


- For flat samples

Powder Diffraction

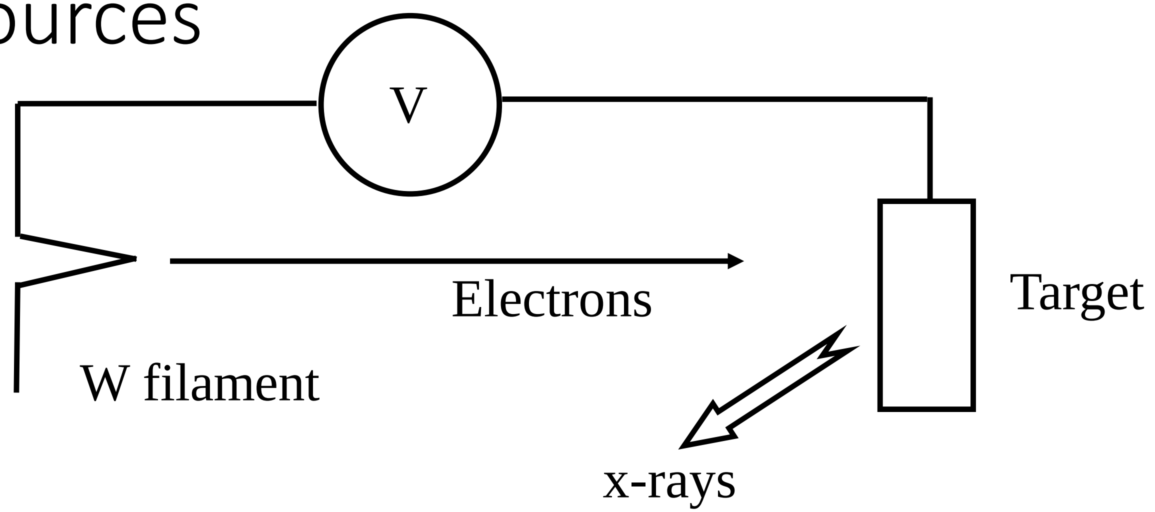
Fine grained powder with random orientation





- All the crystal planes have been diffracted
- Grains have random orientation

X-Ray Sources



- Common target materials:
  - Mo
  - Cu
  - Co
  - Cr
  - Fe

Produce x-rays of different wavelength

- Only ~1% of total energy goes to creating x-rays.

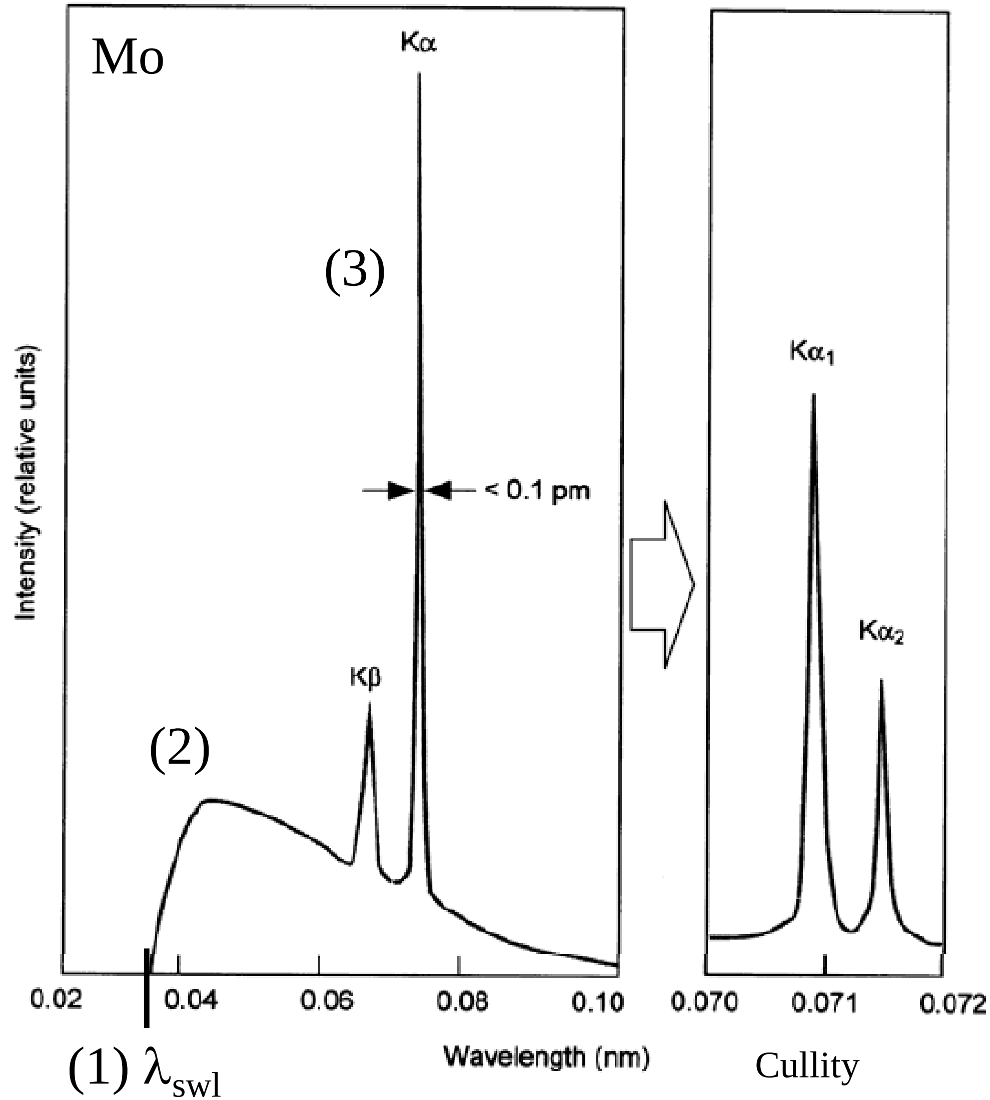
X-Ray Spectrum

1. Short wavelength limit
   - All kinetic energy of electron transferred to photon.

$$eV = h\nu_{\max} = \frac{hc}{\lambda_{swl}} \qquad \lambda \text{ in Å} \qquad V \text{ in volts}$$

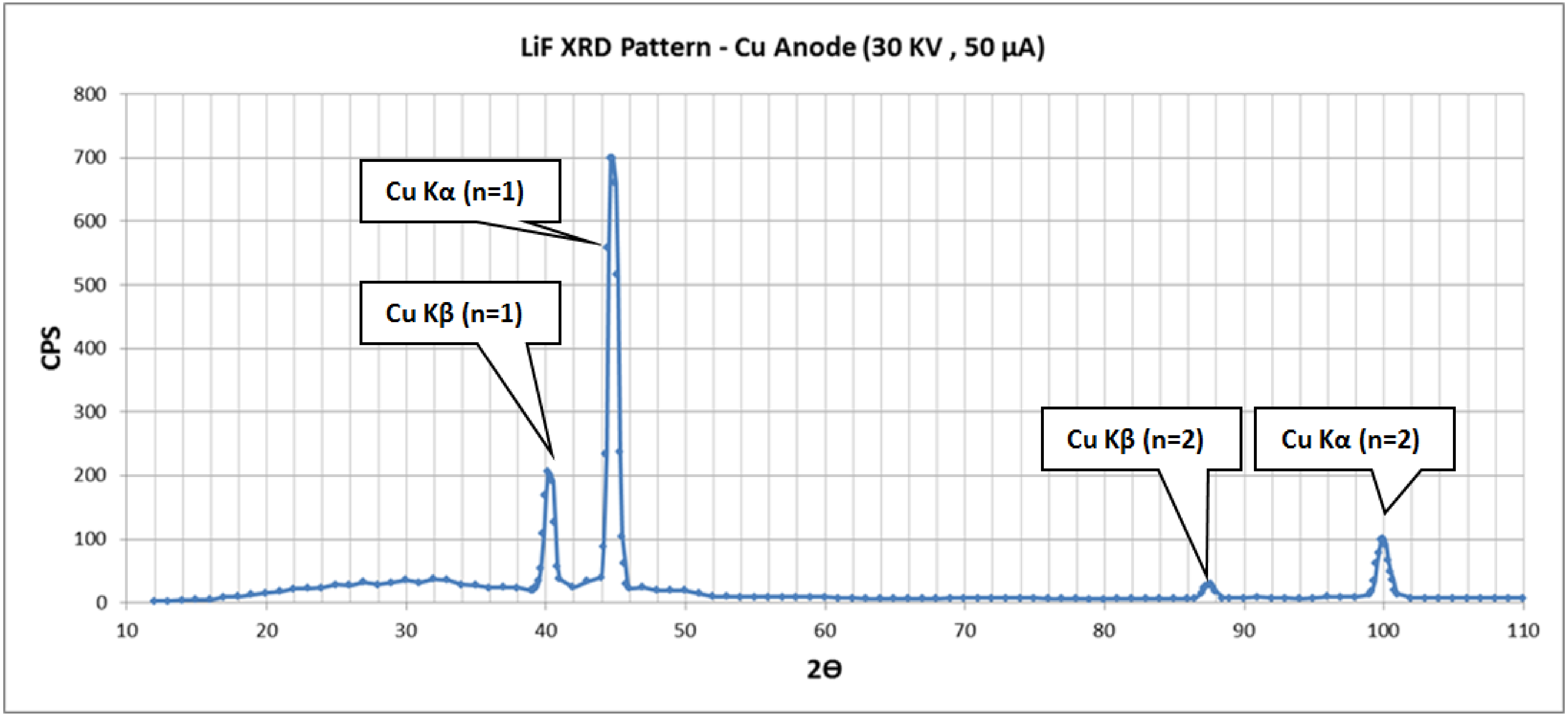
$$\lambda_{swl} = \frac{12.398 \times 10^{3}}{V}$$

2. Bremsstrahlung radiation or "braking radiation"
   - Deceleration of electrons by target atoms results in radiation of x-rays.
3. Characteristic radiation
   - Discrete intra-atomic electron transitions within target atoms.



Cullity

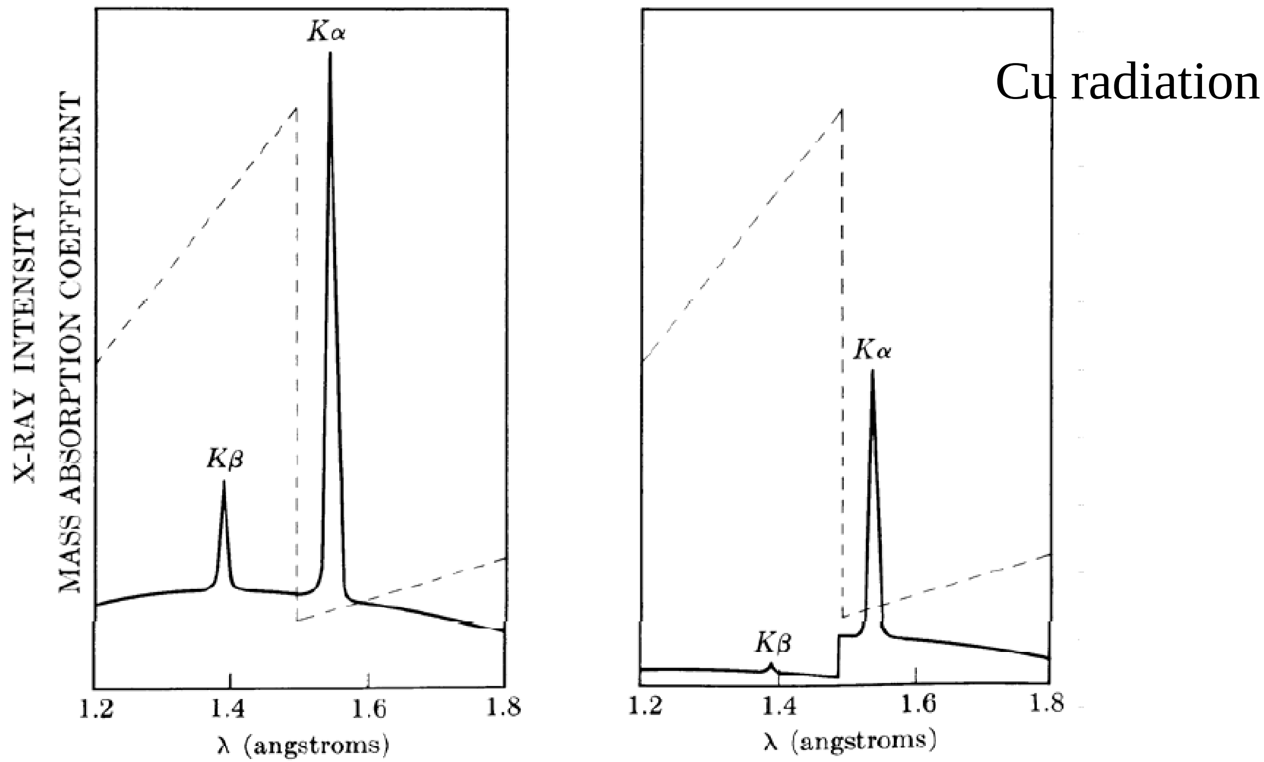
K_α and K_β radiation for the same crystal plane



- For the same crystal plane, two or more diffraction peaks appear
- A filter has to be used to block the unwanted x-ray

Use of Filters

- X-ray diffraction requires monochromatic radiation.
- Use a filter/absorber with an absorption edge just above the undesirable radiation.

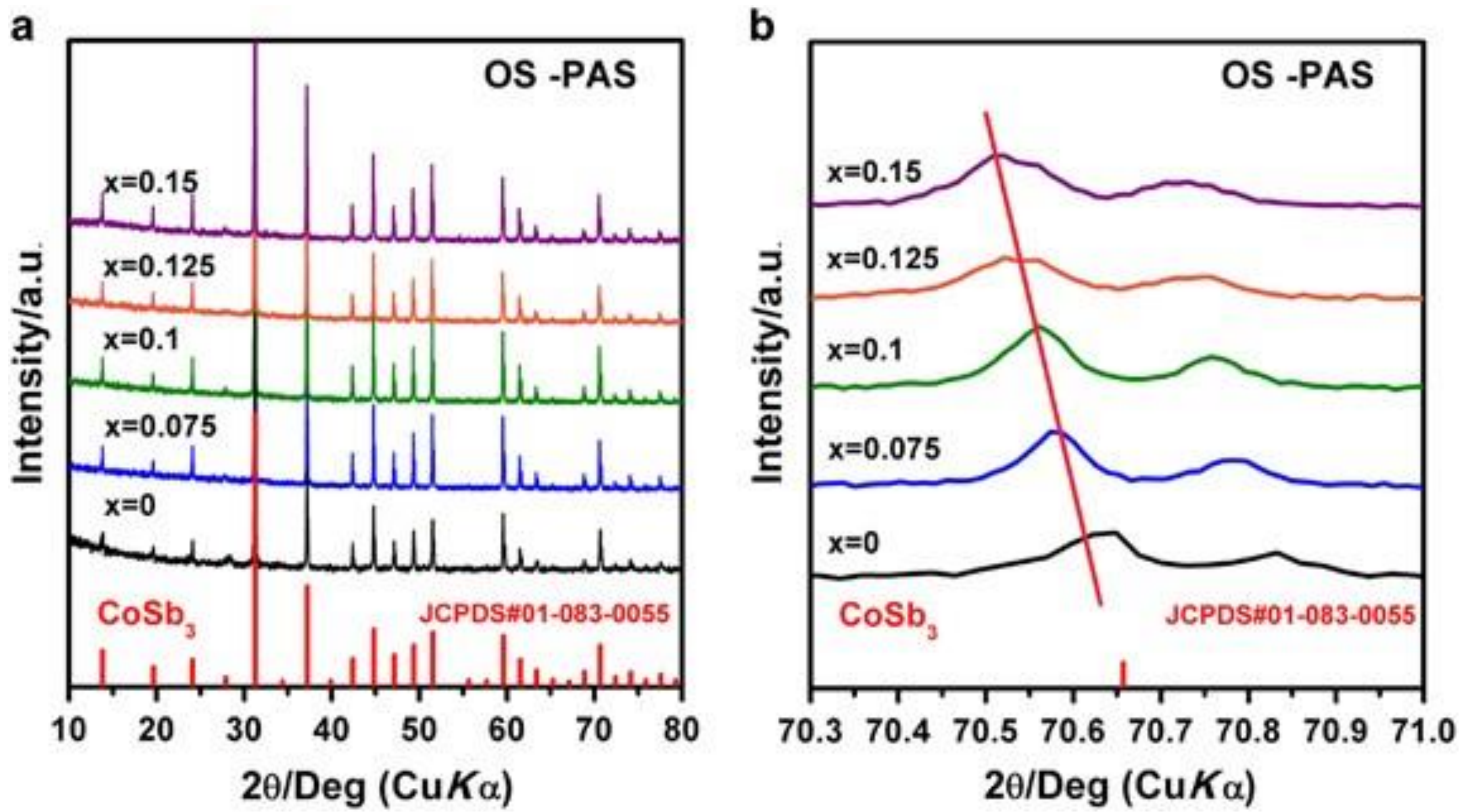


No Filter

Ni Filter

Cullity

Peak shifts



$$2d\sin\theta = n\lambda$$

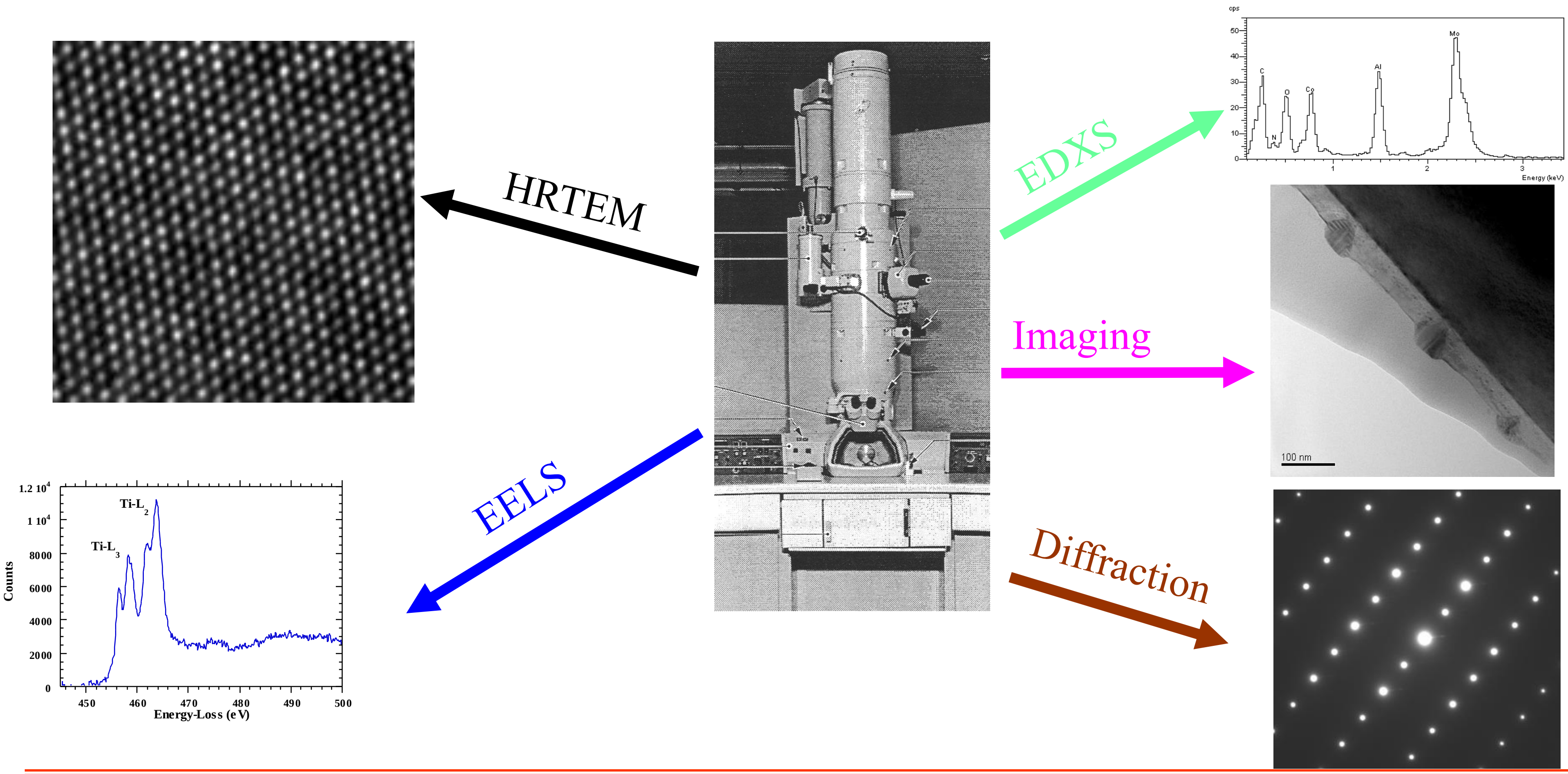
d is associated with lattice constant, a

Peak shift is due to variation of the lattice constant

- ➢ Strain, doping, etc

Transmission Electron Microscopy (TEM)

Transmission Electron Microscopy (TEM)



Why so many terminology !

TEM
Bright-field
Dark-field
SAED
CBED

Imaging
Morphology
Structure
Defect

AEM
EDXS
EELS
Image Filter
EDX mapping

Composition
Chemical state
Spatial distribution

HRTEM

Structural image

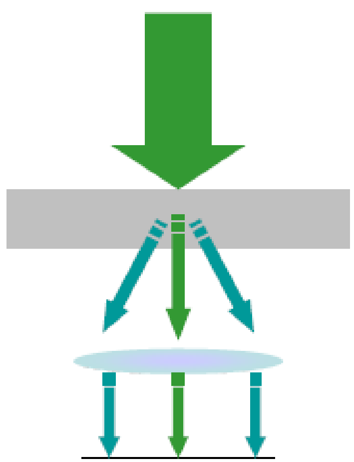
STEM
ADF
HAAD

Image
Z-contrast

TEM vs. SEM

Transmission EM

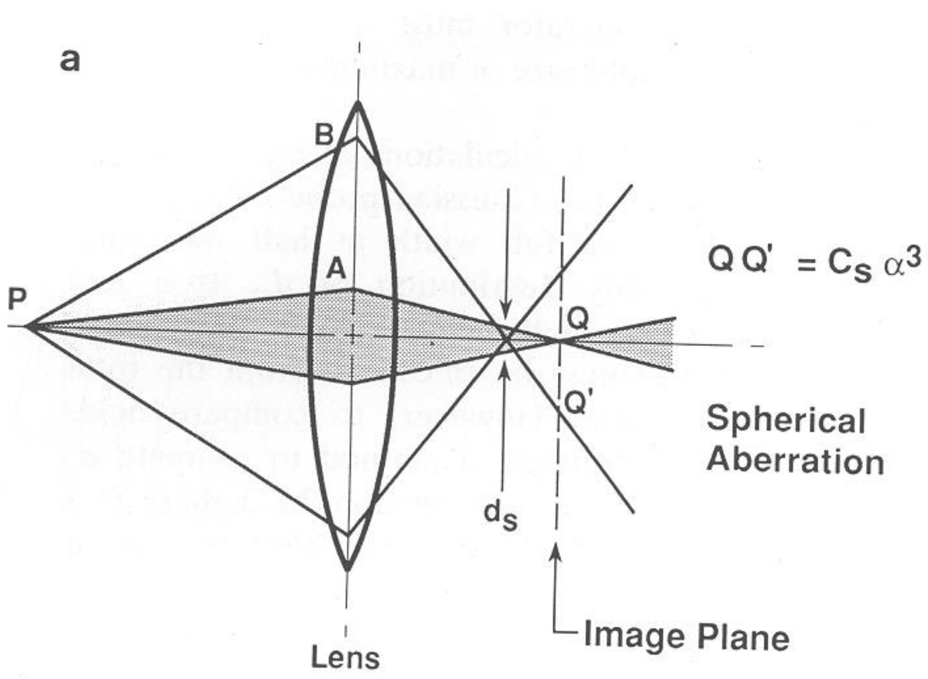
- Projection view
- Volume sensitive
- Diffraction effects
- Higher spatial resolution
- Analytical use (STEM)



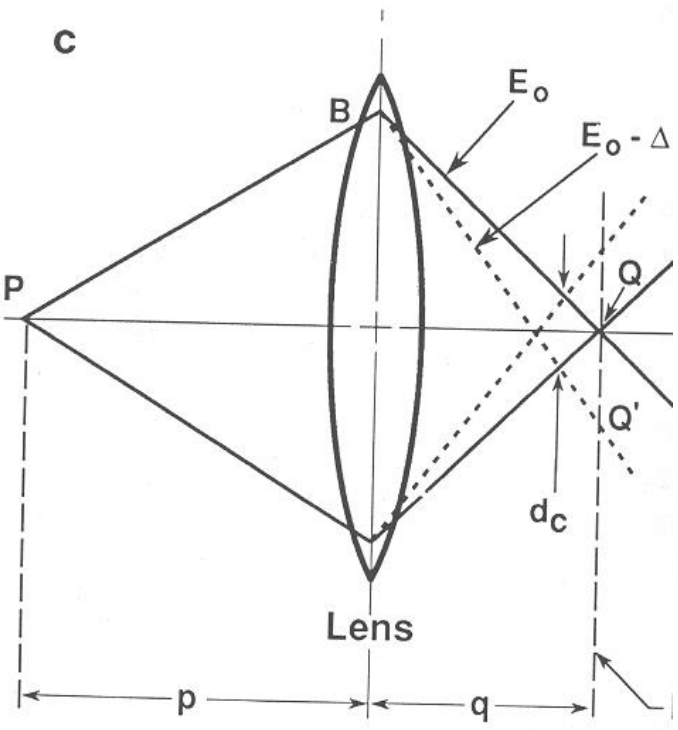
Scanning EM

- Topography effects
- backscattering
- Fewer optical elements
- Ease of use
- Analytical use (scanning mode)

Resolution is limited by lens aberrations



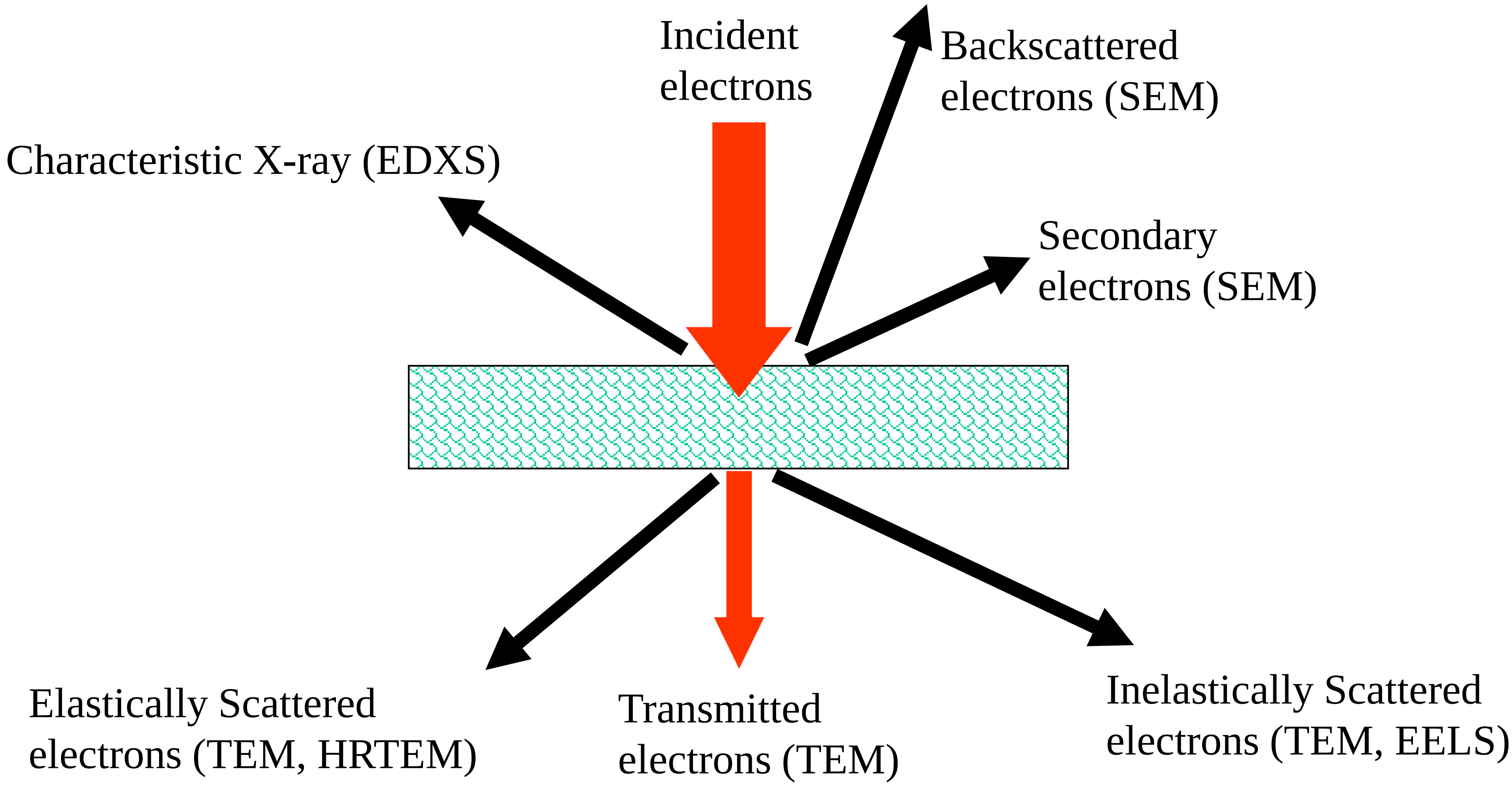
Spherical Aberration



Chromatic Aberration

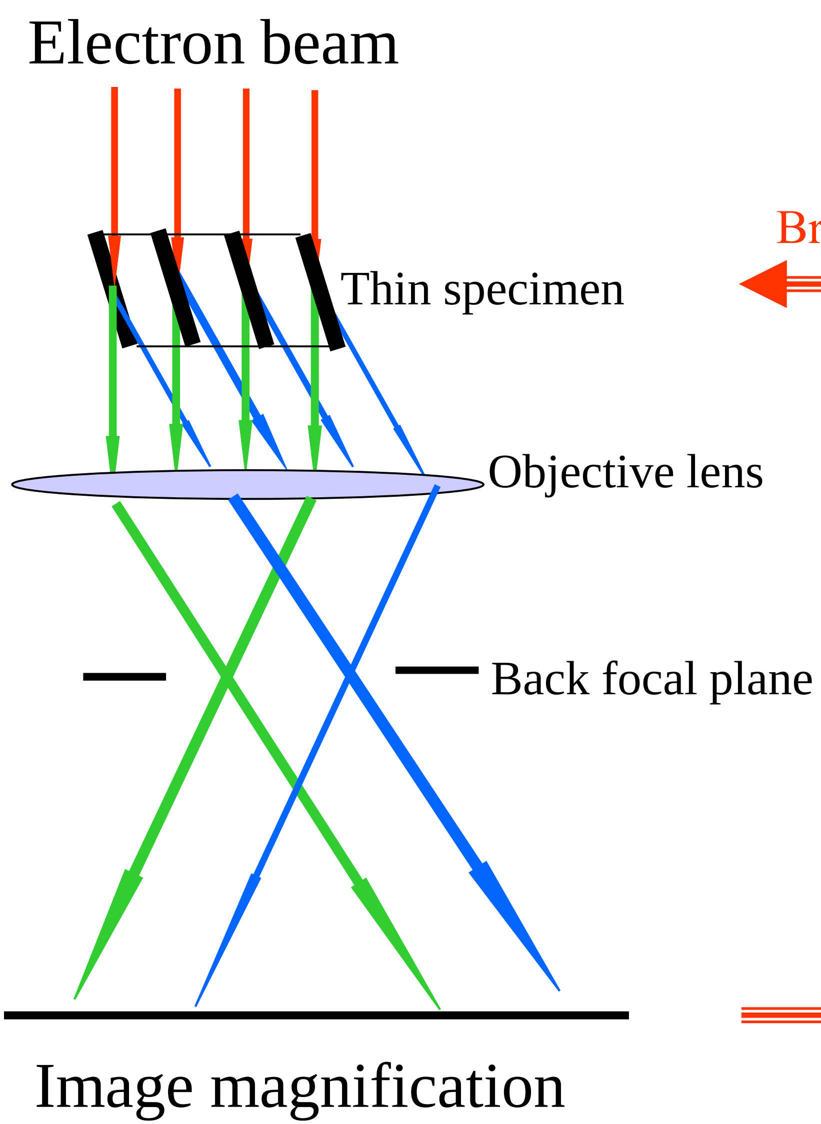
Highest resolution obtained to date is 0.086 nm (example)

Interaction of electron with specimen



Specimen must be transparent for electron ranging from atomic size to 1 μm

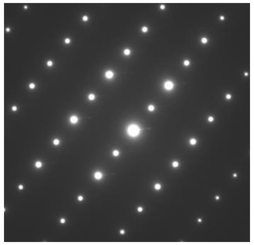
Electron Diffraction and Imaging



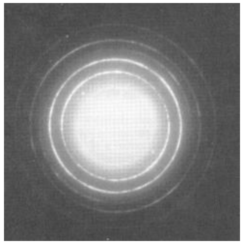
Bragg's Law

$$2d\sin\theta = n\lambda$$

Diffraction

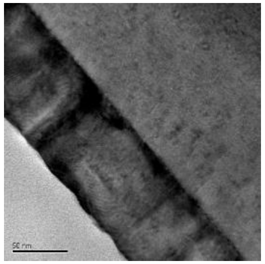


Single crystal



Polycrystalline

Imaging



Image

Limitations of TEM

- Charging of the sample
  → increase conductivity
- Beam damaging (80-3,000 keV electrons)
  → cooling of the sample
- Specimen preparation (electron transparency)
  → specimen thickness ≈ 10-500 nm
- Projection view
- Environment (mechanical, magnetic, electrical)

Beam damage

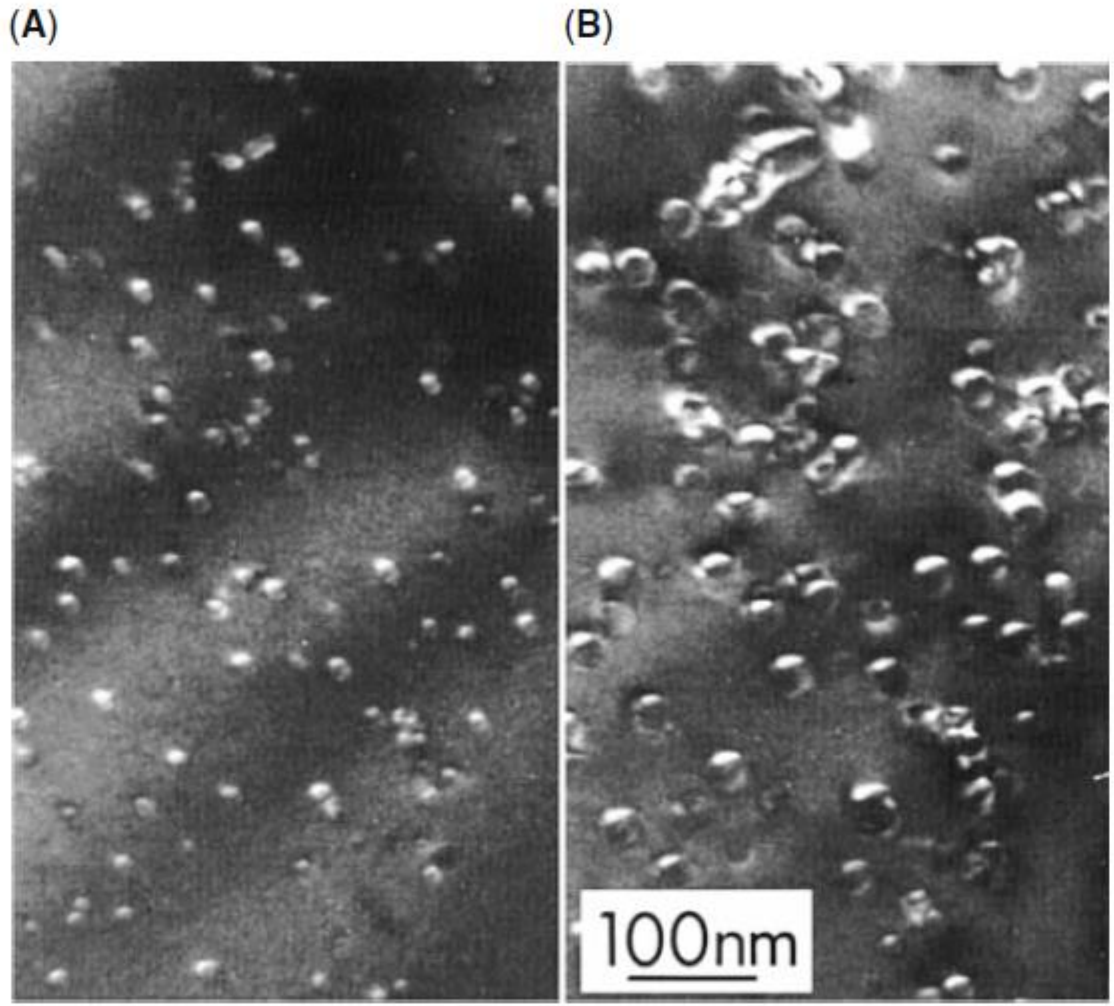
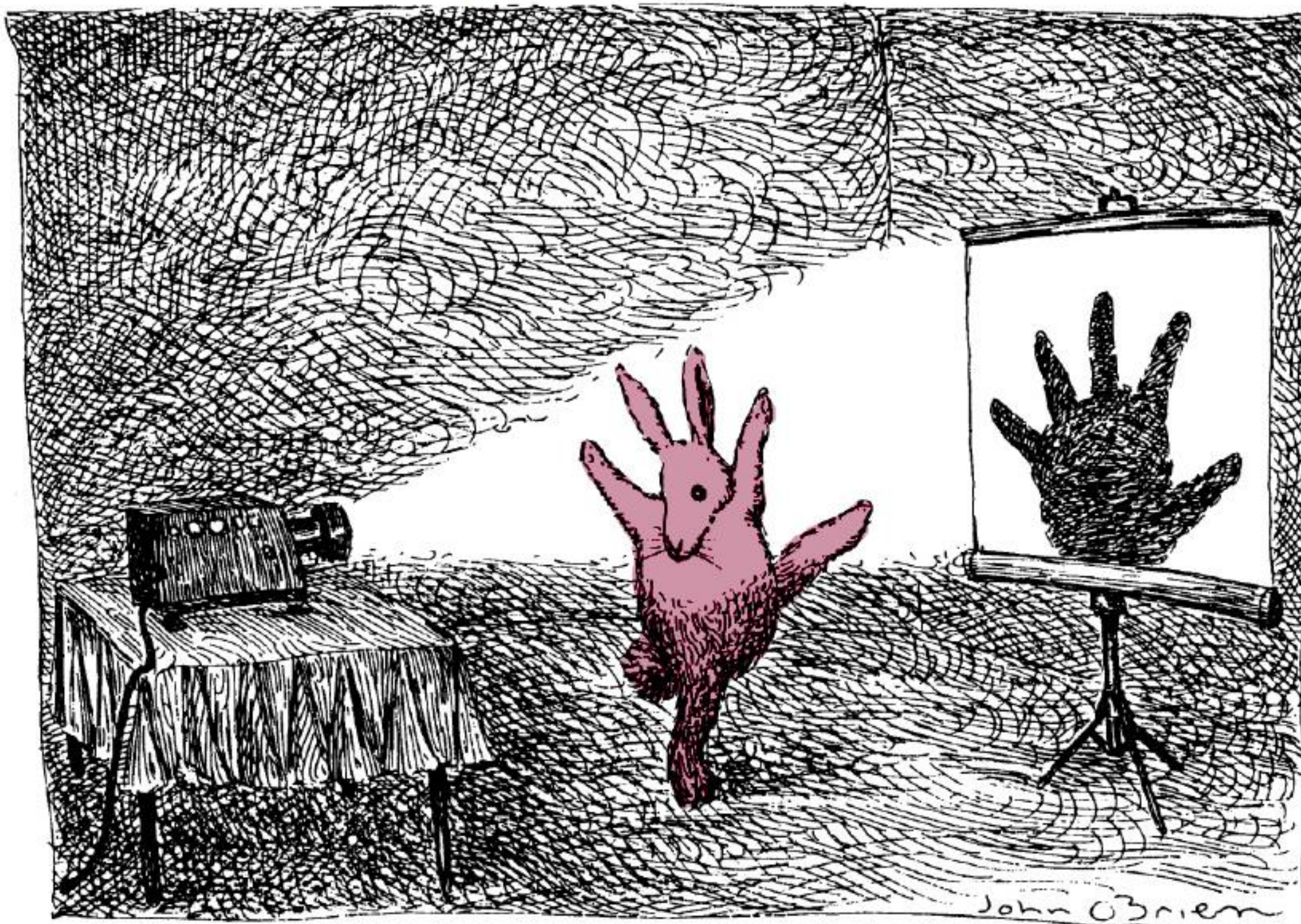


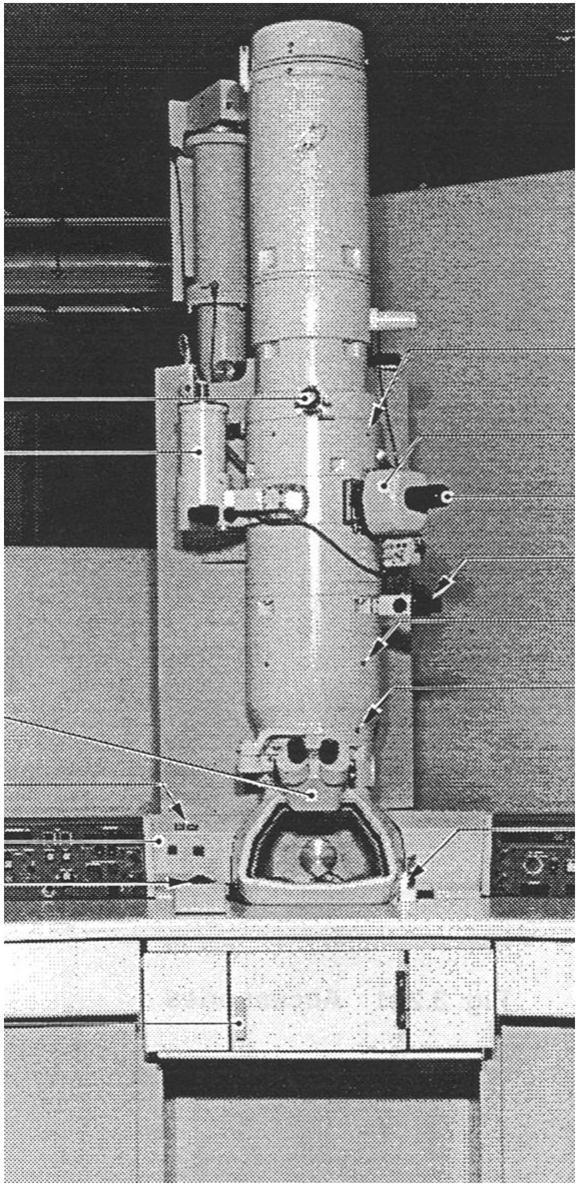
FIGURE 1.8. Beam damage (bright bubble-like regions) in quartz after bombardment with 125 keV electrons. With increasing time from (A) to (B) the damaged regions increase in size.

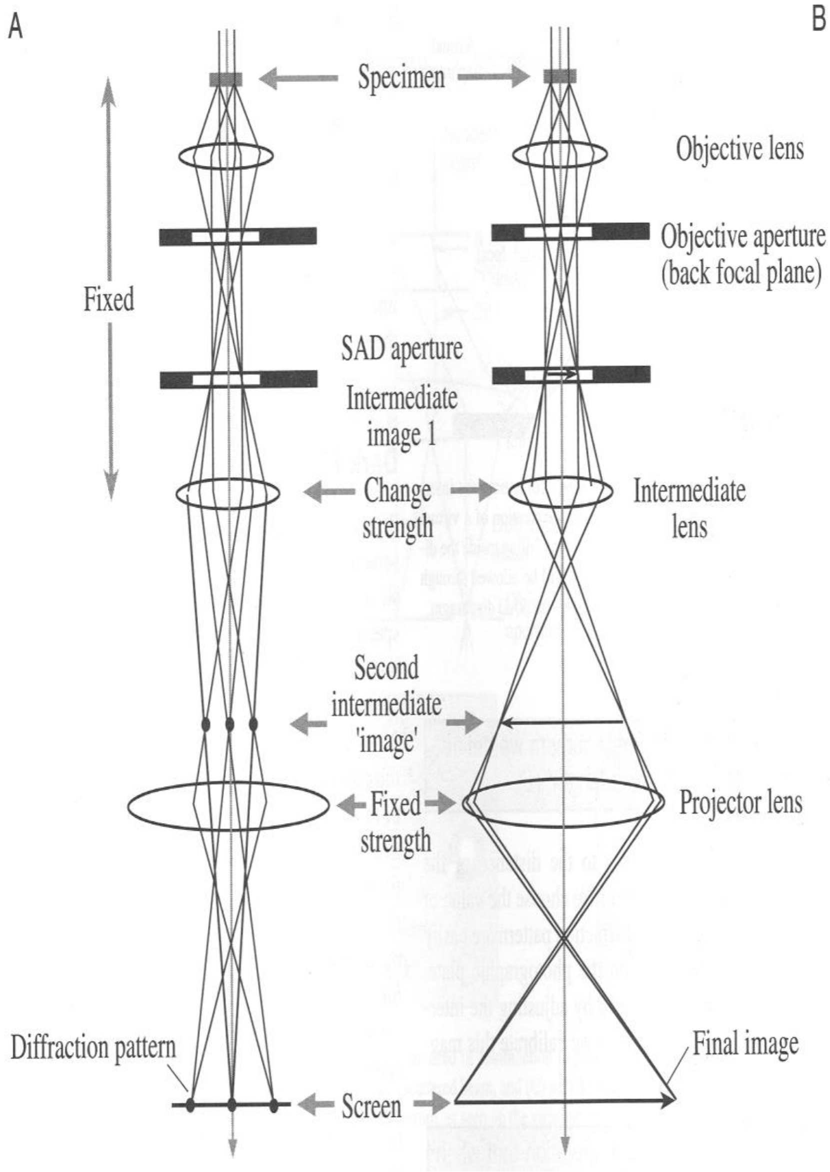
Projection view

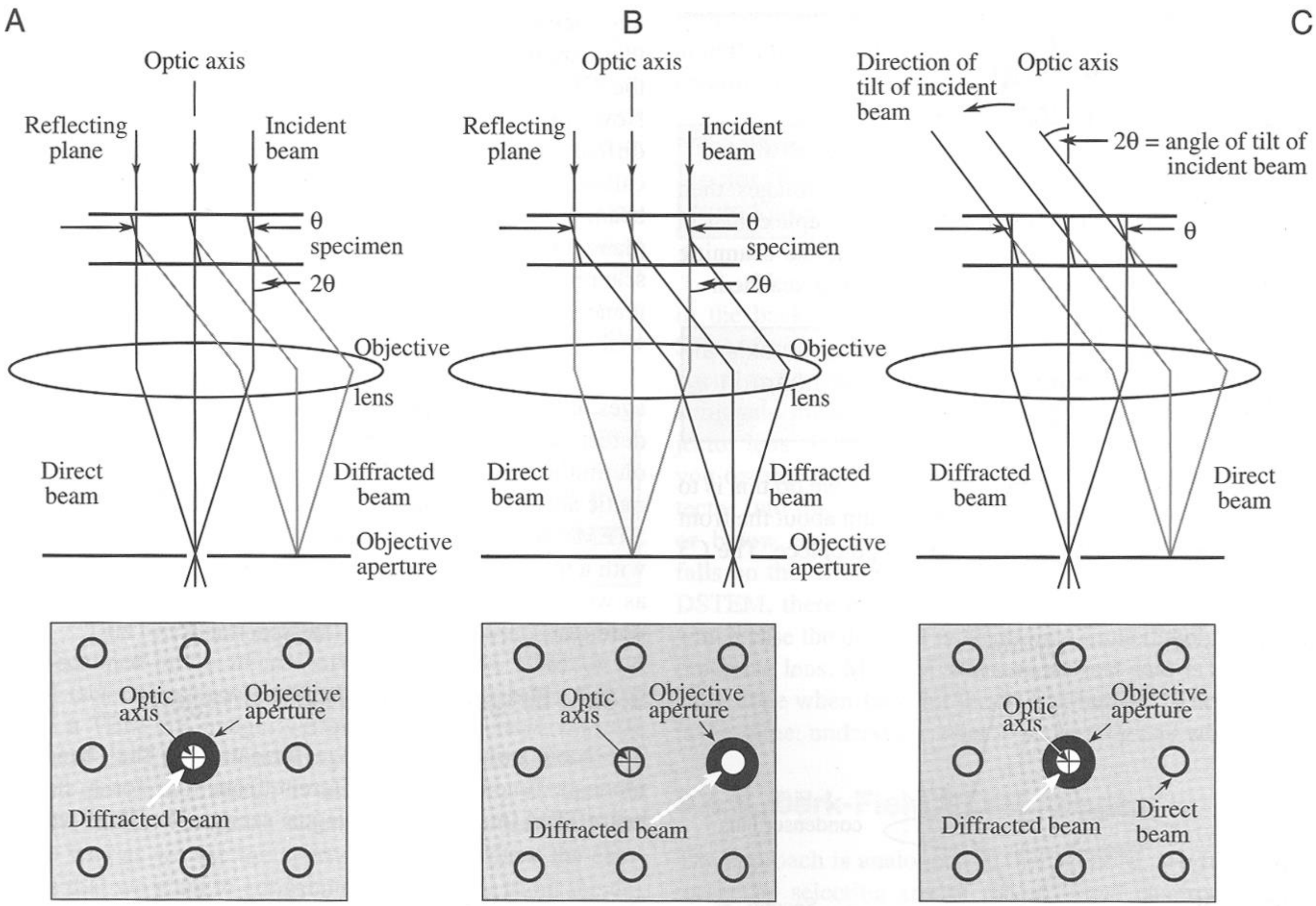


A single projection image is plainly insufficient to infer the structure of an object.

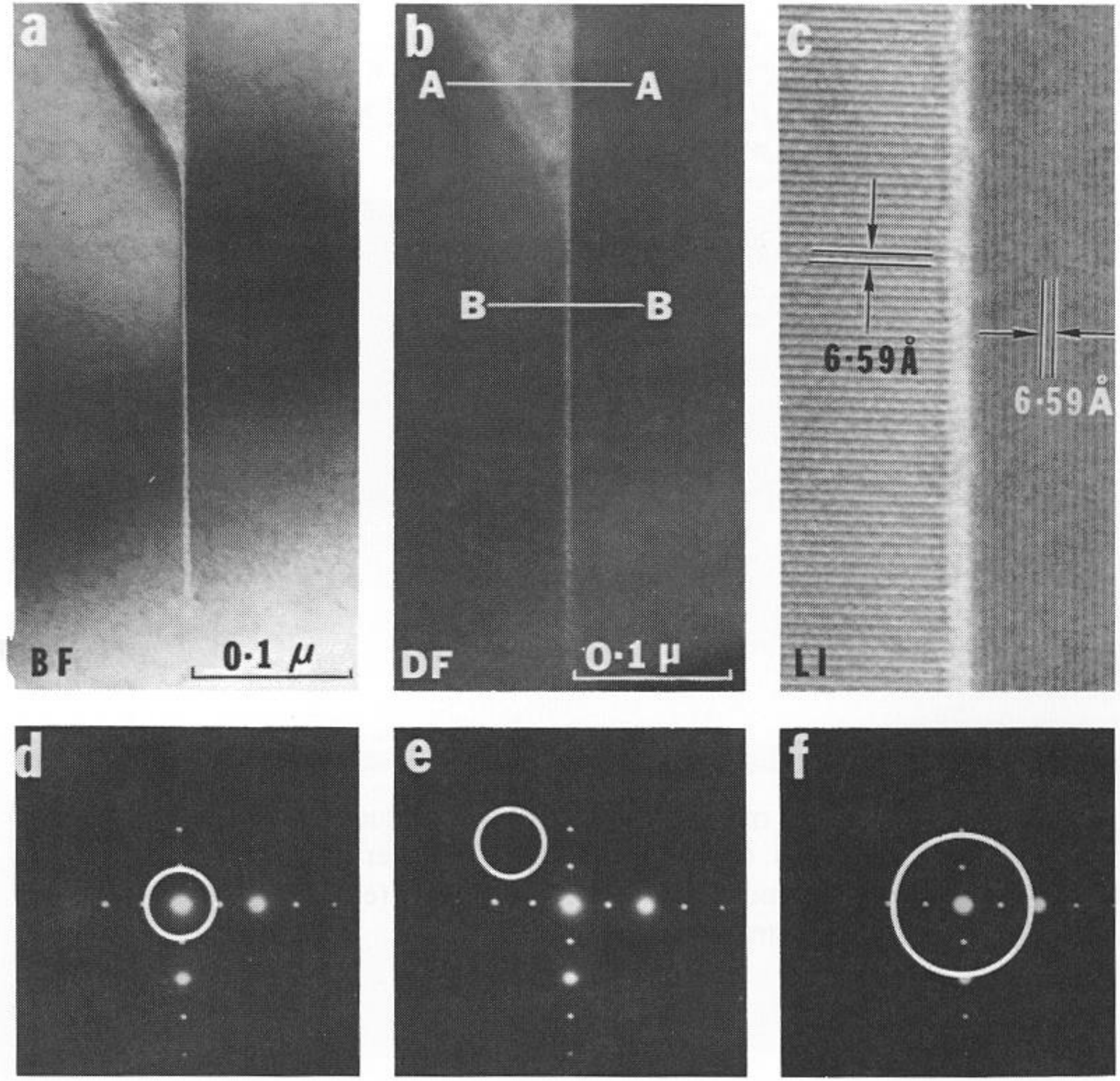
How to manipulate it to get contrast







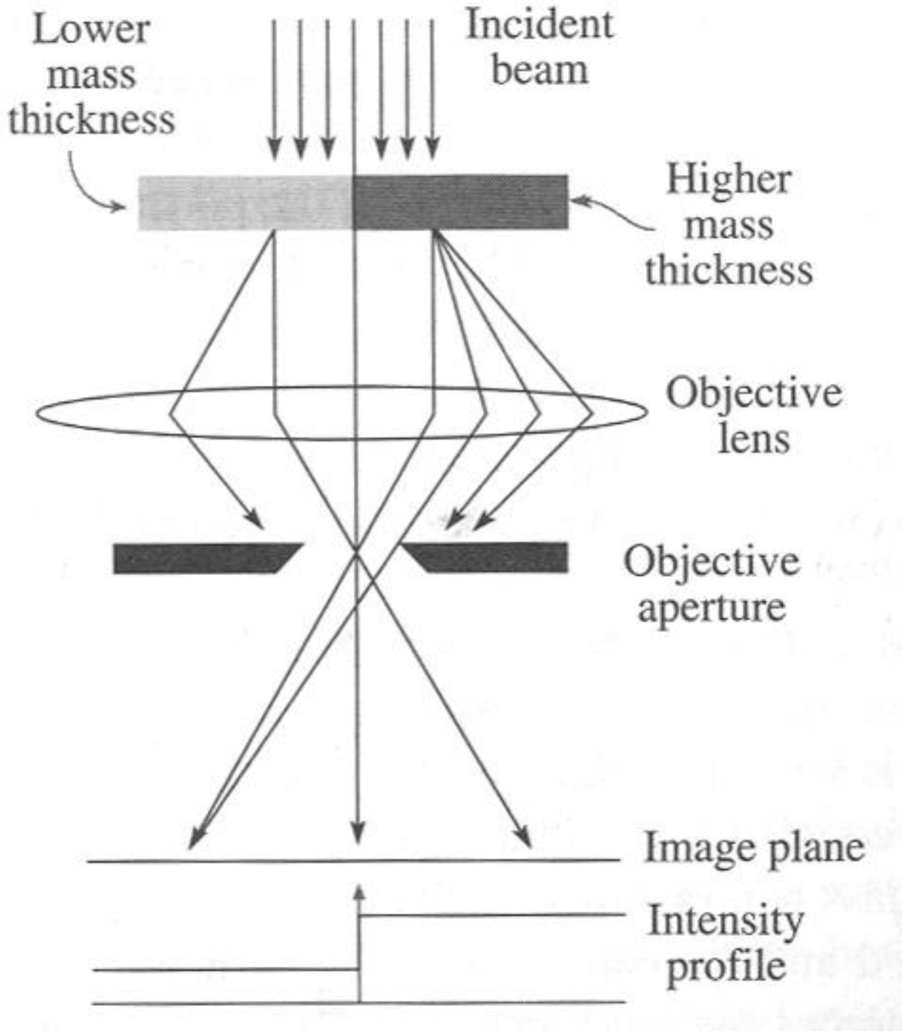
Positioning of Objective aperture to get contrast



Bright field Dark field HRTEM Imaging

(Courtesy G. Thomas, in Image of Materials, Oxford Press, 1991)

Mechanism of Mass-Thickness contrast

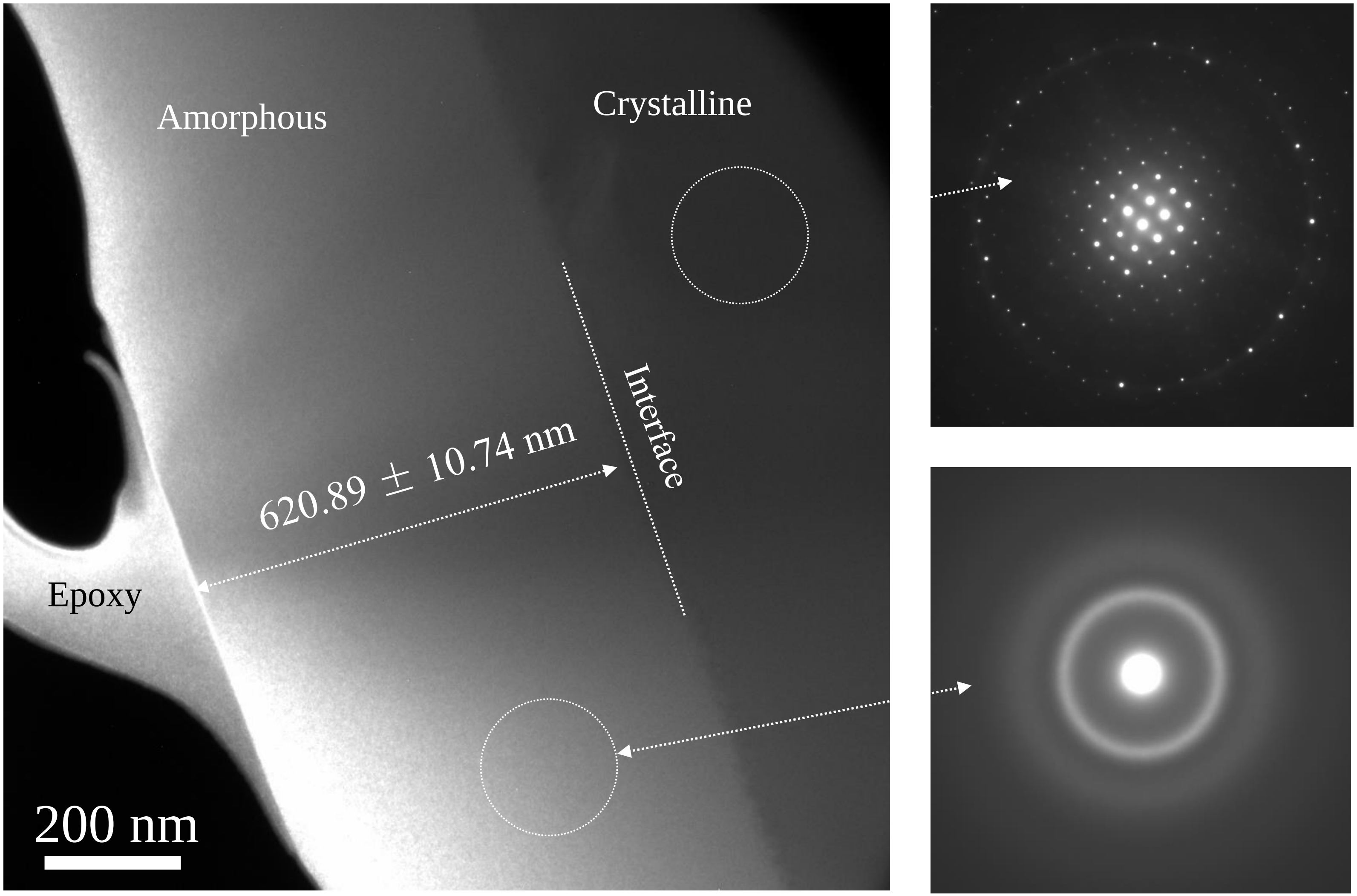


Contrast Mechanisms

Three Contrast Mechanisms

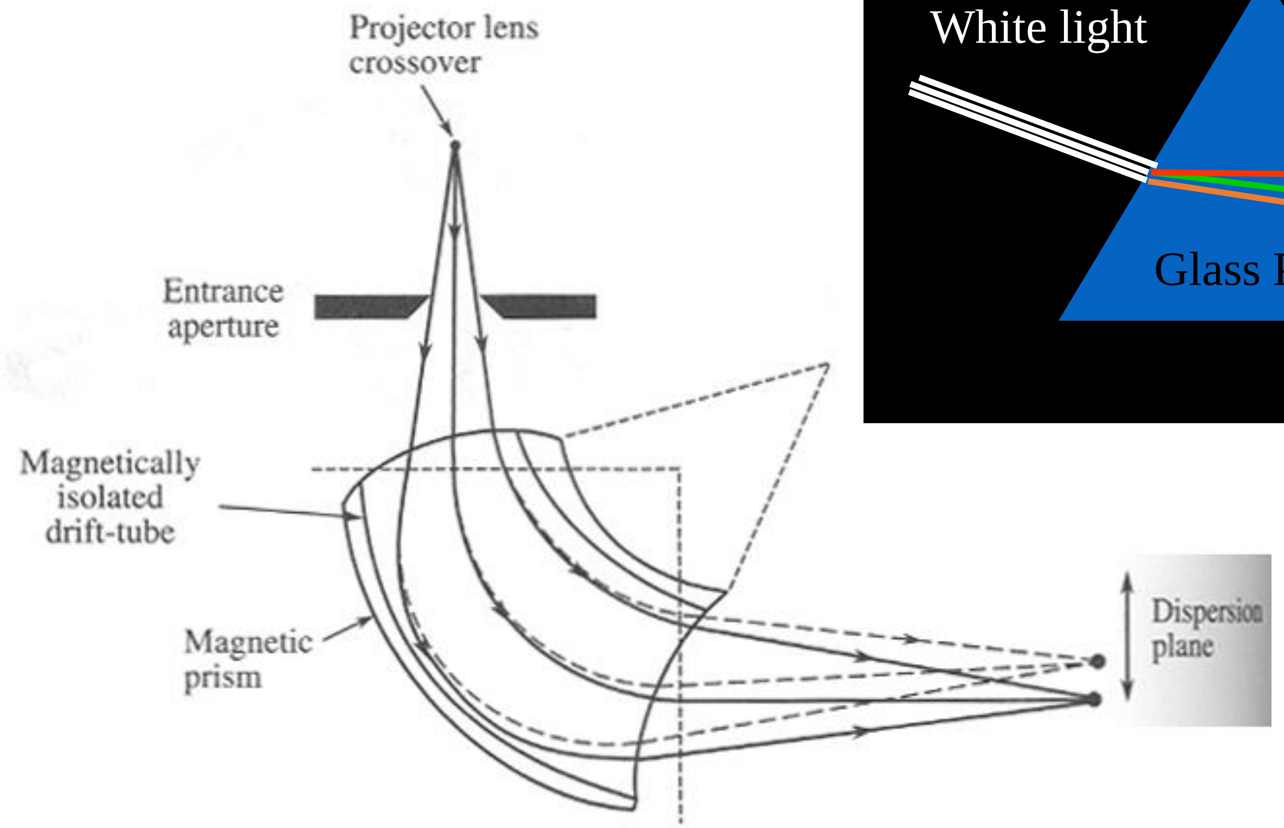
- Diffraction contrast: bright field and dark field
- Phase contrast (HRTEM)
- Mass thickness contrast

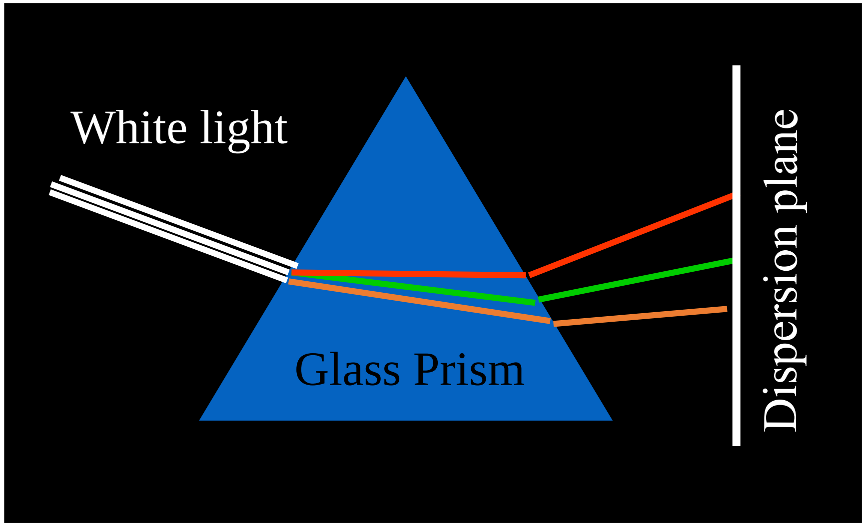
Imaging and diffraction

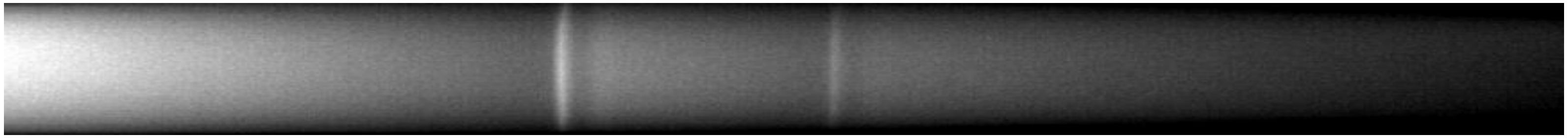


The thickness is an average measured on 4 different regions on the diffuse dark field image. One is these regions is shown above. The standard deviation is regardless the transitional Layer thickness of 20 nm as revealed in the following pictures.

Electron Energy-Loss Spectroscopy(EELS)





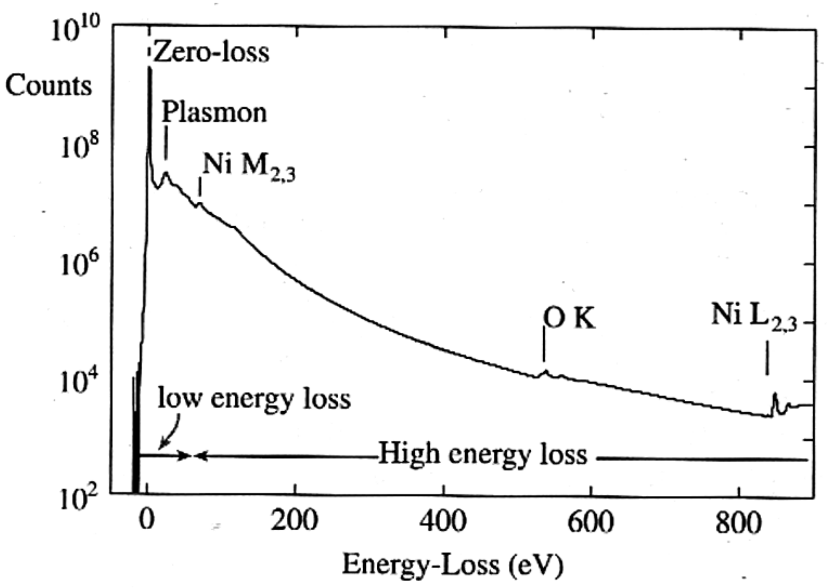


EELS in TEM/STEM

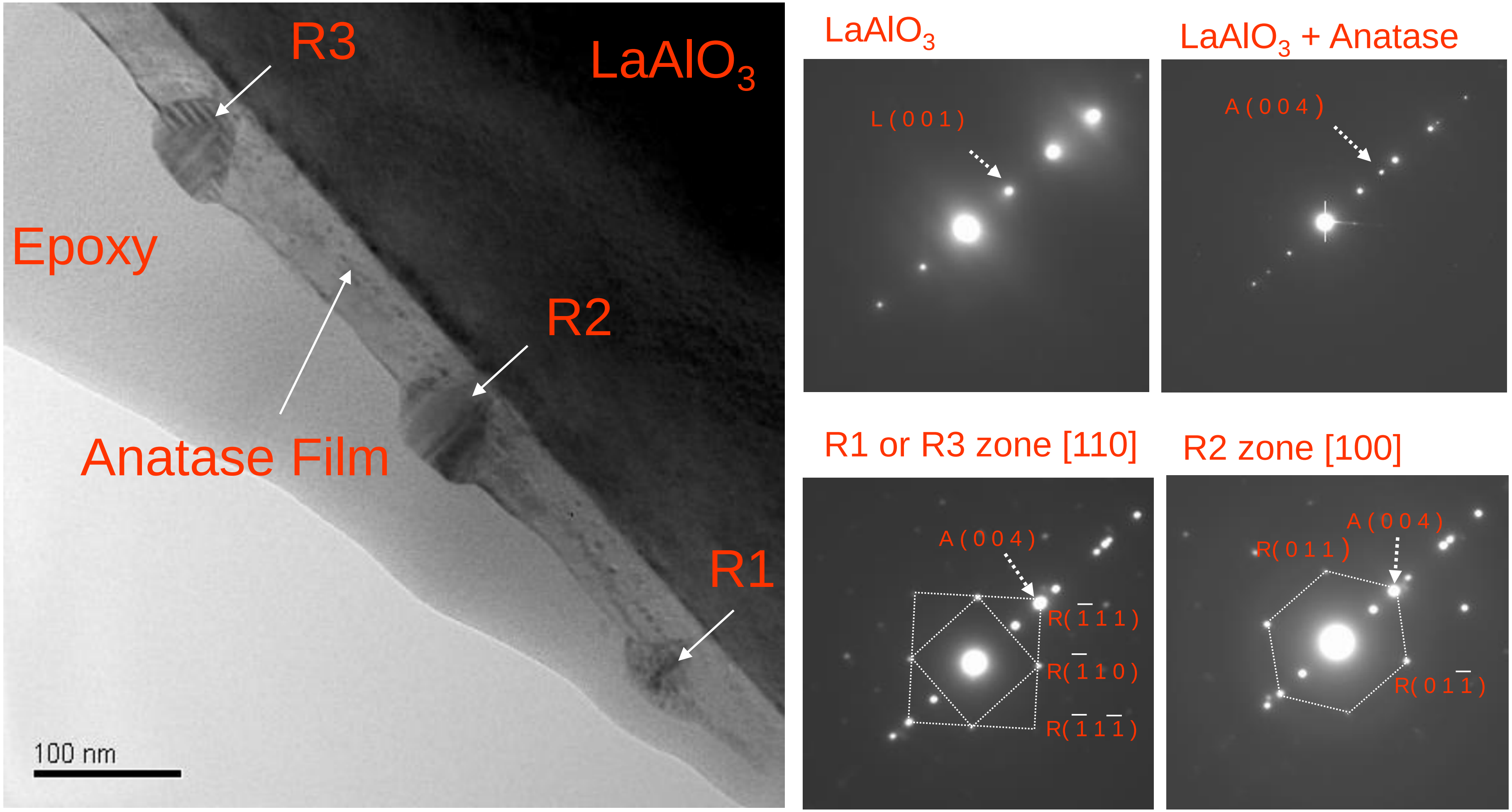
Analysis of the energy loss (and sometimes momentum-transfer) of electrons transmitted through the specimen

- Key Advantages (compare w/EDS):
  - Spatial resolution in STEM ~ d, the electron beam size
  - Detectability ~ complementary and competitive to EDS
  - Any solid/gas/liquid – in principle
  - Qualitative analysis of any element of $Z > 1$!!
  - Quantitative analysis by inner-shell ionization edges of many elements
  - Absolute mass/composition determination
  - Rich signal includes chemical bonding, valence, dielectric function, plasmon and free carriers, effective mass etc.

- Challenges:
  - Need very thin specimen: t <~ 30 nm
  - Stability/quirky experimental settings High intrinsic BG
  - Intensity weak for energy losses ΔE > 1000 eV
  - L- and M- edges not very obvious for some elements
  - Quantitative analysis/interpretation



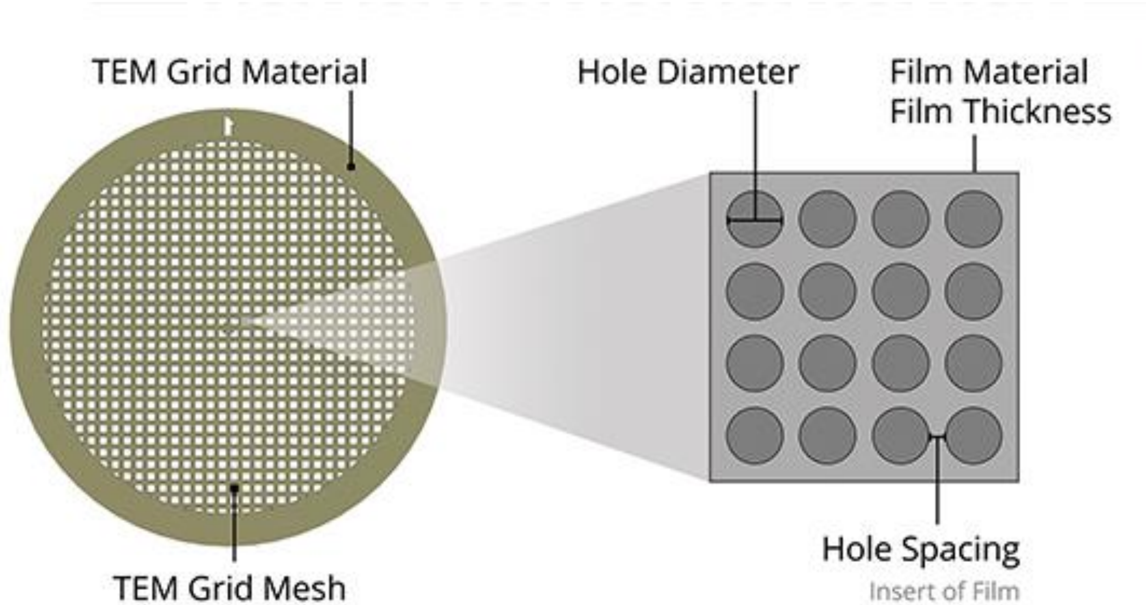
Pure TiO2, high density of precipitates



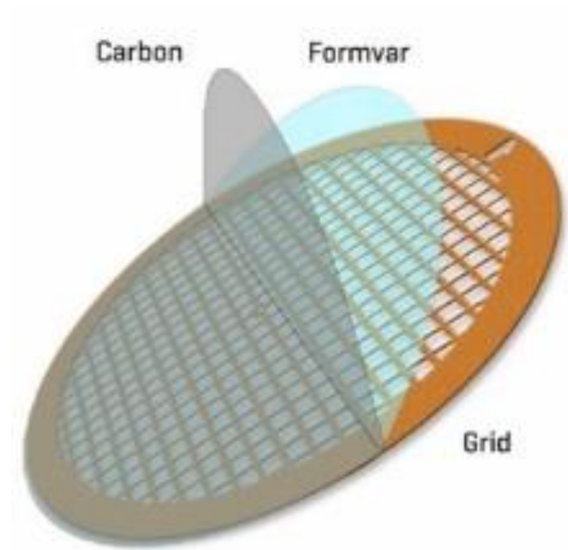
L: $LaAlO_3$ R: Rutile A: Anatase

TEM Sample Preparation

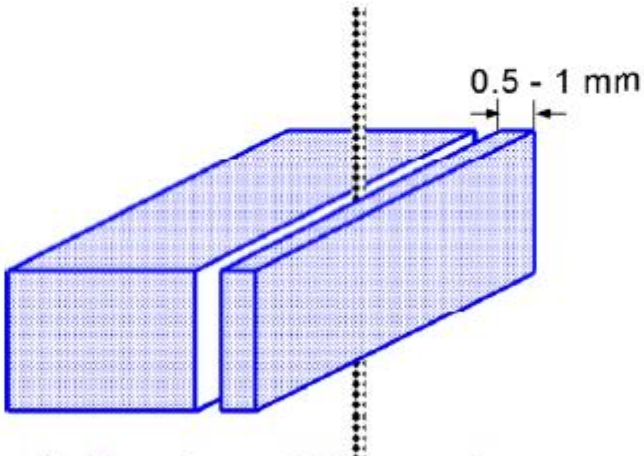
Nanoparticles from solvent



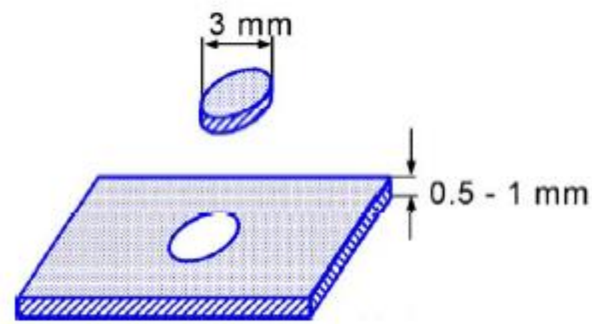
TEM grid



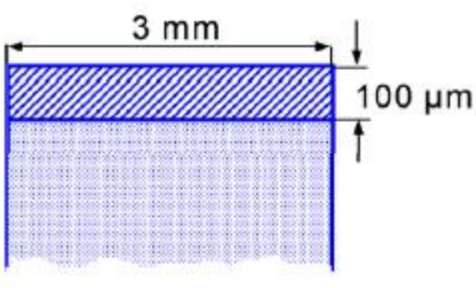
Standard Preparation



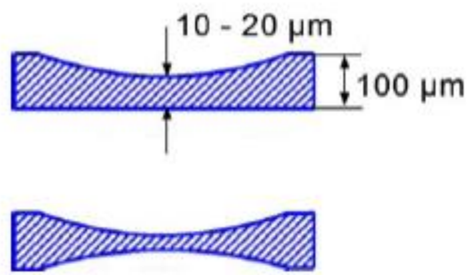
a) first cutting step



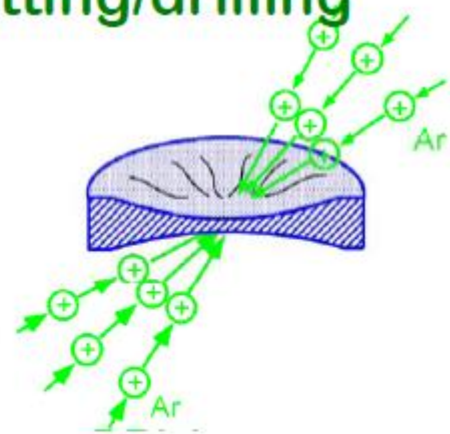
b) ultrasonic cutting/drilling



c) planparallel grinding

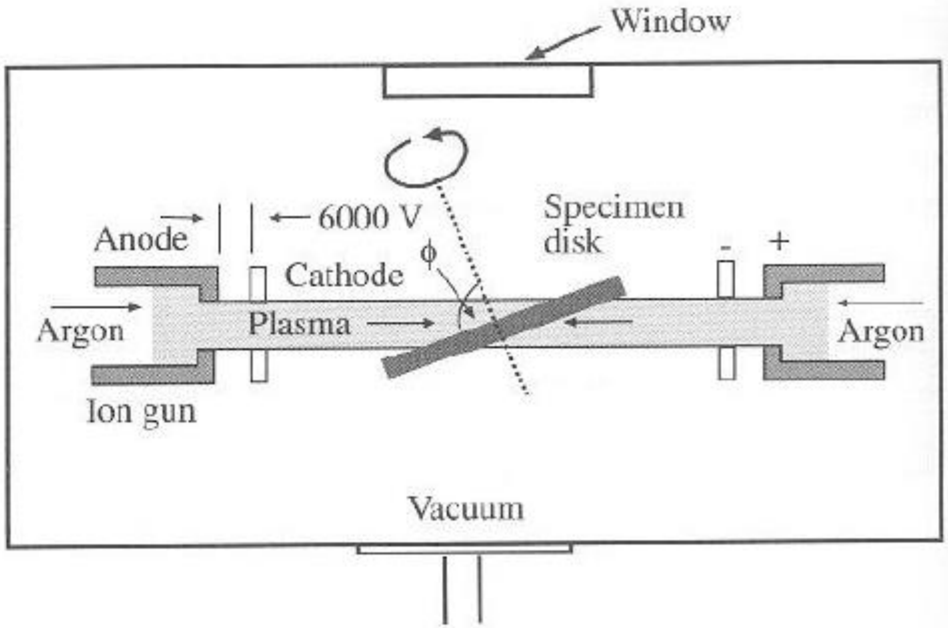


d) dimpling



e) ion-thinning

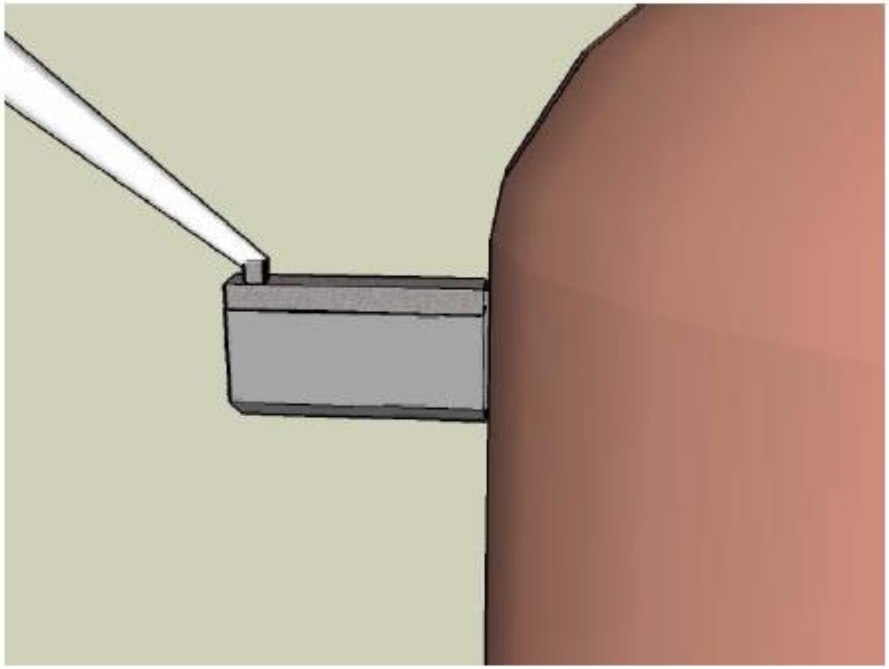
Ion-Milling



- ➢ **Energy: (~1-7 KV)**
- ➢ **Current: (~1-8mA)**
- ➢ **Angle: (~2-20º)**

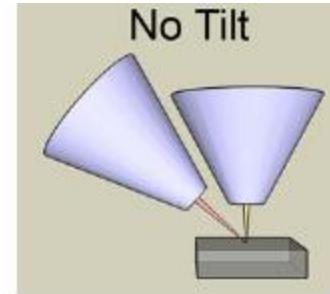
TEM Sample Prep with FIB

- Pt Deposition
- Bulk-Out
- U-Cut
- Lift Out
- Mounting
- Thinning
- Cleaning

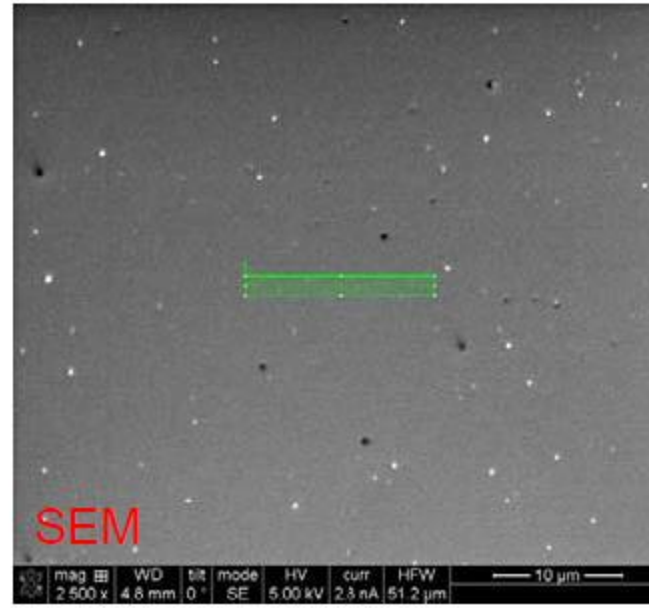


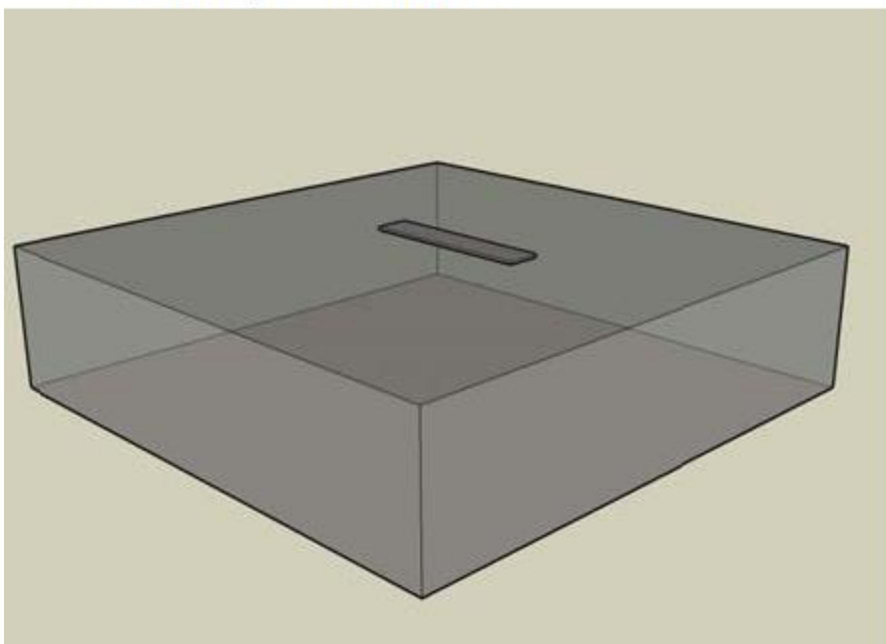
Step-by-step instructions with recipe information and 3-D rendering of each process

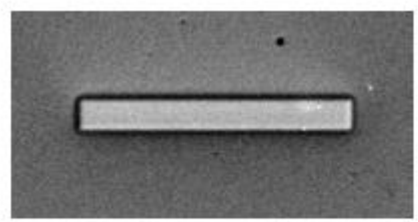
E-beam Pt



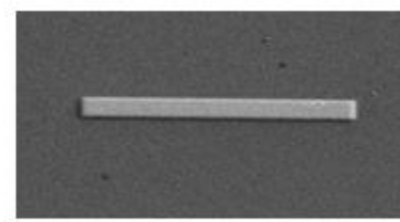
- 0° Tilt
- Application: Pt e-Dep
- Shape: Rectangle
- 15μm (X) x 1.5μm (Y) x 200nm (Z)
- 2-5kV, >1.4nA





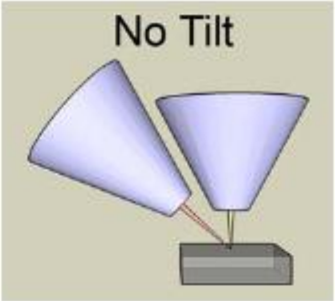


E-beam dep at 0° Tilt (SEM)

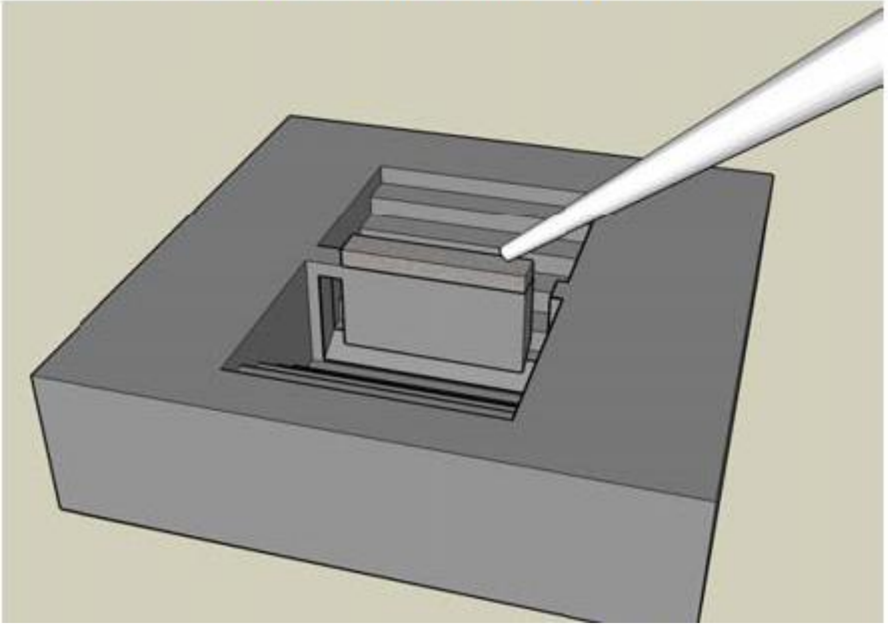


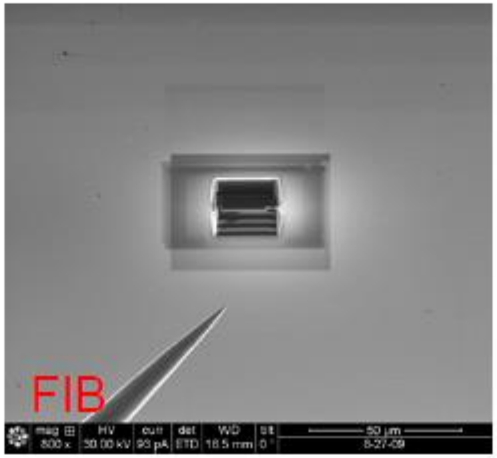
E-beam dep at 52° Tilt (SEM)

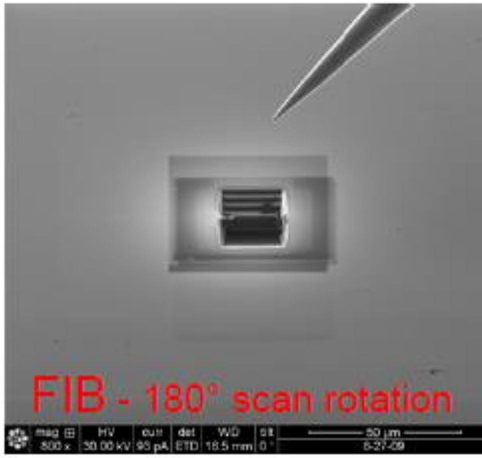
Lift-Out (1 of 4)

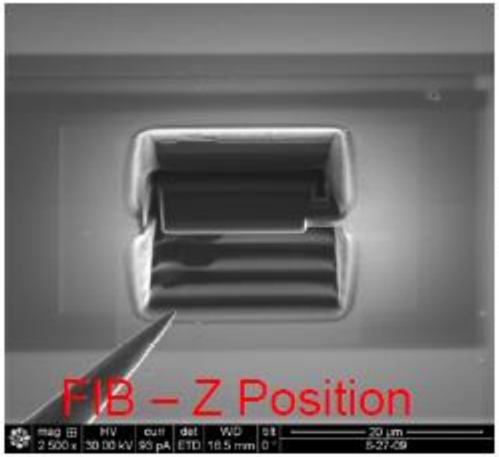


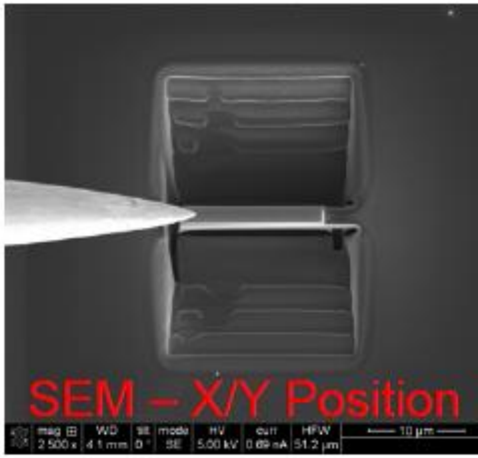
- 0° Tilt
- Insert Omniprobe at Park
- Drive to Eucentric High
- Lower to ~10μm from sample
- Insert Pt GIS needle
- Position tip within ~200nm of sample
- 180° scan rotation is helpful











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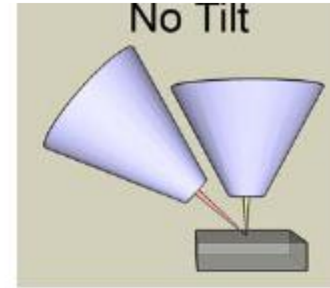
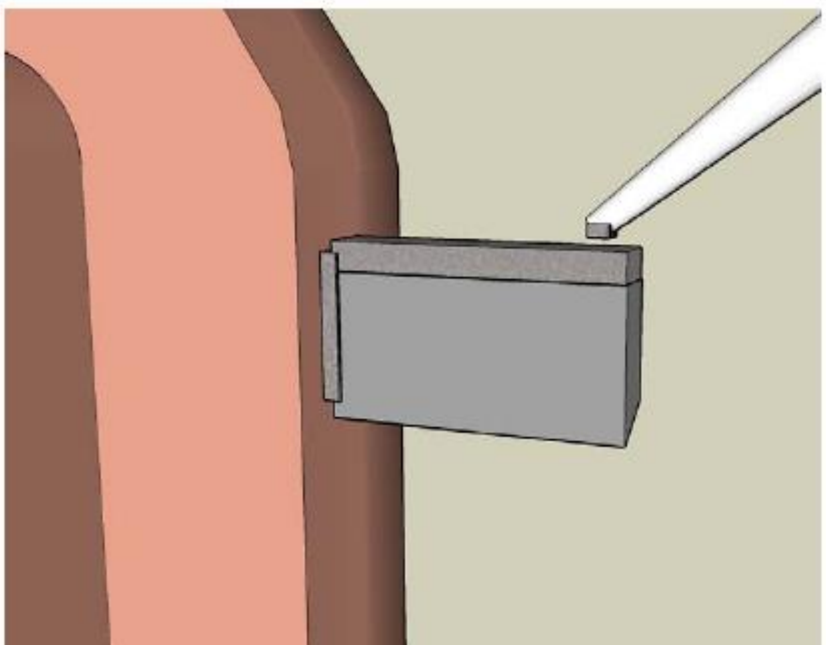
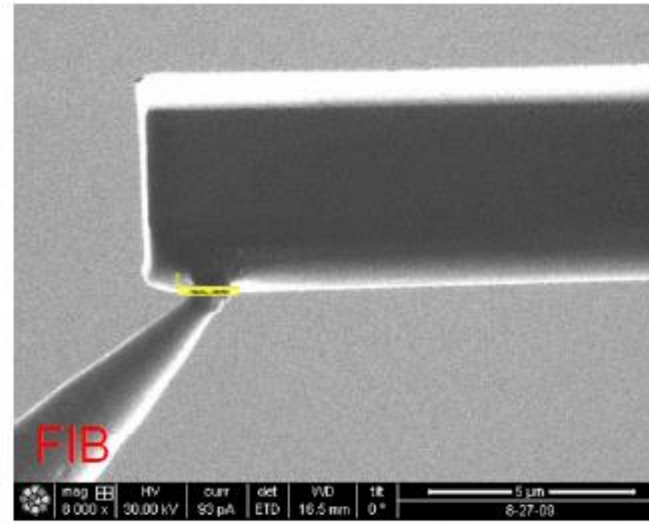
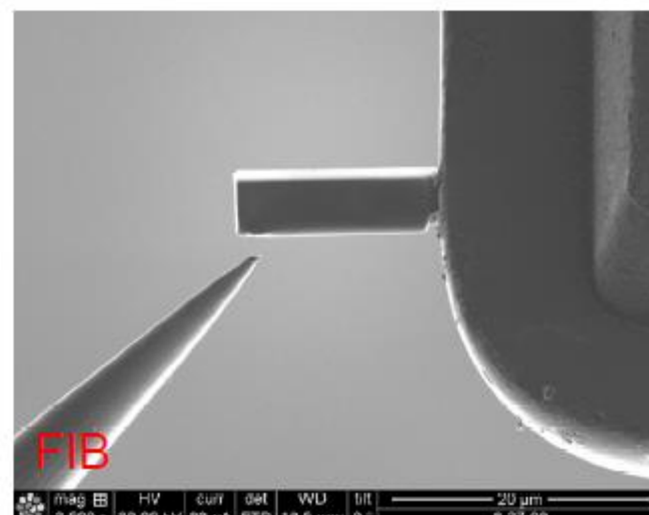
Mounting (3 of 3)



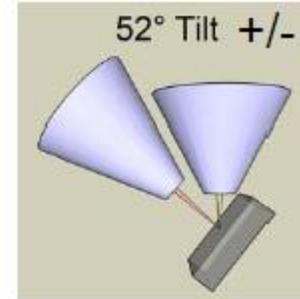
- 0° Tilt
- Application: Si
- Shape: Rectangle
- Size: ~2μm (X) x ~200nm (Y) x ~1μm (Z)
- 30kV, 93pA-2.8nA



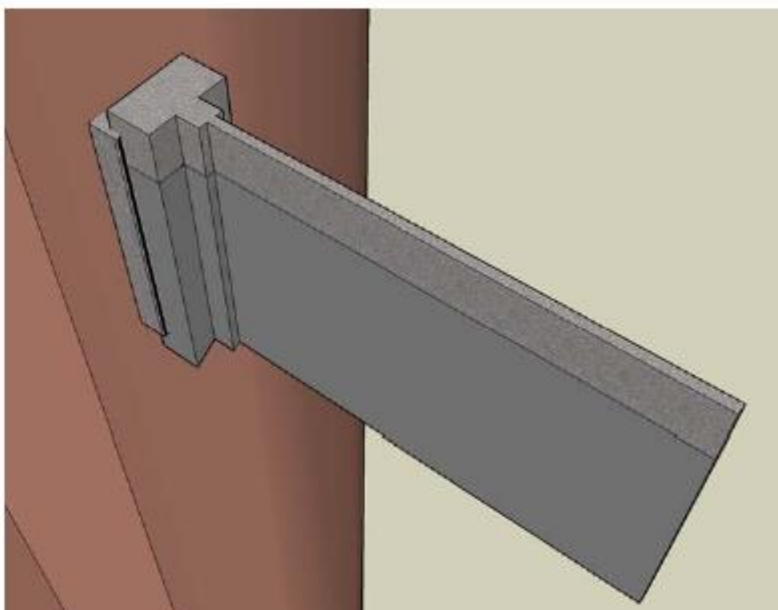


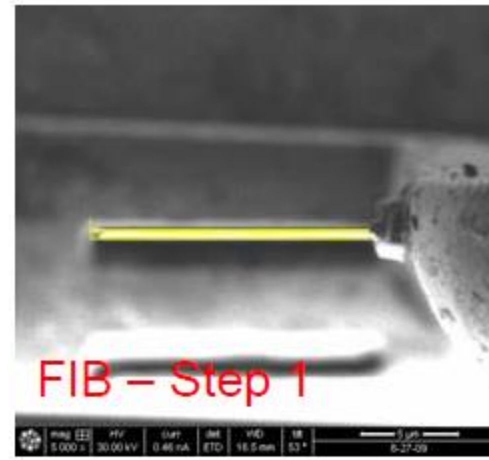


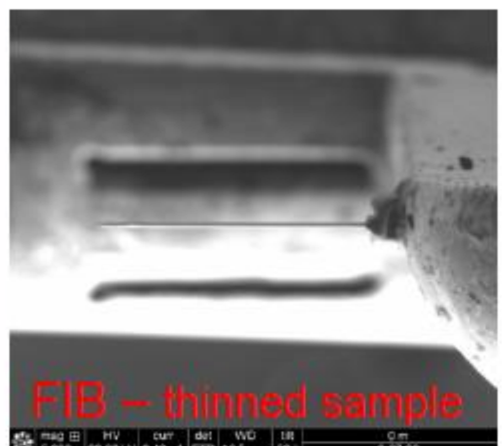
Thinning (3 of 3)

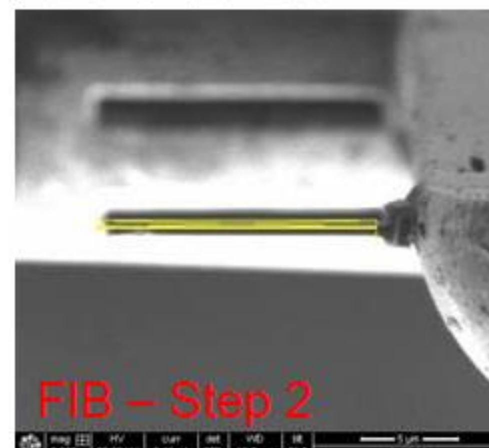


- Repeat thinning steps 1 and 2
- Target final sample thickness +60nm
- Application: Si
- Shape: Cleaning Cross Section
- Size: ~10µm (X) x ~200nm (Y) x ~3µm (Z)
- 30kV, 93pA-0.46nA



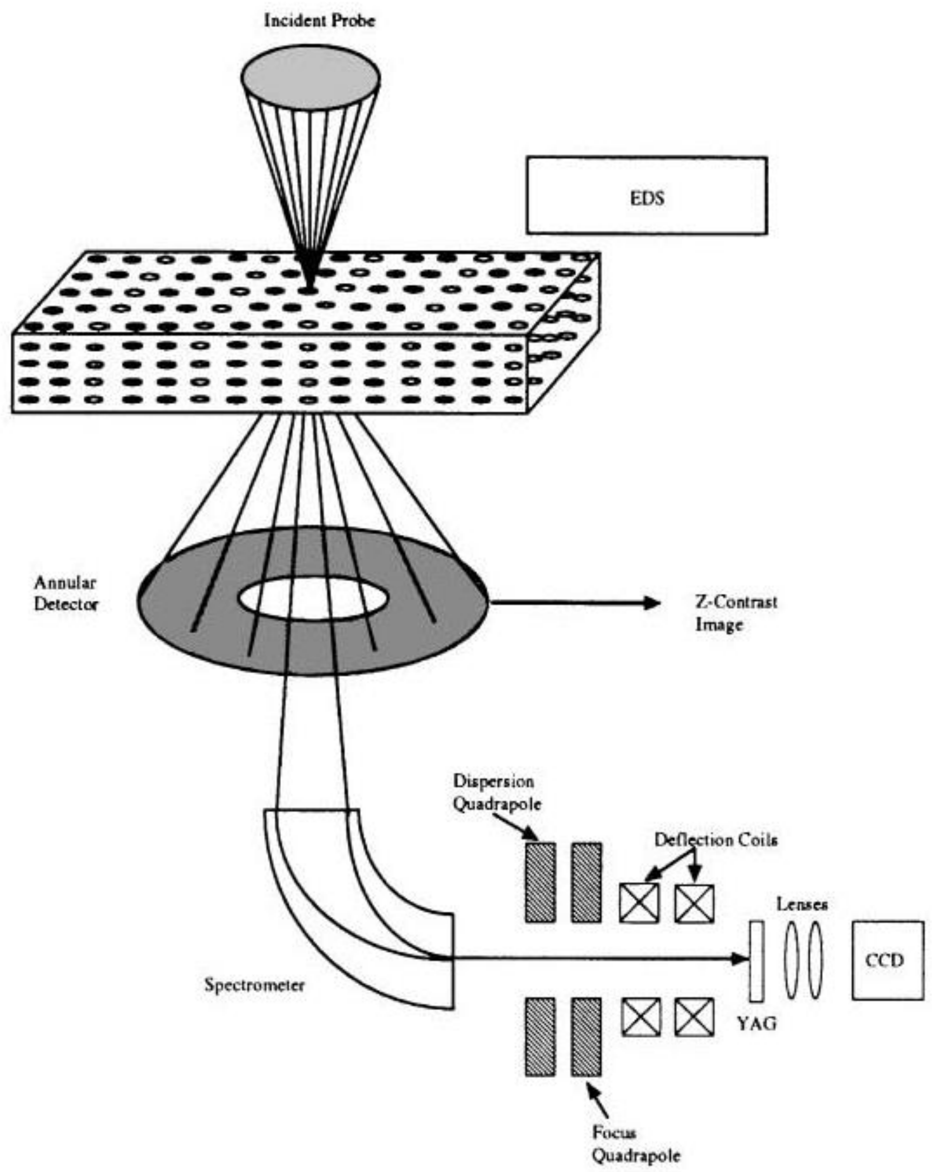




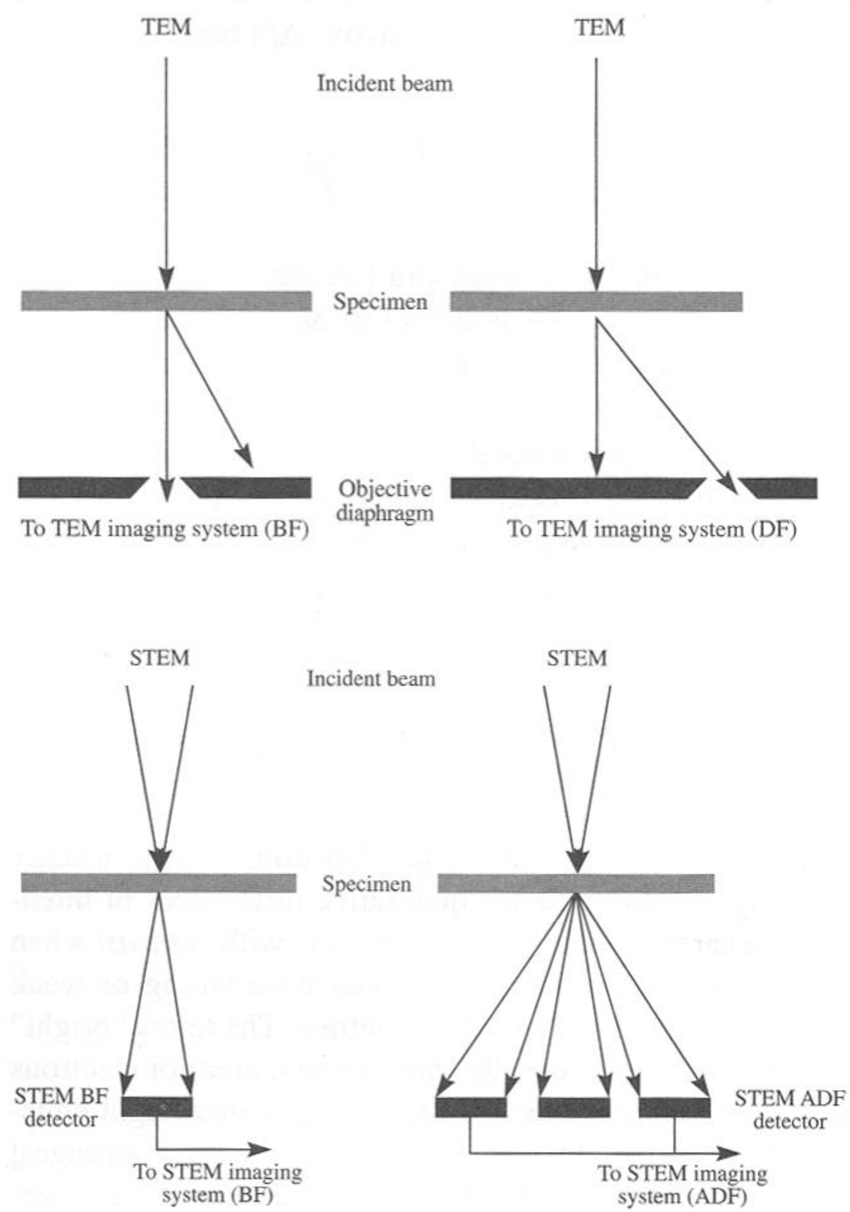


Scanning transmission electron microscopy (STEM)

Scanning Transmission Electron Microscopy (STEM)



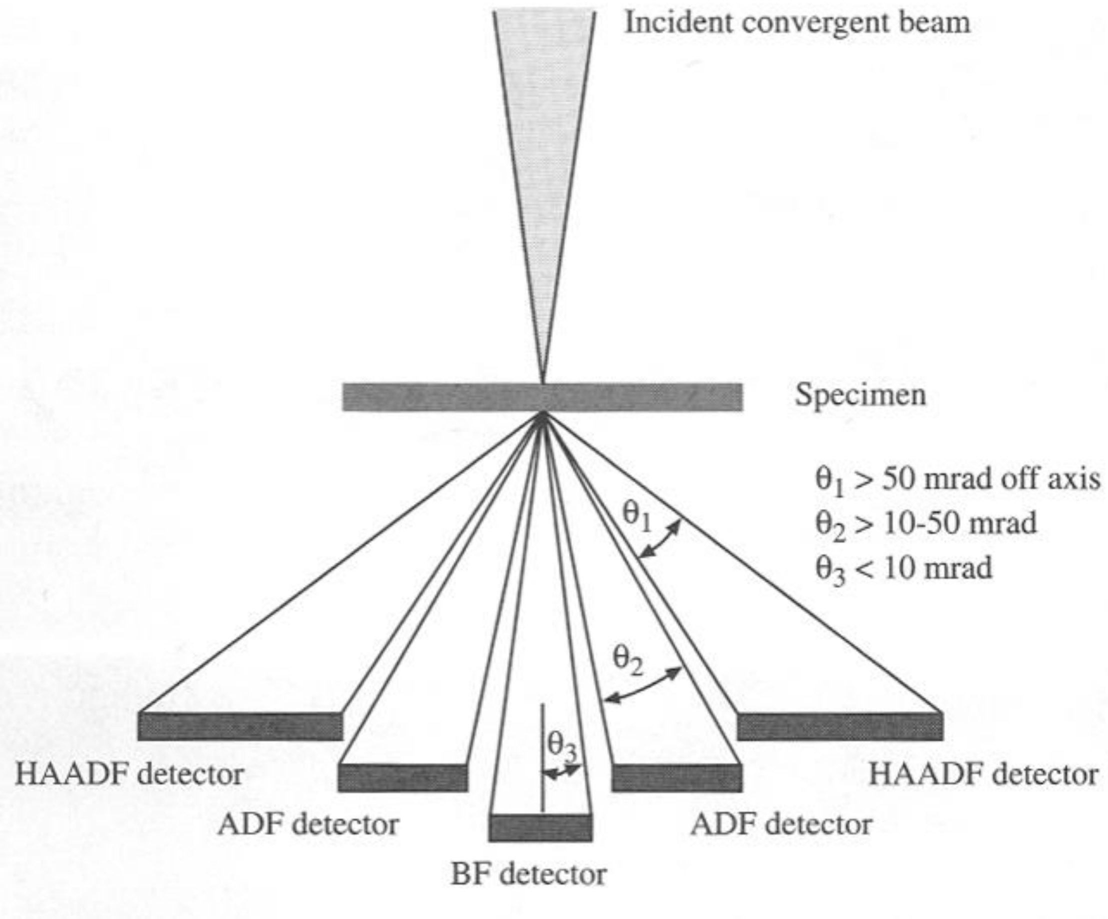
Scanning Transmission Electron Microscopy (STEM)



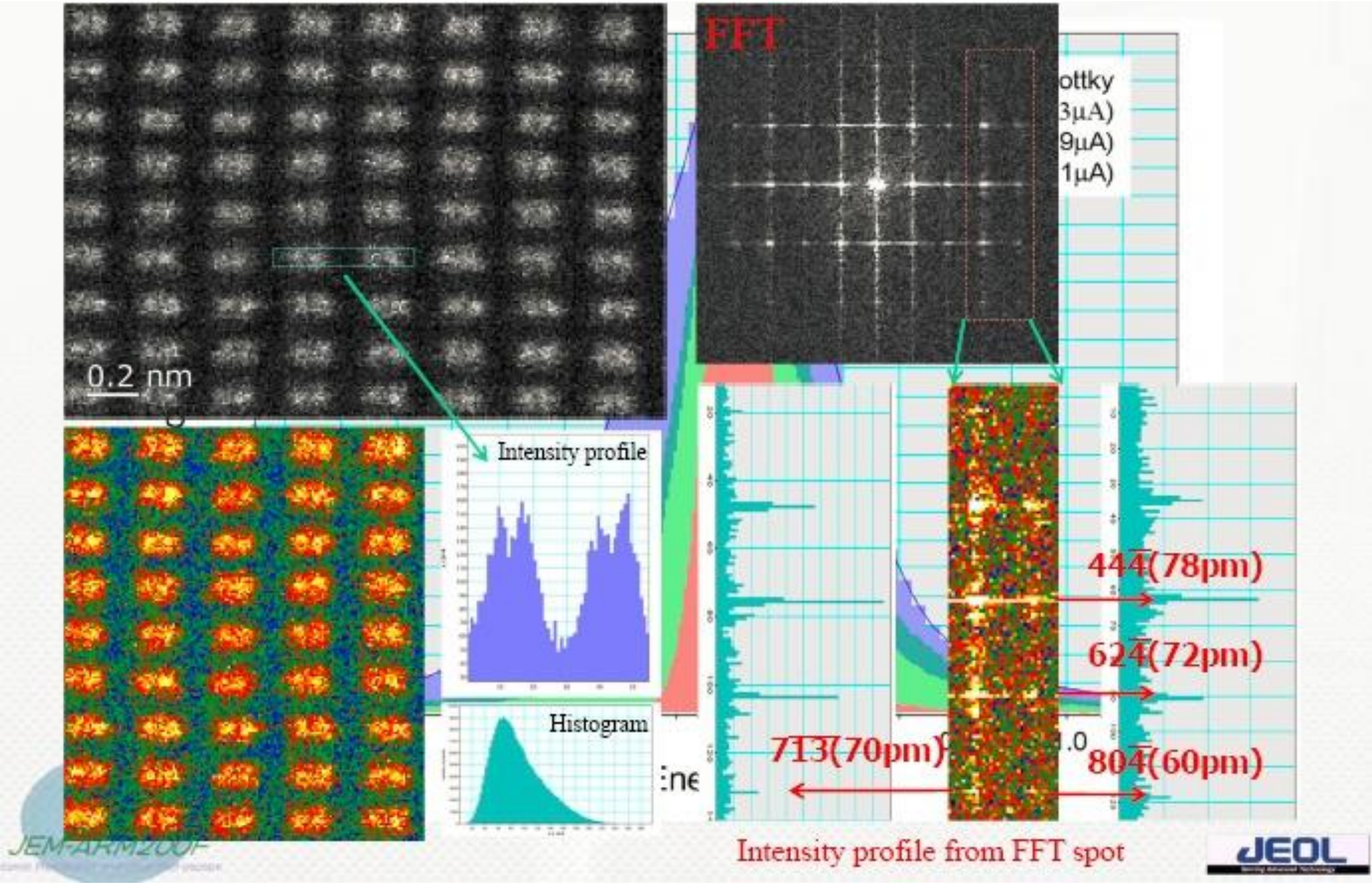
TEM: parallel beam

STEM: focused probe

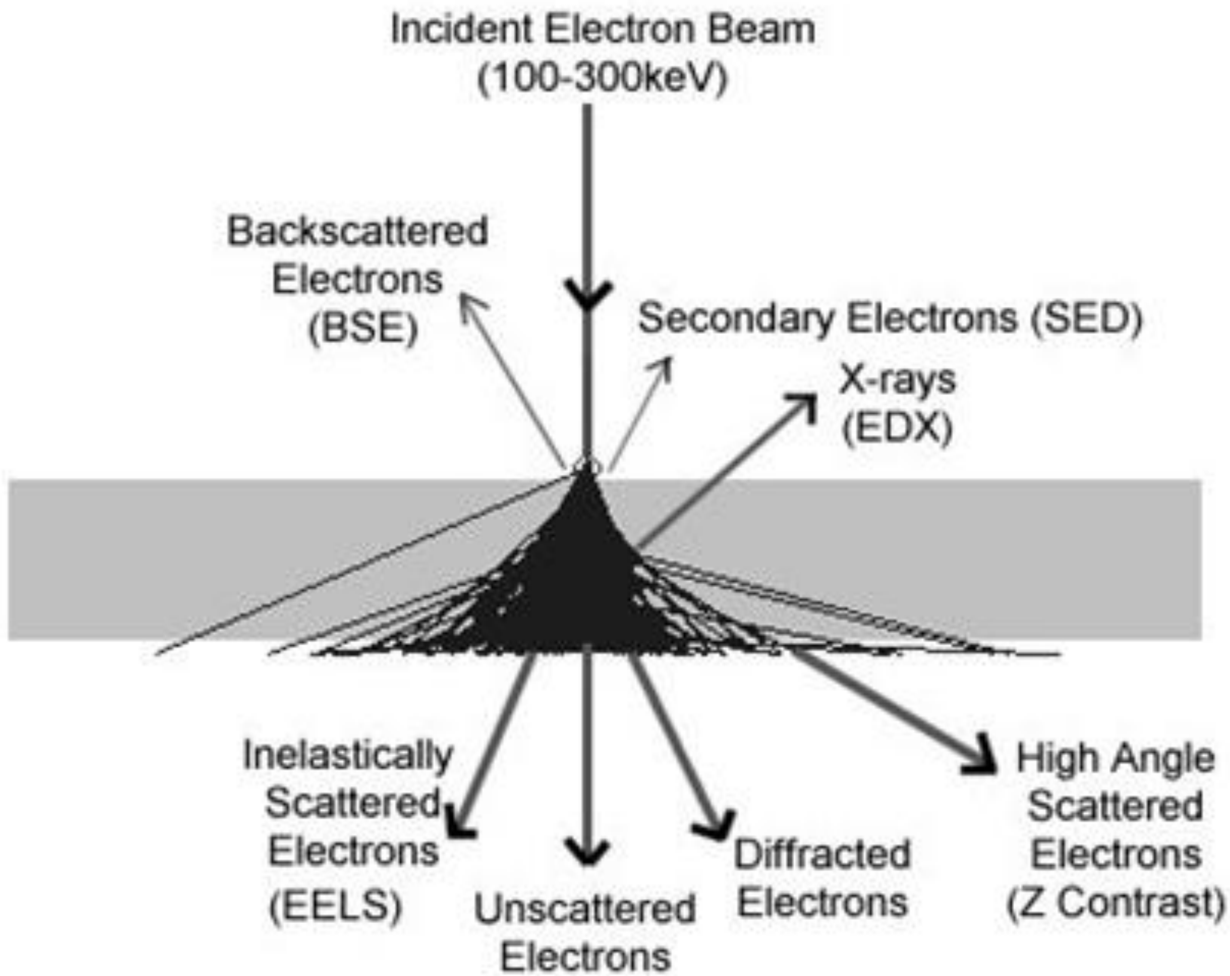
Scanning Transmission Electron Microscopy (STEM)

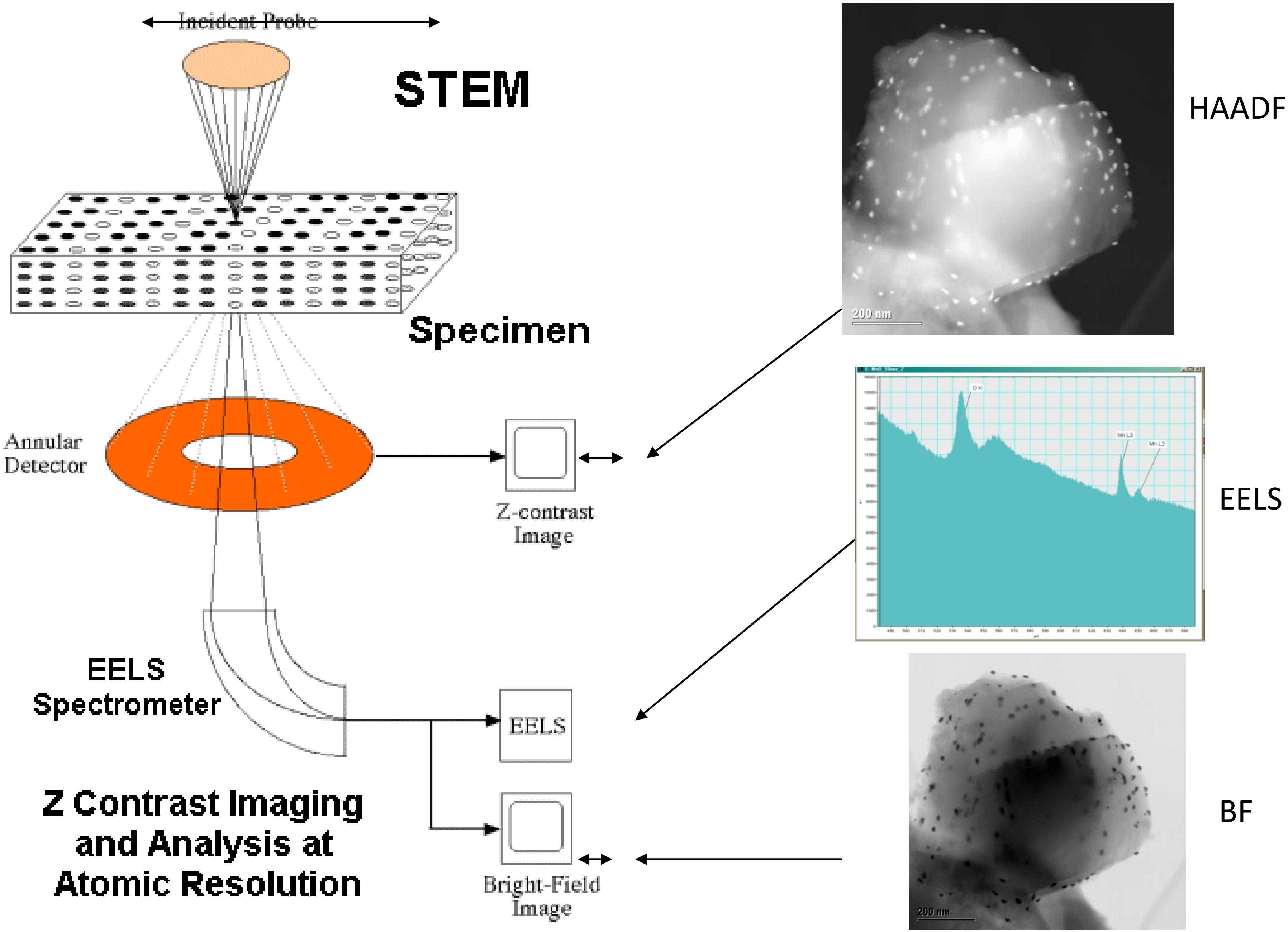


Schematic of the HAADF detector set-up for Z-contrast imaging


FFT
0.2 nm
Intensity profile
Histogram
44$\bar{4}$(78pm)
62$\bar{4}$(72pm)
7$\bar{1}$3(70pm)
80$\bar{4}$(60pm)
Intensity profile from FFT spot
JEM-ARM200F
JEOL

The electron/specimen interaction

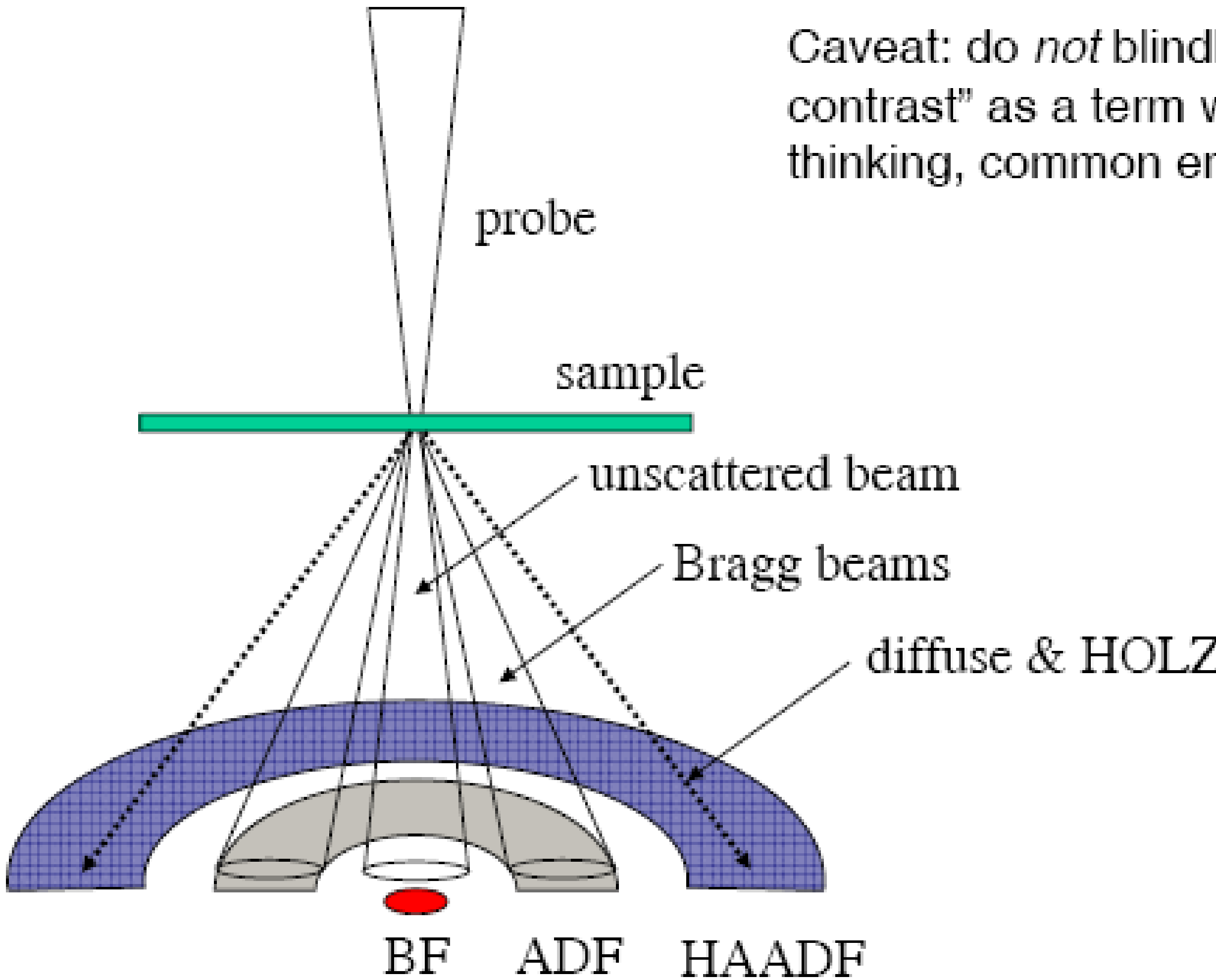




Incident Probe
STEM
Specimen
Annular Detector
Z-contrast Image
EELS Spectrometer
EELS
Bright-Field Image
Z Contrast Imaging and Analysis at Atomic Resolution
HAADF
200 nm
O K
Mn L3
Mn L2
EELS
BF
200 nm

Bright Field & Dark Field

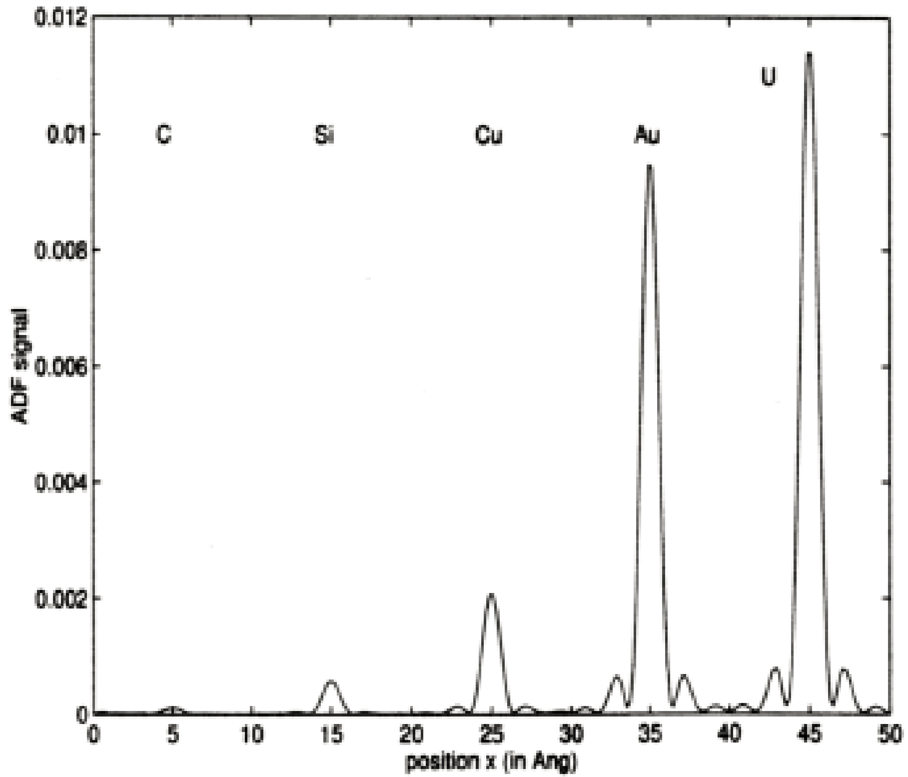
Pick different parts of scattering space with different detectors.



Caveat: do *not* blindly use "Z-contrast" as a term without thinking, common error

Z-contrast

- Scattering scales as $\sim Z^{1.7}$ for common scattering angles
- Images are readily interpretable
- Images contain some chemical information
- Many problems with this simple picture (good enough for Govt work)



E. J. Kirkland, *Advanced Computing in Electron Microscopy* (Plenum, 1998).

Summary

STEM versus TEM

- STEM = TEM, but with the optics reversed (if you ignore inelastic scattering)
- BF, DF, HREM can be done in a STEM almost the same as a TEM
- However:
  - The imaging is serial as against parallel, so in general has more noise and lower signals
  - Scan coils and electronics involved, which leads to noise
  - Some types of imaging can be done easier in a STEM than a TEM (they have all been done in both)
  - STEM needs a small source (demagnified further)
  - Some manufacturers have decided to reinvent the wheel